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The Role of Eye Contact in Multi-Party Human–Robot Interaction

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Abstract

One of the main components of a human conversation is the turn-taking event. However, to give robots this ability to participate naturally in a multi-party interaction remains challenging. Humans use several cues to detect when to take or yield the turn, but two of the main signals are silence and eye contact. The problems arise with accurately detecting eye contact as human gaze behavior is very uncertain. In addition, the interaction quality and how humans perceive the robot during a conversation may be influenced by the timing of the robot's own gaze behavior. This study investigated how gaze detection and the robot's gaze behavior can influence turn-taking performance and the user perception in a multi-party setting.

A 2 x 2 mixed experimental design is conducted with 40 participants. Two gaze detection models are used and compared to each other: an extended gaze detection model that accounts for uncertainty in eye movements and a direct gaze detection model, where participants had to align their heads directly with that of the robot. Next to this, the robot either had a static head orientation towards the middle of the two participants while speaking, or it changed its gaze between the two participants based on timings from human-human interactions (timed gaze aversion). Turn-taking opportunities were detected when at least one second of silence and 0.5 seconds of eye contact were observed. Quantitative measurements are analyzed. For behavioral measures, interaction logs are used that include eye contact moments, turn-take durations, yaw angles of the participants, silence durations, utterance durations, and audio levels. The self-reported subjective measurements were collected through questionnaires that measured the participants' perception of the robot, including likeability, accuracy, warmth, discomfort, competence, habitability, cognitive demand, annoyance, and speed.

The results showed that the extended gaze detection model detected more eye contact moments compared to the direct gaze detection model. Showing that eye contact detection in multi-party human-robot interaction benefits from accounting for uncertainty in gaze behavior rather than relying solely on direct alignment with the robot. However, only a limited proportion of detected opportunities resulted in successful turn-taking. No significant effect was found on the speaking duration and the time needed to take the turn. However, there was a difference in the duration of silence, showing that the silences between the onset and offset of speech were longer in the group of the direct gaze detection model. The extended gaze detection model produced more silences around 1.5 seconds. This suggests that accounting for uncertainty in gaze behavior may enable the robot to identify turn-taking opportunities earlier, reducing prolonged pauses in the conversation. The subjective measurements showed that the participants evaluated the robot significantly more likeable when it used changing gaze behavior when it spoke, resembling natural human behavior. Finally, the combination of an extended gaze detection model together with the timed gaze aversion behavior of the robot had the highest rate of successful triggers for turn-taking. However, the rest of the subjective measurements resulted in a non-significant effect.

Overall, the findings show that the robot's ability to detect eye contact and silence influences the detection of turn-taking moments, and that the gaze behavior of the robot affects how humans perceive the robot as a conversational partners. Unfortunately, the improvements made to the gaze detection and the gaze behavior of the robot did not sufficiently improve the overall quality of the conversation. It highlights how important it is to accurately integrate both gaze detection and changing the robot's gaze behavior together with other conversational cues to get a more natural and human-like interaction.

Keywords: Human-Robot Interaction, Turn-Taking, Eye Contact Detection, Multi-Party Conversation, Gaze Behavior, Silence Duration, Conversational Robots, Floor Management, Speech

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Introduction

During conversations, people must coordinate when to speak or when to listen. Turn-taking is therefore one of the main aspects of human-human interaction. In conversations, people coordinate speech timing, gaze behavior, gestures, and other nonverbal and verbal cues to manage the conversational floor (Skantze & Irfan, 2025). Humans engage in three key floor-management behaviors during conversations: floor-holding, turn-yielding, and turn-taking. Floor-holding refers to the speaker’s maintenance of the conversational floor, signaling the intention to continue speaking. Turn-yielding occurs when the speaker signals that they are done and hands over the floor to someone else. Turn-taking is the process by which a non-speaker takes over the floor (Jokinen et al., 2013). Understanding these mechanisms is important when the interaction is extended to artificial agents. Robots are deployed in several domains where they interact with humans, including collaborative work environments. Robots should be able to understand and respond to individuals’ behaviors. This means that they should understand nonverbal and verbal cues, as these cues are necessary to interact more fluently in collaborative environments. The main non-verbal and verbal cues include gaze direction, speech, and gestures, which closely resemble those used in human–human conversation (Mutlu et al., 2012; Skantze, 2021). Previous research shows that social robots can contribute to group cohesion and productive collaboration (Weldon et al., 2021). However, it keeps challenging to let the robot use the same verbal and nonverbal cues as humans use in conversations. When timings are poorly calibrated in human-robot interaction, conversations can feel unnatural as robots will have delayed reactions, interrupt humans, or maintain gaze in inappropriate ways that can lead to feelings of staring (Skantze, 2021). Robots that make turn-taking decisions base these decisions on simple signals; examples are the absence of speech (silences such as pauses or hesitations), but they also use signals like the head-pose of the conversational partner in that interaction (Skantze et al., 2015). To get a better understanding of the way robots manage turn-taking in conversations, it is necessary to examine how humans coordinate these interactions.

Turn-taking in Human Conversations

Conversations are managed through smooth transitions between participants in a conversation. Turn-taking is defined as the organization of speaking and listening in a conversation so that only one person speaks at a time (Sacks et al., 1974). To do that smoothly and efficiently, people have to take into account verbal and nonverbal cues (Jokinen, 2010). A key concept within the turn-taking mechanism is the transition relevance place (TRP), which refers to the moment the current speaking turn can be considered complete, allowing other participants to take the floor (Hara et al., 2019). According to Hara et al. (2019), turn-taking can be described in a two-step process in which the listener judges whether the current turn can be completed (TRP) and then decides whether to take the turn. A more detailed description is provided in a study by Sacks et al. (1974), who established three rules for turning in conversations. These three rules describe how the next speaker is chosen at a TRP:

1. The current speaker chooses who speaks next.
2. A listener may self-select as the next speaker.
3. The current speaker continues speaking.

If the speaker continues (per rule 3), the same rules apply again at the next TRP. These rules help ensure smooth coordination and efficient turns, minimizing overlapping speech.

Turn-taking relies on predictive processing, which happens when people understand what the speaker is going to say and when cues at the end of the turn signal the start of speaking (Barthel

et al., 2017)). This is often needed in human-human conversations as the gaps are short between the speakers. Turn-taking has to be appropriately timed because that would prevent unnecessary cut-ins, which are moments when people overlap in conversations, or when they want to take the floor while another speaker still has the floor (Spiegels, 2015). Furthermore, it reduces silences (less uncomfortable and awkward situations) and reduces overall cognitive load (Spiegels, 2015). It remains challenging to implement turn-taking behavior that is similar to humans into robotic systems. In a multi-party setting, multiple people can be addressed, making it even more important to have appropriate turn-taking behavior. Two of the main cues that signal the next speaker are speech and gaze. That is why it is useful to examine their roles in turn-taking in more depth. That is why it is useful to examine their roles in turn-taking in more depth.

Speech in Human Conversations

Speech helps with regulating conversational floors and plays a big role in spoken communication. In human-human interactions, the reaction time between the speaker and listener is rapid. Typical gaps between speakers are 200 ms, suggesting that listeners anticipate turn endings before silence occurs (Skantze, 2021). It requires some time to produce speech, which is why listeners have to rely on predictive mechanisms that help them to prepare a response even when the speaker is still talking (Barthel et al., 2017). The predictive mechanisms often make use of syntactic and semantic patterns in the utterance of the speaker, but it is not uncommon to also rely on the prosodic signals at the end of the utterance.

When someone is speaking, silences occur naturally between utterances. However, a silence can have multiple meanings and is therefore seen as an ambiguous cue to signal turn completion. Silence can be a pause that a speaker uses to search for words, organize their thoughts, or it happens when the speaker hesitates. These reasons do not explicitly mean that the speaker is ending their turn, and therefore it can be an ambiguous signal for turn-taking. According to Spiegels (2015), speaker hesitations can extend to around 550 ms. These short pauses do not have to indicate that the speaker yields the floor, which makes silence cues alone an unreliable signal for turn-taking.

Speakers use pauses, fillers, and repetitions to maintain the floor while they are planning their next utterance, often producing fillers like "uh", "well", or "um" (Awang et al., 2022; Shokirjonovna & Muxtarovna, 2025). Speakers try to signal that they don't want to finish the turn, with the use of these short utterances, and that they want to continue speaking. These short utterances show that the speaker is processing and thinking, while it also softens the abrupt transitions and makes the spoken conversation more natural (Shokirjonovna & Muxtarovna, 2025).

A challenge is to get the appropriate timing of these cues, because these cues are often context-dependent and can have multiple interpretations. A short pause may indicate a hesitation of the speaker, whereas a long pause may indicate a TRP where others can take the turn. In conversations, the participants of a conversation can make use of fillers and backchannel responses to signal hesitation, understanding, and engagement to the listener. These backchannel responses are short verbal or non-verbal utterances, like "uh-huh" or head nods, that are often produced by the listener to signal that they are still paying attention to the speaker (Heinz, 2003). The backchannel responses overlap a lot of times with the speaker's turn, so it is easy to misinterpret them as attempts to take the turn. This is also one of the reasons why it is challenging to know the exact moment when the speaker ends a turn. Humans can naturally interpret these signals to anticipate the end of a turn. However, it is still a complex process for robots to understand the exact moment for a floor change in conversations.

Gaze in Human Conversations

The main nonverbal channels are gaze, gestures, and facial expression that organize conversations and provide feedback (Jokinen, 2009). In conversations, gaze is used as a nonverbal signal in managing the conversational floor. Gaze refers to when you direct your eyes towards a person or an object, including how long you maintain this visual attention (Hessels, 2020). Gaze is not only a good indicator of attention, but it also helps with turn-taking.

According to Admoni and Scassellati (2017), eye gaze can be categorized into several types, four of which are:

- *Mutual Gaze*: occurs when one person directs their eye gaze toward another person's eyes or face, and that person returns the gaze.
- *Referential Gaze*: is a gaze that is directed to an object.
- *Joint Attention*: happens when multiple people direct their eye gaze to the same object.
- *Gaze Aversions*: occur when the eyes shift away from the main focus of attention.

During interactions, the speakers and listeners systematically coordinate gaze with speech. Gaze works together with speech timing, prosody, and contextual cues. Speakers tend to avert their gaze from the listener during long utterances, mainly indicating speech planning and cognitive processing. Gaze can also be used to yield the turn to someone else by making mutual gaze with the listener. Furthermore, gaze also plays a role in facilitating coordination and attention, making the conversation smoother (Mutlu et al., 2012; Patel & Rollins, 2025; Xu et al., 2016). During a conversation, the listener who wants to take the turn often directs their gaze towards the speaker. This signals the speaker that the listener is ready to take the turn. After getting the turn the new speaker averts their gaze to signal that they want to hold the floor (Skantze et al., 2015). The study by Jokinen et al. (2013) showed that the recognition of turn-holding and turn-yielding actions in a conversation is significantly improve with the use of gaze.

The timing of gazing is important; when the gaze is too long or poorly timed, it can result in feelings of staring and non-responsive behavior (Xu et al., 2016). To maintain natural interactions, it is crucial to regulate the gaze duration correctly. In multi-party settings, gaze becomes even more complex. Speakers might shift attention between listeners to give turns, meaning they will exclude or include individuals. So, gaze functions both as a cue for interpreting others' intentions and as a means of regulating one's own conversational behavior. However, it remains technically challenging for robots to accurately measure eye gaze in multi-party interaction.

Turn-taking in Robotic Systems

Humans have the capability to predict the TRPs naturally with the use of context, syntax, prosody, and gaze. However, robots do not have the ability to understand these cues as humans do. Robotic systems have to rely on sensors and algorithms to detect when it is appropriate to take a turn. If the timing of the robot with respect to these cues is inaccurate, it may interrupt the current speaker or respond too late. This will negatively affect the perceived naturalness of the conversation. Robotic systems often use models for turn-taking that rely on silence cues. The robot decides to take the turn after a certain silence threshold. Waiting for a certain amount of silence in the user's speech can lead to interruptions or late responses (Pinto & Belpaeme, 2024; Skantze & Irfan, 2025). In addition, robots use gaze cues to improve the turn-taking behavior. Prior work by Andrist et al. (2013, 2014) has shown that gaze aversion can positively influence floor management and that using the appropriate timings for gaze aversion results in better human-robot interaction compared to badly timed gaze aversion. These studies make a turn-take model that includes gaze aversion timing based on patterns observed in human-human interaction. However, they do not

compare different onset timings of gaze aversion after the robot takes the floor. Moreover, in the study of Andrist et al. (2014), they rely only on silence cues as a signal to avert the gaze of the robot. They do not incorporate the direction of human gaze as a signal for turn exchange. This limitation becomes relevant in multi-party settings where the robot has to understand who is addressed. Robots also have several technical limitations, like restricted body movements or limited sensors (Koller et al., 2023), which makes it even more complicated to apply robotic systems in a multi-party setting.

Robotic Speech

Previous studies have shown that appropriately timed and well-calibrated responses have a positive influence on the perceived naturalness and social engagement in human-robot interaction (Blik, 2020; Van Rossum, 2025), which is why backchannel responses are often implemented in robotic speech. Indicating that accurate speech coordination is crucial in conversational robots.

Robots typically rely on fixed silence thresholds, which can create situations where the robot takes the turn and produces an utterance even when the current speaker is not finished. As people create silence for multiple reasons, like hesitations, cognitive load, and planning. It does not always signal the end of the speaker’s turn, so with the fixed silence thresholds, it can lead to unnecessary interruptions. Having shorter silence thresholds can increase the responsiveness of the robot, but they risk interrupting the current speaker. Using longer silence thresholds might reduce interruptions, but that can mean that the response of the robot will be delayed and will feel less natural. Therefore, it remains crucial to determine valid cues that robotic systems can use to identify appropriate moments to take the turn in conversations. As silence during speech can have different interpretations in conversations, researchers also explored gaze direction to improve the timing in turn-taking. It is even more complex for robots to determine when to take a turn in multi-party interactions because robots not only have to determine when to speak, but they also have to understand which participant they have to address through gaze direction.

Challenges of Gaze in Robotic Systems

Interactions with robots are often smoother when they display human-like behaviors. Verbal cues, such as speech, are most often used in the development of social robots; however, 60-65% of conversations are nonverbal (Weldon et al., 2021).

During conversations, the type of eye contact that a robot makes depends on the context (Admoni & Scassellati, 2017). Eye gaze in robots has the same purpose as in human-human interaction, as it can allocate turns to a specific person and it helps shape the roles during conversations as addressee (the person who is engaged in the conversation and to whom the signal is addressed), bystander (participant that is engaged but to whom the signal is not addressed), overhearer (other actor that is known to the speaker who is not engaged in the conversation but will receive the signal), and a eavesdropper (a participants not known to the speaker and not engaged in the conversation but who will receive the signal) (Bohus & Horvitz, 2010; Mutlu et al., 2009). Using eye gaze can also be useful to direct someone’s attention from one place to another (Hoque et al., 2011).

This has shown why eye gazing plays an important role in conversations with robots. Despite its importance, eye contact detection in a human-robot interaction is technically demanding. The main way studies capture eye contact with robots is through head orientation, instead of eye movements. That is because this way is more convenient, but it does not accurately show where the person is looking. Humans can move their eye gaze without moving their head, indicating that someone’s head orientation estimates might lead to wrong interpretations. Subtle eye movements can provide information about the visual attention, which is not given with complete head turns because individuals can keep their head oriented toward one interlocutor while directing their gaze towards someone else (Hietanen, 1999). This distinction highlights that eye gaze provides

information about the location of the individual’s attention, which cannot be seen by head pose alone. There are several reasons why robots have trouble estimating the precise eye movements. Some robots have low-resolution cameras, and it is recommended to have high-resolution, narrow field-of-view cameras to capture these eye movements (Palinko et al., 2016). As the cameras of robots are most often not of good quality, they are easily influenced by light and shadows, which affects the capture of eye movements. That is why some studies use external hardware to capture eye movements; humans have to wear special glasses or helmets that can capture these movements (Palinko et al., 2016). These are some of the reasons that robotics systems determine the visual attention solely on head orientation. This is convenient, but it neglects the visual attention of the eyes. Quick eye movements, saccades, or short gaze aversions may be missed when using head orientation only. These eye movements can give additional information besides the attentional focus of the person. Research has linked these quick movements to cognitive load, indicating that they are connected to the underlying cognitive processes (Nordmark, 2025). This means that the eye movements can provide information about the person’s conversational state (Nordmark, 2025). This information can be useful to determine when the speaker is ready to speak, give away the turn, or pause their speech. In addition, when the robot can detect eye gazing correctly, it is also important to generate correct gaze behavior that looks like human gaze. If a robot timed its eye contact incorrectly and kept eye contact for too long, it can be perceived as aggressive, and it can reduce persuasion (Stanton & Stevens, 2017). When a robot averts its gaze, it can reduce feelings of discomfort and prevent the negative associations that arise from prolonged staring (Andrist et al., 2014).

A study done by Johnson and Cuijpers (2013) demonstrated that the relationship between head orientation and gaze can be modelled with a simple linear relationship, indicating that humans’ gaze can be interpreted from the head pose yaw, and pitch. Head yaw refers to the head rotation from left to right, while the head pitch is the horizontal head rotation from up to down. The head orientation is easier to detect compared to eye movements. The essential idea is *gaze = head turn + eye turn*. They show that with this relationship, which is explained in more detail in the Methods section, it is possible to guess the gaze based on head turn alone. In other words, the model compensates for the unknown contribution of eye movements, allowing gaze direction to be approximated without directly tracking the eyes.

Multi-party Interactions

Turn-taking in dyadic interactions already raises some challenges, but these challenges become even more complex in multi-party interactions. In these interactions with multiple participants, a robot has to determine which participant to address and when to shift gaze between multiple individuals. As explained earlier, gaze behavior plays an important role in signaling attention and managing the floor. Incorrect gaze allocation may lead to confusion about who the robot is addressing or listening to. This could negatively affect the flow of the conversation.

Multi-party settings introduce perceptual challenges in robotic systems. In these settings, the robot does not only have to detect when silence or speech occurs, but it also has to determine which participant is addressed and which participants produce the utterance. Identifying the speaker and the addressee is important for appropriate turn-taking and for how speaking turns shift between the participants. However, to do this, it requires the detection of speech signals, identifying the source, and inferring the addressee (Bohus & Horvitz, 2010). That means that a robot does not only have to detect the cues for turn-taking, but they also have to interpret them to decide how to respond. These decisions are often made under uncertainty due to delays in processing and noisy signals (Bohus & Horvitz, 2011). Bohus and Horvitz (2010) highlighted that implementing coordinated conversational behavior in a robot requires the robot to manage several interactional events. For example, the robotic system should manage its own behavior during the turn of someone else, but it should also get the attention of that speaker when eye

contact cannot be established. Furthermore, the decision framework for turn-taking proposed by Bohus and Horvitz (2011) estimates the dynamics in conversations. Think of whether the floor is released by the speaker, or when someone is likely to speak. These estimates are then used to decide whether the system should take the floor or if it should wait. The system then weighs the costs of different outcomes, like successful turn transitions or delays, to reduce the number of interruptions and unnecessary long gaps. Even though nonverbal cues are implemented in the system to infer conversational dynamics, it is not explicitly examined how the robot’s gaze can influence the conversation.

Creating robotic systems capable of understanding human gaze and floor management is one part of the problem. The other part is that these robotic systems should be able to incorporate these floor-management principles into their own gaze behavior. This enables them to regulate turn-taking that aligns with human-human interaction norms. Studies tried to find the optimal time for a robot to gaze away, which should improve turn-taking efficiency that enhances the interaction by improving both mutual understanding and the perceived naturalness (Butter, 2025). From those studies, it became clear that when the robot looked away after 3 seconds, it would not increase the engagement, but having well-timed moments where the robot looks away in a human-to-robot interaction, they would be more successful in managing conversational floor (Andrist et al., 2014; Butter, 2025). These ”well-timed” gaze aversions are based on a study by Andrist et al. (2013) that captured how humans timed their gaze aversion. However, this is not yet incorporated into robots that have to operate in a multi-party setting.

Research Questions and Hypotheses

This study builds upon the Master Thesis Project (MTP) conducted by Spiegels (2015). During that study, a conversational state machine was built that enabled the NAO robot to participate in a multi-party conversation. The state machine made it possible for the robot to regulate turn-taking based on silence thresholds and head orientation. The state machine helped the robot to take turns during a conversation. However, there were several issues with the face detection model, and the effect of the assumption that people would look directly at the robot to address it was not examined. As a result, it remains unclear whether this assumption influences the robot’s ability to capture moments for turn-taking.

The present study will extend that research by improving the state machine and the detection and use of gaze cues in a multi-party setting. Specifically, addressing the issue that humans do not always look directly at each other when making eye contact, but also rely on eye movements. To account for this, a region of eye contact based on the linear relationship between gaze direction and head orientation developed by Johnson and Cuijpers (2013) will be implemented. Making it possible for robots to infer eye contact even when participants are not directly facing the robot. In addition, this study will also focus on how the timing of gaze aversion during robot speech influences the quality of the conversations between humans and robots. While previous research has shown the importance of gaze aversion (Andrist et al., 2014) in human-robot interaction, it has not been examined in a multi-party setting, and it was not combined with gaze detection and silence cues. Building on the work of Andrist et al. (2013, 2014) and Spiegels (2015), the present study will examine how the participants’ gaze behavior and their silences can be used to identify opportunities to take the turn in a multi-party human-robot conversation. In addition, this study investigates whether initiating gaze aversion immediately or after a delay influences how humans perceive the robot. Integrating the participants’ gaze with silence cues and comparing them to different onset timings of gaze aversion in a multi-party setting, this study will fill the gap in previous work by Andrist et al. (2013, 2014).

These problems in human-robot interactions have led to the following main research questions in this study:

- How should a robot detect eye contact from human gaze behavior when eye movements cannot be reliably measured?
- In addition, how does the robot's gaze behavior after taking the turn influence conversational fluency and perceived interaction quality?

To answer the first research question, the behavioral state machine will be extended with the previously proposed gaze model to improve the detection of eye contact. The robot will be programmed so that it can perceive eye contact even if the participants are not directly oriented towards the robot. Furthermore, a measurement of fluency and perceived naturalness will be done through questionnaires that capture the subjective experiences. Based on previous research that showed that gaze can signal turn-yielding and that it can help with managing the conversational floor (Jokinen et al., 2013; Skantze et al., 2015), it is expected that, by implementing a more accurate eye detecting model into robots, the interaction will be rated as more fluent and a better perceived naturalness. To answer the second research question, we will manipulate the timing of the robot's gaze relative to speech. The robot will avert eye contact either immediately or after a short delay when getting the turn. In addition, the probability distribution of turn-taking time between both participants and between the robot and participant will be analyzed. These distributions are compared to examine the differences between humans and between robots and humans in how gaze timing influences turn-taking behavior. In the research done by Skantze (2021), it was shown that turn-taking between humans could be very quick, indicating that people predict turn endings before it occurs. Therefore, it is expected that turn-taking between humans will be shorter and more consistent. Furthermore, it is expected that the quality of the interaction will improve when the robot looks away for a short delay before making eye contact again, as these patterns closely resemble human-human behavior (Andrist et al., 2013). Next to this, participants are expected to report higher comfort levels and experience a smoother conversational flow when the robot uses the timing for gaze behavior that is found in human gaze behavior, compared to continuously looking away from the participants. Previous research has shown that appropriately timed responses can reduce discomfort (Andrist et al., 2014).

From a Human-Technology Interaction (HTI) perspective, this project contributes to understanding how robots can align their behavior with that of humans. During the project, both technological development and human factors will be addressed. The technology will include the programming of the robot's behavior, for example, the robot will detect eye contact from a human based on implementing different head orientation thresholds. This study will also address human factors because it might improve the perceived naturalness and comfort of the conversations in a human-robot interaction, and it can make the robots more human-like when the robot has a more accurate detection system. This research contributes to HTI science by providing insights into how robots should behave in a multi-party setting where turn-taking is crucial. The findings of this project can help align the design of conversational robots better with how humans behave during conversations with multiple participants.

Robot Behavior Framework

Turn-Taking Mechanism

All the conditions combined the silence and eye contact detection before the robot could take the turn. Meaning both cues had to be present simultaneously for the robot to initiate speech. It was designed so that the order did not have to be fixed. This resulted in two scenarios that could happen:

1. Eye contact before the silence occurred.
2. Eye contact after the silence period was started.

The system was designed to continuously monitor speech activity and gaze behavior during the conversation. The robot took the turn when both conditions appeared in the interaction period. The silence period was considered a valid turn-taking cue after 1000 ms of non-speech. This silence cue duration was chosen because spoken dialogue systems often make use of a silence cue duration between 700-1000 ms (Skantze, 2021). So, using the 1000 ms duration would reduce the likelihood of interruptions (cut-ins) because short speaker silences could be hesitations and not a signal to yield their turn. Participants' gaze direction was estimated from head pose. Depending on the condition, either direct-gaze thresholds or broader gaze thresholds were applied. When eye contact was detected before sufficient silence occurred, the robot kept monitoring until the silence threshold was reached. Also, when silence was detected first, the systems waited for eye contact to occur during the active silence period. Only after both cues were present did the robot transition to a turn-taking state and initiate speech. This built further on the theory that described that turn-taking, especially in a multi-party setting, depended on the combination of silence and gaze (Patamia et al., 2025). The program for the robot's behavior builds further upon the multi-party turn-taking framework developed by Spiegels (2015). The behavioral state machine, turn-taking mechanism, and subprograms are used from this earlier study, but several components were modified for the current study. To investigate how different turn-taking strategies influenced the conversations, two components of the turn-taking system were manipulated. Firstly, the eye-contact detection model, which determined how the robot interpreted participants' gaze behavior when identifying opportunities to take the turn. Secondly, the robot's gaze behavior after acquiring the turn was manipulated. Each component consisted of two variants. The eye contact detection model consisted of a direct gaze model and an extended gaze model. The robot's gaze behavior consisted of a static gaze aversion and a timed gaze aversion. The combination of these components allowed the investigation of both turn-taking detection and gaze behavior during the robot's speaking turn.

Gaze Detection Models

Extended Gaze Model

Eye contact is often made before people look directly at each other. In the extended gaze model, the results of the study by Johnson and Cuijpers (2013) were implemented. The linear relationship that Johnson and Cuijpers (2013) found between gaze and head pose is best described with the following equation:

$$y = 0.38x + 1.11 \tag{1}$$

where:

x = yaw angle of the gaze direction

y = measured yaw angle of the head pose

Next to this linear relationship between gaze and head pose is the Region of Eye Contact (REC). The REC defines the range of gaze angles within which an observer still perceives eye contact. It is important because it determines how precisely the robot must estimate the gaze to create the impression of eye contact. So, using the REC creates a bottom and upper boundary when implementing the linear relationship between gaze and head pose. This range covers an area on the observer's face and is 7° over a distance of one meter when viewing images of human faces (Cuijpers & Van Der Pol, 2013). For the extended gaze model, the upper boundary for the broad gaze model was calculated at 19.84° by using angles of the task setup in Equation 1 (Figure 5). To account for measurement noise and small inaccuracies in the face detection program, the upper boundary was rounded up to 20° . The minimal range by REC would be 3.5° around the face, but this was extended to 5° to improve robustness against small inaccuracies in gaze estimation. These inaccuracies can occur due to factors like camera resolution, lighting conditions, and natural variations in head movements. To measure whether the participants looked at the robot, the absolute values of the yaw angle are used. This means that any head orientation with an absolute yaw angle smaller than 25° will be seen as eye contact. However, the original yaw angles remained as normal, with negative values indicating orientations to one side of the robot and positive values indicating orientations to the other side. The detection is illustrated in Figure 1, where the extended gaze model is approximately 15° broader on each side compared to the direct gaze model that has a range from -10° to $+10^\circ$. In Figure 1, the light gray area is the range in which eye contact is detected for the extended gaze model. Allowing the robot to detect eye contact even when participants do not face the robot directly. The robot could also determine whether participants directed their gaze towards the robot or towards the other participant based on predefined head-orientation thresholds. For the left participant, gaze was classified as human-directed when the yaw angle was between (30°) and (60°) , whereas for the right participant, this range was between (-60°) and (-30°) . In both cases, the pitch angle had to remain within $(\pm 20^\circ)$. Gaze that did not meet the human-directed criteria and fell within the eye-contact thresholds of the corresponding experimental condition was classified as robot-directed gaze.

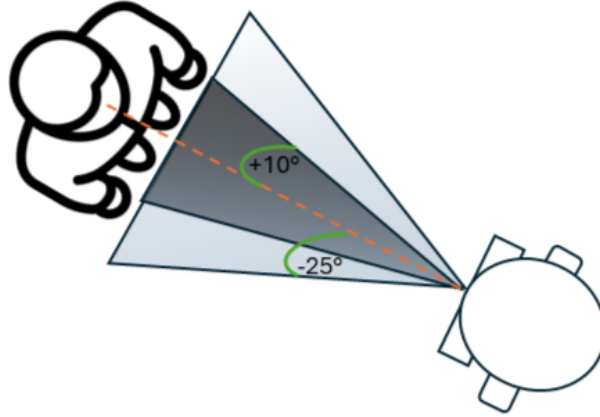


Figure 1: Illustration of the extended and direct gaze detection model. The dark grey area with $\pm 10^\circ$ is the range where eye contact is detected in the direct gaze detection model. The light grey area shows the area that will detect eye contact in the extended gaze detection model. The orange dashed line is the line when participants look directly at the robot with a yaw angle of 0° .

Direct Gaze Model

The direct gaze condition captured the measurements when the turn-take behavior was based on strict accuracy. With this direct gaze model, a turn was initiated when a participant looked directly at the robot in combination with a silence cue of at least 1000 ms. This condition aimed

to minimize ambiguity in identifying turn transition points. For this model, the same REC of 5 degrees was used as in the extended gaze model. To keep distinct gaze regions between the two gaze models, the eye contact region for the direct gaze model was with the REC set at -15° to 15° . However, the boundaries used for the direct gaze condition were set at 10 degrees to account for potential systematic biases in head-pose estimation, where the detected head orientation might deviate from the true orientation. As these deviations can vary across participants, a narrower eye contact region could reduce the risk of false-positive eye contact detections. This narrow boundary was chosen to represent only the gaze that was directed precisely toward the robot. Figure 1 clearly shows that the dark gray area, which represents the direct gaze model, is narrower than the extended gaze model.

Robot's Gaze Behavior

Static Gaze Aversion

In addition to the manipulation of the eye contact detection models, the experiment varied the robot's gaze behavior after taking the turn. The turn-taking mechanism remained the same and was still based on the silence and gaze cue. However, when the robot took the turn, the gaze behavior would differ. When the robot detected an opportunity to take the turn based on the cues, it averted its gaze to the middle, where it did not face any participants. During the speaking turn of the robot, it maintained this neutral position and did not establish eye contact with any participant. This should have isolated the effect of gaze aversion on the conversation. Figure 2 shows a scenario that could happen when the robot avoids eye contact when it gets the turn during a conversation. If the robot gets the turn based on the silence and gaze cues that are detected, it will immediately gaze away from the person who spoke last and keep its gaze towards the middle between the participants. That is also the reason why human gaze is not there when the robot takes the turn because it cannot be detected at that moment.

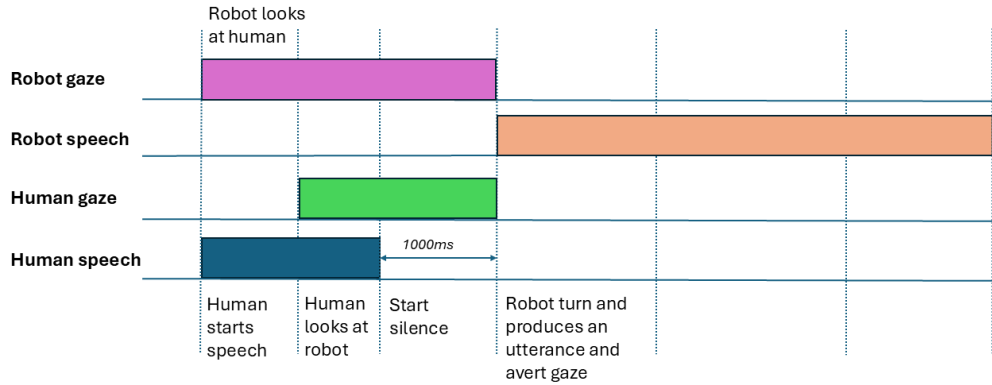


Figure 2: Schematic illustration of the signals responsible for triggering a turn take from the robot with the behavior of only averted gaze. It illustrates that when no speech is detected by the robot for at least one second, and it detects eye contact from the humans, it initiates a turn by producing an utterance and by averting its gaze.

Timed Gaze Aversion

In contrast to the robot's gaze behavior, where it only averted eye contact when it got the turn, the timed gaze aversion behavior varied the robot's gaze behavior. After obtaining the turn, the robot alternated between periods of gaze aversion and eye contact based on timings derived from human-human interaction research. When the robot acquired the turn, it would first avert its gaze toward the neutral position (facing the middle), which signals that it starts its turn. This

gaze aversion holds for 1.27 seconds, which is based on the timings for gaze aversion from human-human interaction. The duration for gaze aversion of 1.27 seconds was chosen as it corresponds to the average duration that turn-taking gaze aversions continue after speech onset in human-human conversations reported by Andrist et al. (2013). That study also reported an average duration of gaze aversions of 2.30 seconds. However, 1.03 seconds of these gaze aversions typically began before the speaker started talking. This robot can only initiate its gaze behavior after detecting an opportunity to take the turn and cannot predict exactly when another speaker will finish their turn. That is the reason why only the average duration of gaze aversion after the turn taken is implemented. If the robot looked away for 1.27 seconds, the robot had a 50% chance of re-establishing eye contact with the left or right participant. Then the robot held eye contact for 4.75 seconds, which corresponded to the average interval between gaze aversions during speech reported by Andrist et al. (2013). Human-human conversations showed that gaze aversions during speech play a role in floor management (Andrist et al., 2014). That behavior of looking away was incorporated into the gaze behavior of the robot to see whether it would affect the perception of the participants. The second scenario, illustrated in Figure 3, depicts the implemented timing of the robot's gaze behavior. It shows that when the robot takes the floor after both gaze and silence cues are detected, it averts its gaze for a fixed duration of 1270 ms while producing an utterance. After this, gaze aversion of 1270 ms, the robot makes eye contact with one of the two participants and switches to the other participant after 4750 ms. The robot takes a turn only when there is a silence of at least 1000 ms, and eye contact is detected. If one of the conditions is not met (no eye contact during sufficient silence, or eye contact during insufficient silence), the system classifies the situation as ongoing speech of the participant. Humans can also avert their gaze and not look at the robot, which is visible in the illustration in Figure 3. The robot can determine whether the participants are making eye contact with the robot or if they are looking at the other participant in the conversation.

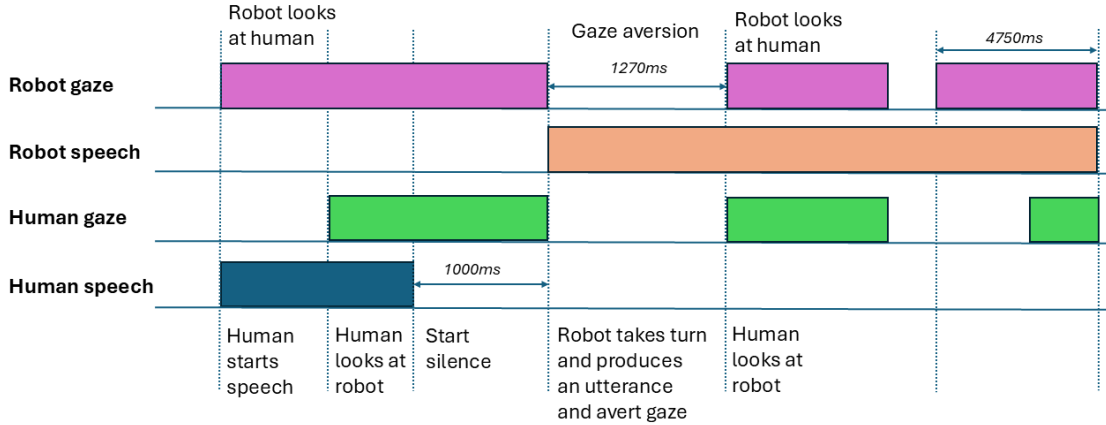


Figure 3: Schematic illustration of the signals responsible for triggering a turn take from the robot with the behavior of timed gaze aversion. After the robot detects both the silence and gaze cue, it produces an utterance and gazes away for 1.27 seconds. After this time, it looks back at one of the participants for 4.75 seconds before directing its gaze to the other participant.

Behavioral State Machine

A finite state machine (FSM) was developed, which worked as the main component of the whole program. This finite state machine defined the behavior of the robot and enabled the robot to make inferences about the interaction and select appropriate behavioral responses. With the use of the audio and visual programs, which were discussed in the Software section, it obtained information on silence duration, speaker location, and eye contact. The robot's behavior was developed as a rule-based model, where transitions between states depended on the input from the audio and visual modules. It depended on the state, but the robot could turn its face toward participants, provide feedback and information, and produce speech. The system worked both autonomously and controlled, which was done for the flexibility of the program during the experiment.

The behavioral state machine consisted of 13 individual states that worked together to govern the flow of the interaction between the robot and the participants. These individual states included conversational states regarding waiting for speech, speaker tracking, regulating gaze, backchanneling, turn-taking, information utterances, verdict handling, and the termination of the experiment. The transitions between the states were made through conditional rules that used sensor input from the discussed visual and audio programs. The individual states will be discussed in detail, and they are visualized in Figure 4.

State 0: Initialization & Calibration

The first state of the FSM was used for initializing the experimental environment. It collected the participants' names and IDs, which the robot could use to address participants during the conversation. In addition, all logging files that were needed for the experiment were created. These logs captured speech activity, gaze behavior, and RMS values. After the initialization, the microphones could be calibrated. This was done to determine the speech detection thresholds for both speakers. First, the silence threshold was determined. This created a boundary that labeled sounds as silence if the RMS value was lower than the determined threshold. Secondly, the speech threshold would be determined. This created a boundary that the speech RMS value has to pass before being labeled as speech. The RMS values that fell between the silence and speech threshold were labeled as both speakers, indicating cross-detection. The operator had to manually do the calibration and had to give an input whether the calibration was successful or if it had to be done over.

If the initialization phase was completed, the robot would introduce itself as "Misty". Here, it welcomed the participants and asked for their names individually. This was done together with orienting its head towards that participant. It exhibited this behavior to show the participants what it is capable of and to make clear what the role of the robot is during the conversation. When this was finished, the robot asked if the participants were ready and started the introduction to the task. Here, it explained what the participants had to do and made it clear what would happen. After completing the introduction phase, the dialog stage was initialized and transitioned to state 10.

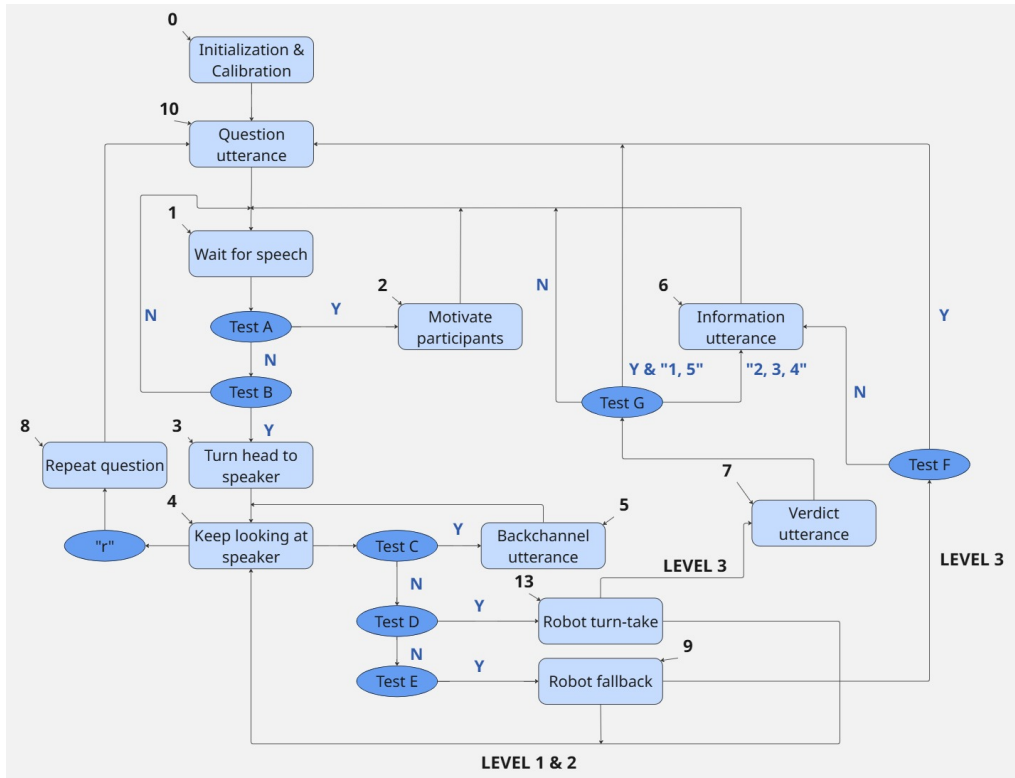


Figure 4: Diagram that illustrates the behavioral state machine of the robot, including the states and tests. Showing how everything is connected and how the robot decided how to respond during the interaction.

State 10: Question utterance

The next state of the behavioral state machine was state 10. This state was responsible for managing the questions throughout the interaction. It could be seen as the main state, as multiple states returned to this one. It happened after a successful verdict or when the interaction advanced to a new question following input from the Wizard. The state was very dependent on the dialog manager, from which it returned the current dialog stage. If the dialog stage was between stage 1 and stage 9, the robot would move on to the next question and produce this question in artificial speech. The questions that would be asked depended on the chosen script: holiday planner scenario or building their dream house scenario. When the robot was done asking the question, the finite state machine transitioned to state 1.

When the 10th dialog stage was reached, state 10 combined all chosen options from the participants for each question into a final summary. After this, the robot thanked the participants for participating and instructed them to fill in the questionnaire. It moved on to state 12, to close the program and save all the logs. However, this state was only created to save the logs and terminate the program, and was therefore not included in the behavioral state machine in Figure 4.

State 1: Wait for speech

In state 1, the robot monitored the interaction. It used the speech detector subprogram to continuously analyse the audio streams from both microphones. This analysis determined which of the two participants was speaking, and it logged the duration of the silences. This caused the robot to be in a passive mode, which waited for a response from the participants. Next to

the speech detection, this state also updated the gaze orientation of the robot. If the left or right participant spoke, the robot prepared itself to rotate its head to this speaker. There was also a programmed cooldown period, which continued the robot monitoring phase. However, the autonomous turn-taking was disabled to prevent the robot from reacting too quickly after finishing its own utterance. This was because the robot's utterances lasted slightly longer than indicated by the code execution shown in the terminal, the robot would pick up its own utterance through the microphones of the participants. In the condition where the robot rotated its head towards the participants during its own utterance, the robot maintained its final gaze direction to signal a turn-taking cue towards that participant. Furthermore, it did two tests to check for inactivity and speech.

Test A: Inactivity detection

The first test of the behavioral state machine (test A) detected whether there was a silence for at least four seconds. This happened when both participants did not say anything, for example, when they finished their discussion, or if they did not know what to say. The program detected this as inactivity, and to keep the interaction going, it transitioned to state 2.

Test B: Speaker detection

The second test (test B), tried to detect speech from the participants. The participant who sat left or right of the robot and spoke for at least 500 ms is stored as the active speaker. If speech was detected before the silence threshold of four seconds is reached, the behavior state machine transitioned to state 3; otherwise, it stayed in state 1.

State 2: Motivation after inactivity

If during test A, a silence of four seconds or more was detected the behavior state machine transitioned to state 2. This state was developed to prevent the conversation from stopping, and when participants did not know what to discuss anymore or what to say, the experiment could still continue. This meant that when no participant had spoken for four seconds, the program would break the silence with a random utterance. These random utterances were chosen from a list of motivating sentences that should encourage the participants to keep discussing. When the robot was done with the motivational utterance, the behavioral state machine returned to state 1. It was programmed this way to keep the robot from immediately taking over. This would instead give participants the opportunity to start a new discussion. The motivational utterance was produced with the function: `robot_speak_and_wait()`.

Function: robot_speak_and_wait()

The function `robot_speak_and_wait()` was used by states in which the robot had to produce an utterance. This function not only produced speech, but it did much more. It also logged the speech for later analysis. First, it checked whether a participant was speaking, and if this was the case, it logged this as a robot interruption. This helped to understand the differences between normal robot turns and where robots spoke too early and created a cut-in. Secondly, before the robot spoke, it could center the robot's head depending on the condition the participants were in. When this happened, it stopped the microphones and RMS logging to prevent the system from detecting the robot's voice as the participant's speech. Furthermore, the robot's speech time was measured based on the number of words and letters, to get the most accurate speech time, because the audio status of the robot itself was often not reliable. If the robot spoke and used this function, it also changed its gaze depending on the condition. In condition A, the robot kept its head centered when it was speaking. In condition B, it averted its gaze for 1.27 seconds, and it gazed between the participants every 4.75 seconds. In condition B, the robot kept its gaze towards the participant whom it looked at when it finished its turn. In condition A, the robot would center

its head in the middle again. The starting time of the robot’s speech was logged, together with the end time of the robot’s turn.

State 3: Turning head toward the speaker

The behavioral state machine transitioned to state 3 when test B detected one of the participants speaking. In this state, the robot would turn its head towards the participant who was speaking, which could be the one who was sitting left or right of the robot. After the head turned, it moved to state 4.

State 4: Keep looking at speaker

This state was the main monitoring state. This state caused the robot to keep looking at the participant who was speaking, and it checked for turn-taking cues. It checked who was speaking, if there was a silence, if the participant made eye contact with the robot, and if there was overlap in speech between both participants. The robot first detected the participant who was speaking and the duration of the silence. Furthermore, to make eye-contact moments more stable, a small buffer was programmed. This buffer only detected eye contact when enough frames indicated that a participant was looking at the robot, and not when only one frame detected eye contact. If only one frame was used for eye contact, then fast head movements that accidentally passed the robot’s face would be falsely detected as eye contact. In state 4, the logging was done of who the speaker was, duration of silence, yaw, pitch, eye contact, participant looks at participant, head position, and human-human speech overlap. Also, to keep the robot active during the conversations, this state updated the robot’s head direction based on who was speaking. Lastly, three tests were done before moving on to the next state.

Test C: Long-turn acknowledgment

Test C tested if a participant was speaking for at least four second. If this were the case the behavioral state machine transitions to state 5.

Test D: Silence + eye contact

If the system detected a silence of at least one second together with a stable moment of eye contact, the robot interpreted this as a turn-yield cue and transitioned to state 13.

Test E: Silence without eye contact

It happened that there was silence without eye contact. When this was the case, the robot did not immediately interpret this as a turn-yield cue, but it transitioned to state 9. Where a fallback strategy is created.

State 5: Backchannel utterance

If test C in state 4 detected a participant who was speaking for more than four seconds, it moved on to state 5. This state was created to indicate to the participants that the robot is still listening. To do this, the robot would provide short backchannel utterances. The robot would randomly select one of the short responses. These responses were short words that are commonly used by humans as well. Think of ”uh-huh” or ”okay”. During this utterance, the robot would not center its head and instead kept looking at the active speaker. When the utterance is completed, the behavioral state machine would return to state 4.

State 13: Robot turn-take

When the speech duration was shorter than four seconds, it moved to test D. If both eye contact for a brief moment of time and a silence of 1 second were detected, it transitioned to state 13. This was interpreted as a strong signal that the turn was given to the robot. The state consisted of three levels. This was done to keep the conversation going and create more interactions between the robot and the participants.

1. The first level, meaning the first time the robot detected both turn-taking cues simultaneously, it gave a prompt about one of the possible options. This option was randomly chosen, and the participants were asked their opinion on that option.
2. The second time the robot detected both cues together at the same time, it used a switch-turn so the other participant could say something and contribute to the conversation.
3. The last level, when the robot got the turn for the third time, it stopped stimulating the discussion and transitioned to state 7. This was done as the participants could be done with discussing after multiple turn-taking attempts.

Levels 1 and 2 returned to state 4, so the discussion can proceed. Level 3 moved on to state 7.

State 9: Robot fallback

The behavioral state machine moved to state 9 when test E returned True. If there was a 4-second silence, and it was detected when the participant was not looking at the robot. It would not be labeled as a turn-yield. The reason for this was that both cues had to be detected. However, awkward moments could occur when the participant faced the robot to make eye contact, and the robot did not react. To account for the fact that participants were looking at the robot, but the system did not recognize it as eye contact, the system should have a strategy to continue the conversation. State 9 had three levels to manage this situation. If it happened the first time when it detected only silence and no eye contact, it went to level 1. If it happened for the second time when the same participant spoke, it moved to the second level. Finally, when it happened three times in a row, it moved to the last level.

1. The first level of this state had two options. It could choose to address the last speaker directly by asking to elaborate on his or her opinion. Secondly, the robot could choose to switch the turn to the other participant. This was done to increase the turn-taking moments during the conversation and give both participants the opportunity to elaborate on their opinions.
2. For the second level, the robot did not use an utterance to minimize redundancy; it now only gazed at the other participant to invite them to the discussion.
3. The final level asked if they wanted to go to the next question and continue the conversation.

Levels 1 and 2 returned to state 4, and level 3 went to test F. What happened in level 1 was that the robot switched turn-taking between the participants. This function caused the robot to redirect the conversation toward the other participant by turning its head to the listener and prompting an utterance. This created a visual and verbal cue that encouraged the conversation to continue. If the direct turn-taking option was chosen, then the robot produced the verbal prompt toward the active speaker.

Test F: Continuing conversation

When level 3 was activated, it moved to test F. This tested whether participants wanted to move on to the next question or if they wanted to keep discussing. If the participants were done

and wanted to proceed, the system returned to state 10 and proceeded to the next question. Now, no option was selected during the verdict state, and the program would automatically fill in an option for that specific question, so the conclusion was complete. If the participants did want to discuss a bit longer, the system returned to state 6, and the robot offered extra information on one of the options.

State 6: Information about an option

During the conversation, participants could ask for additional information about certain options. One limitation of the robot was that it didn't understand speech. Meaning that the Wizard had to select the option to provide information. If the Wizard saw that the participants asked for additional information, it would select the option, and the robot would provide some information to them. After the information was given, the behavioral state machine would return to state 1, where it waited for speech and allowed the participants to discuss further.

State 7: Verdict

After discussing the options, the participants had to decide which option they wanted to choose. This state gave the operator 5 types of verdicts to choose from, which were: single, multiple, none, unclear, or speedup. These types were already explained in detail in the Section: Dialog Manager.

Test G: verdict confirmation

The operator had to select one of these five types. If the verdict single or speedup was chosen, the behavioral state machine moved to the confirmation and returned to state 10, letting the robot ask the next question. When the other options were chosen, the behavioral state machine moved to state 6, where it provided additional information about one of the options to stimulate further discussions. The option could be confirmed by the Wizard with an input of "yes" or "no". If it was "yes" the behavioral state machine transitioned to state 10, where it asked the next question. If the input was "no" the robot apologized and asked the participants to clarify their option, and moved to state 1. When the Wizard confirmed the chosen option for the last question, the system moved to state 10 and gave the participants the summary of their conversation.

State 8: Repeat question

State 8 let the robot repeat its question. There could be moments when participants did not remember the options they could choose from and wanted the robot to repeat itself. So, the Wizard could press the button "r", which repeated the questions for the participants.

Dialog Manager

The robot had to act as a travel agent and as a designer during the two tasks. To enable the robot to be verbally interactive, it was crucial to have utterances that could be used by the system. Two scripts were implemented that were both designed to maintain a conversation for around 15 minutes. One script was used to build a holiday, and the second script focused on building a dream house. Both scripts had extended sentences compared to the original ones, so that most sentences had a speaking time of at least 10 seconds. This would cause the robot to rotate its head to the left and right multiple times (condition B), making this behavior better distinguishable from when the robot did not perform these gaze shifts (condition A). An overview of the constructs is provided in Appendix F, which should clarify how the dialogs worked.

The dialog manager for both scripts had a total of 11 dialogs in a pre-defined order. The first dialog gave an introduction to the task, which gave a short explanation of what the participants

should do in each scenario. This introduction was followed by 9 questions. The robot provided a set of answers with every question, from which the participants could choose. To control the duration of the conversation, the robot asked for an opinion from both participants. It could do this in four different ways:

1. By switching between participants and asking one participant to comment on the response of the other participant.
2. By asking a participant to elaborate on their own response.
3. By referring to a specific option from which the participants could choose.
4. By using confirmation questions to speed up the discussion and guide the interaction toward a decision.

There were situations where the participants could not come to a mutual agreement, or they wanted to know certain options better; the robot would provide information about the set of answers or repeat the question when they didn't understand it completely. The dialog could speed up the discussion with the use of the so-called speed-up function. This function was used when participants were unable to reach a decision or had finished their discussion. In such cases, the robot posed a final question to conclude the current dialogue state and let the participants wrap up the discussion. There was also an option to skip directly to the next question. In such cases, the system automatically selected an option to ensure a coherent and logical conclusion at the end of the interaction.

When a question had been asked, and the participants came to a mutual agreement, the dialog manager came with a verdict utterance. This verdict was a conclusion that reflected back on the participants' discussion and would signal to the participants its understanding of their chosen option. The verdict statement occurred after the robot took the turn for the second time, making the conversation flow better and keeping the interaction more natural. However, the verdict depended on the Wizard's input. It went to the next question due to the input from the Wizard, which ensured the continuation of the conversation. There were five verdict options from which the Wizard had to choose:

1. **SINGLE**: When there was one clear option, the dialog manager would acknowledge their discussion, and the Wizard chose the preferred option.
2. **MULTIPLE**: When multiple options were preferred, the dialog manager would provide additional information for one of the discussed options to continue the discussion.
3. **NONE**: When no options were selected, it would give extra information about one of the answer options that they did not discuss to stimulate further discussions.
4. **UNCLEAR**: When it was unclear which option was chosen, it stimulated further discussion about a random option.
5. **SPEEDUP**: This option helped to speed up the conversation by prompting utterances that stimulate ending the discussion.

The second time it came to the verdict state for the dialog state when it was unclear, no option, or when there were multiple options it randomly selected an option when it was still unclear, it chose the option that they did not discuss when there was no option that fit their preference and it would select one of the option which they were discussion when they had multiple options that would fit their preference. This whole process would be repeated for every question.

To let the robot direct to one of the participants, it had to connect that utterance with a head turn. This would initiate a turn-take from the participant who was listening. However, to

reduce the repetitiveness of the conversation, there were several options from which the robot could choose, making the utterance of the robot less repetitive. The interaction could be influenced when the utterances were redundant across all dialog states, so to minimize this, the robot randomly selected one of the options per interaction construct. If participants had nothing to say, the robot could break the silence with a selection of motivational sentences to stimulate further discussion. In Appendix A, the whole script and utterance options of both topics (holiday planner and building a dream house) are provided. Showing all the options from which the robot could choose.

The Wizard needed to provide correct input during the conversation to maintain the dialog flow. In this system, the turn-taking timing was done automatically based on silence duration and gaze. The timing of the utterance could only be affected in the verdict state when the Wizard had to select one of the verdict options to influence the progression of the dialog states. To minimize waiting time, the Wizard only needed to press a number between 1 and 6 corresponding to the available options. The rest of the dialog manager did not depend on the Wizard's input, which minimized the influence on the response time of the robot.

Speech Recognition

The task required the robot to speak and respond to what was being said. The robot could detect the direction of speech, but implementing speech recognition software would pose some risk. If the robot were to miss words or have recognition errors, it could negatively influence the experiment. The study was designed to get a better understanding of how a robot should behave within a multi-party conversation, and the risk of implementing speech recognition software could have a negative impact on this aim. That was the reason why implementing a speech recognition software was not chosen, but instead a Wizard of Oz technique was used. This technique lets the experimenter select the speech for the robot, which would result in correct responses from the robot during conversations. This was possible because the experimenter followed the conversation closely through cameras from inside the control room.

Additional Features

The robot should appear lively and more realistic to the participants. To create that impression, three features were added to the robot's behavior. The first feature was that the robot gave the impression of eye blinking. Using the LED eyes, it was possible for the robot to briefly turn them on and off. This should simulate the behavior of eye blinking. However, this feature was not programmed manually for the experiment, but it was already implemented in the robot's default behavior.

Secondly, the speed of the head movement from the robot was set at a fixed level. The maximum speed was not used to turn the robot's head to reduce the feelings of aggressiveness. However, it was set at 90% of the maximum speed. This was chosen to make the head movement noticeable and responsive. This was crucial as the main focus of the study was the timing and visibility of the robot's gaze.

Lastly, the participants had different heights when they sat. To account for this and create eye contact correctly, the robot adjusted its head based on the participant's length. The participants had to measure their torso-to-eye length, which was used to increase the robot's pitch accordingly. This should improve the eye contact between the participants and the robot. Also, it helped to keep the faces of the participants within the camera frame of the robot, which was important. Otherwise, a lot of data would be missed, and the robot would not behave correctly

Methods

Research Design

A humanoid robot (Misty II) was used to facilitate interaction in a multi-party setting by engaging in a conversation with two participants. An adapted version of the conversational state machine developed by Spiegels (2015) was implemented. This state machine organizes turn-taking into different states, allowing the robot to regulate turn-taking behavior based on sensory input. A Wizard-of-Oz approach was used because the robot was not capable of understanding speech. Hereby, the experimenter monitored the interaction and selected the appropriate responses that the robot had to give to maintain the flow of the interaction.

The effects of different gaze-based turn-taking strategies were investigated using a 2×2 mixed-subjects design. The first independent variable was the gaze detection model, which was manipulated between subjects. Participants were either assigned to the extended gaze model, in which instances were captured when participants were not looking directly at the robot, or a direct gaze model, which only detected instances where participants should look straight at the robot. The second independent variable was the timing of the robot's gaze aversion relative to turn-taking, which was manipulated as a within-subjects design. In one condition, the robot disengaged eye contact after taking the turn and maintained gaze aversion throughout its speaking turn (only gaze aversion). In the other condition, the robot initially averted its gaze and re-established eye contact with participants according to timings derived from human-human interaction research (timed gaze aversion). The gaze detection model was manipulated as a between-subjects design because participants could learn and adapt to the robot's eye contact detection behavior over time. If participants are exposed to both gaze models, it could influence the behavior of the participants in the second condition and reduce the naturalness of the interaction. The robot's gaze behavior was manipulated as a within-subjects design because it provided the opportunity to compare conditions within the same participants while reducing the influence of individual differences in conversational behavior.

The system and the interaction were evaluated through both objective behavioral measures and self-reported measures. This combination provided a broader and more accurate understanding of the interaction, as behavioral measures captured the observable interaction outcomes, and the self-reported measures captured the perceptions and experiences of the participants (Muckler & Seven, 1992). This allowed for evaluating both the performance of the system, through the amount of turn-takes and interruptions, and participants' subjective experiences, including perceived fluency, naturalness, and likeability.

Participants

The sample size was determined based on the effect sizes reported in prior studies. The effect size was determined based on the Bachelor's End Project of Köhler (2025). An effect size of 0.44 was found for the relationship between silence duration and the presence versus absence of eye contact in the study by Köhler (2025). The effect size for this project is determined by the standard error (SE). When the measurements increase, the SE will decrease. That means doubling the measurements would reduce the SE by a factor of 2. The experiment done by Köhler (2025) had only five dialogues per conversation. For this project, the number of dialogues per conversation increases to nine, which is almost double. This will reduce the SE by $\sqrt{\frac{5}{9}}$. The effect size of 0.44 will increase to 0.59, which is calculated as $0.44 \times \sqrt{\frac{9}{5}}$.

The current study used a between (2 types of eye contact detection) x within (2 timings for robot gaze behavior) design. The hypotheses of interest focused on the within-between interaction effect between the type of eye contact and the gaze timing. To estimate the required sample size, an a priori analysis was done with a power of 0.9 and an effect size of 0.59. This resulted in a sample size of 36 participants for a 2 x 2 repeated measures ANOVA, with a within-between interaction. The results of the power analysis, which was conducted using G*Power (Faul et al., 2007), are presented in Table 1.

In the study by Spiegels (2015), a total of 237 turns were observed during conversations with the conservative turn-take model. These conversations had an average duration of 13 minutes and 27 seconds. Every turn-take could be seen as an individual measurement. These turn-taking decisions were made with the turn-taking model that was based on a gaze and silence cue that was similar to that applied in this study. This meant that with the 7 groups Spiegels (2015) used for this turn-take model, approximately 34 turn-take measurements occurred during the conversations. Based on this observed data and the expected 15-minute conversations, it is expected to measure a total of 38 turn-takes per conversation. With the mixed-subject design, participants completed two dialogues, resulting in an expected total of 76 turn-takes per group. Across all 18 groups, this yielded an estimated total of 1,360 speaker turns.

Table 1: Pre-experiment Power Analysis

Analysis Input		Output Specification (G*Power)	
Effect size f	.59	Noncentrality parameter λ	11.84
Alpha error probability	.05	Critical F	4.13
Power	.90	Numerator df	1
Number of groups	2	Denominator df	34
Number of measurements	2	Total sample size	36
Nonsphericity correction ϵ	1	Actual power	0.92
		Conversations per factor	18
		Participants per factor	18

Participants were recruited by sending an invitation to the JFS participant database. An email was sent to this database where people could sign up for the experiment by clicking on the link to the online booking schedule. This booking schedule was used to let people reserve a time slot. Participants were paid to compensate them for their time.

Experimental Task

There were two tasks that had to be completed. The first task was similar to the "Holiday Planner" from the work of Spiegels (2015), and the second task was an extended version of the task "Designing a Dream House", which was used in the Bachelor End Projects of (Butter, 2025; Köhler, 2025; Van Rossum, 2025). Because both tasks are used by several Bachelor's End Projects and a Master's Thesis, it has proven to be suitable for facilitating a multi-party conversation between a robot and two participants (Butter, 2025; Köhler, 2025; Spiegels, 2015; Van Rossum, 2025). The first task assigned to the participants was to collaboratively plan a holiday. Where the robot (Misty II) acted as a travel agent, and it would ask different questions about the holiday. During the second task, the participants had to design their dream house. The robot acted during this task as a moderator, guiding the participants through the questions. For both tasks, the questions had several options from which the participants could choose. After the robot asked the question, the participants had to discuss the answer and provide it to the robot upon agreement. The robot could give extra information about the topic and was capable of producing backchannel responses to keep the participants engaged and let them know that the robot was still paying

attention. Furthermore, it could motivate participants by switching the attention between the two participants or by directly asking the other person to tell their opinion about the topic.

The duration of each task was approximately 15 minutes, where participants had to find an agreement on 9 questions. Examples of the questions in the holiday planner task included the preferred destination, activities to do during the trip, and the available budget. For the task in which participants were asked to design their dream house, questions were posed about aspects such as interior, location, and size. The questions were in a pre-defined order that makes the next question logical based on the answer to the previous question. This sequence of questions helped them to define their hypothetical holiday or their hypothetical dream house. After each question, the participants had time to discuss. This created opportunities for them to exchange turns, while also giving the robot a chance to intervene. The combination of the pre-defined questions and the open discussion gave a naturalistic setting for studying turn-taking behavior in a multi-party human-robot interaction.

Materials

Experimental Setup

The experiment was conducted with a specific setup, in which the social robot, named Misty II developed by Misty Robotics served as the main component (Misty Robotics, n.d.). The setup was as follows: the Misty II robot was placed on the table, and the participants had to sit on the left and the right of the robot. This meant that they did not sit in a dyadic setup where they face the robot. The two participants were instructed to take a seat at opposite ends of the table, which made them face each other. This would help to make more of an open interaction that would encourage the participants to discuss with the robot, but also with each other. The position was fixed for the participants, ensuring that the same head orientation thresholds for eye contact could be used for all participants. Both participants had their own directed microphone that could signal who was speaking. Placing the participants opposite each other removed challenges related to multi-face tracking and minimized cross-detection from the microphones because each participant was positioned farther from the microphone on the opposite side. The setup is illustrated in Figure 5.

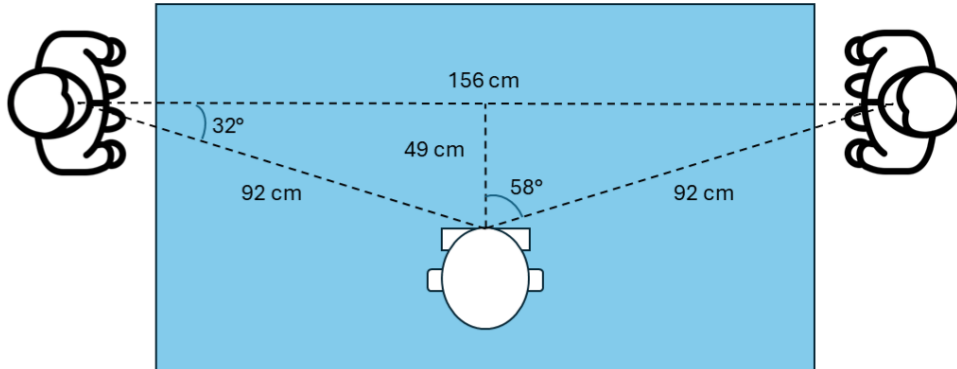


Figure 5: Illustrated top view of the setup during the experiment. Showing the positions of the robot and participants, the distances between them, and the angles.

Hardware

A robot that was capable of interacting and communicating with people required several components to enable this capability. First, the Misty II robot was used together with a wired network router. Misty II was programmed in Python 3.10 using libraries such as OpenCV, PyAudio, and NumPy. Besides this, there are some custom modules for speech detection, MediaPipe framework for face detection, and to control audio sound levels. To capture the speech of the participants, two directional microphones (M5 paired pairs small-diaphragm condenser microphones) were used and connected to the Rednet X2P audio interface. This audio interface converted the audio from both microphones into digital signals that were used to process participant speech. The audio signals were transmitted to an HP laptop running Windows 10 via a USB-C-to-Ethernet connection. A detailed description of the hardware used to capture the speech signals can be found in Appendix B.

Robot Subprograms

The robot's behavior was based on a behavioral state machine and small subprograms. The software behind the behavior of the robot depended on the behavioral state machine and subprograms. These subprograms were developed to process speech and camera input into a format that could be used by the state machine. However, this data had to be changed to understandable and usable data, which was done with the subprograms. This section will give a detailed explanation of the subprograms used.

Speech Detection

The laptop needed specific software to receive the audio signals from the Rednet X2P audio interface. First, a license for Dante Virtual Soundcard was obtained to enable the use of the software. The Dante Virtual Soundcard was used as an interface between the network and the laptop. It enabled the laptop to receive audio streams as if they came from a physical sound card. The Dante Virtual Soundcard had to be turned on before the Dante Controller could be used. The Dante Controller was the second program that had to be installed before the laptop could receive the audio signals. The Dante Controller was crucial in this experiment because it could bring the audio signals from the Rednet X2P audio interface to the laptop. With the use of the Dante Controller, it could easily be set up to determine which input channel on the laptop received the audio signals from the left participant and which input channel received the audio signals from the right participant.

The first subprogram was developed to calibrate the microphones for each participant to compensate for cross-detection and noise. PyAudio was installed because it lets Python record audio. The recording of the audio was done continuously in parts of 20 ms and at a rate of 44.1 kHz. To determine the loudness of the microphones, the Root Mean Square (RMS) was determined. The Python program also separated the channels so that each participant was recorded individually, but this was not necessary anymore, as the Dante Controller did this already. Each microphone was set up as a stereo pair through the Dante Controller, but only one channel was used per participant. If both channels were used, they had to be averaged, which could reduce the loudness and influence the speech detection. Therefore, it was chosen to let the microphones only come in at one channel to ensure a consistent representation of the audio signal. The Python program kept the splitting function, as it would make it easier to replicate the study with different audio setups. For every microphone, it recorded the silence and speech for three seconds each before setting a threshold. The silence was recorded to estimate the sound level of the background noise, and the speech was recorded to determine a threshold relative to that of the silence. The program classified sound as speech when the RMS values exceeded the speech threshold for consecutive frames. This prevented short moments of sound from being mistakenly interpreted as speech. It classified the sound as silence when the RMS value fell below the silence threshold. Cross-detection was minimized by letting the RMS values that fell between the silence and speech threshold be classified as

neither speech nor silence. Furthermore, the robot would only turn towards a new speaker if they spoke for at least 0.5 seconds. In addition, peaks were filtered out because coughs or touching the microphone should not be detected by the robot. This would minimize unnecessary short head movements of the robot.

Face Detection

The second subprogram focused on human gaze behavior. A real-time face detection system was implemented with the use of MediaPipe. This program estimated the orientation of a participant's head, where the orientation was represented by three angles.

- Pitch: vertical movement (up/down)
- Yaw: horizontal movement (left/right)
- Roll: tilt of the head.

In this study, yaw was the main indicator of head orientation. Pitch was also used but less strictly, only to exclude extreme upward or downward orientations, as these do not correspond with eye contact. Roll was not included because it was not relevant to the setup of this study. The system used MediaPipe's Face Landmarker model to detect facial landmarks in real time from the camera that was mounted on the head of Misty II. This model gave back the spatial coordinates of the three facial angles. It worked by capturing an image using the camera, which was then processed by MediaPipe to detect the facial landmarks. These landmarks were used to estimate the head orientation and calculate the pitch, yaw, and roll. These values were then used to indicate whether a participant was looking at the robot. The estimation with the landmarks was done with OpenCV with the function "solvePnP", which estimated the orientation of an object in 3D space from a 2D image. It mapped key features, like the nose and mouth, to determine where they appeared in the image. Two different thresholds were determined for the conditions, which were used to label face orientations as eye contact. For the first condition, the threshold was very strict and set at a yaw angle of 6° or less. This forced participants to look directly at the robot before it detected eye contact. The second condition sets its threshold based on Equation 1 and the REC. Based on the measured distances between the robot and the participants, the corresponding angles were calculated, resulting in a yaw threshold of 18.67°. As explained earlier, the threshold was set at 25° for the extended gaze model as it accounted for the region of eye contact and noise. The pitch angle was set at 30° to exclude extreme values. For the robot to register eye contact as a valid cue, participants were required to maintain eye contact for at least 0.5 seconds. This threshold was implemented to prevent glances at the robot, which may not indicate a signal to take the turn. A sanity check was conducted to verify if the face detection model reports plausible yaw and pitch angles for different head orientations. The angles were found to be consistent with the expected direction of head movements.

Audio–Visual Processing for Turn-Taking Decisions

The subprograms were executed separately from each other next to the behavioral state machine. The audio recording regulated how the microphones picked up the sounds and speech from the participants. The speech detection program processed the two microphones and returned the current speaker, while the face tracking program updated the latest head-pose estimation. The behavioral state machine used the data from these subprograms to make a decision whether to respond or to keep listening during the conversation. If the behavioral state machine did not have these input values, it could not act and react appropriately in a real conversation because these programs acted as the eyes and ears of the robot.

System logging

The data collected during the interactions had to be logged correctly to support the analysis. The logged events were saved to CSV files, stored in a session-specific folder, along with the participant ID and condition. The logging consisted of seven files, each capturing a different aspect of the interaction.

Table 2: Overview of the generated log files and their purpose

Log File	Purpose
RMS log	Records microphone activity and sound levels during the interaction.
Gaze log	Has information about the head pose of the participants, providing details about how the participants behaved related to head orientation.
Speech log	Contains turn-taking and speech activity data of participants and robot.
Timeline log	Chronological record of interaction events, used to reconstruct conversations and debug issues.
Operator log	Captures actions performed by the Wizard during the experiment.
Face-pose log	Stores the head pose data of the participants.
Summary log	Provides the total number of interaction events during the conversation.

During the interaction, several logs were created (Table 2). A lot of events were logged because they can show how the interaction went, but it was also important to have this much data if something went wrong. Making it possible to look back to find where the error or problem occurred. A complete and detailed overview of all the variables that were logged can be found in Appendix C.

Measures

Objective Behavioral Measures

The evaluation of the effect of turn-taking based on gaze and silence during an interaction was done by collecting several quantitative measurements derived from the systems logs. The main measures focused on different aspects of the conversation, including turn-taking behavior, speech timing, and gaze behavior.

Turn-Taking Measures

The turn-taking behavior was measured through the variables frequency of turns and the interval between participants' speech, which were extracted from the speech log, summary log, and timeline log. The frequency of turns is the number of turn switches during the conversation, and could give insight into how many times the participant or the robot took the floor. Turn transitions were classified as human-human, human-robot, or robot-human. The turn-taking timing is the interval between the end of a speaker's utterance and the onset of another speaker's utterance. Indicating how long it took for someone to start speaking reflects on how fast the robot or participants respond. It was recorded as positive values, indicating a silent gap between utterances. This measurement could give better insight into how quickly speakers responded to each other during the conversation. The turn-taking time interval is visualized in Figure 6, which shows a schematic overview of an interaction that can happen between two speakers, and how each variable is defined within a conversation.

Speech Measures

Speakers' utterances were measured through speech duration, speech overlap, and silence duration. These measures were extracted from the speech and timeline log. The duration of an utterance is the total time that a speaker spoke and is measured between the start (onset) of the speech and the end of the speech (offset). The measurement of speech overlap refers to moments when one speaker has the floor and speaks while someone else who didn't have the floor tries to take the turn and speak. These overlaps ranged from short backchannel responses to longer periods of simultaneous speech. Silence duration was determined separately for each participant's microphone. This measure was calculated from timers that kept track of the interval between the offset of one utterance and the onset of the subsequent utterance from the same speaker. These speech-related variables are visualized in Figure 6.

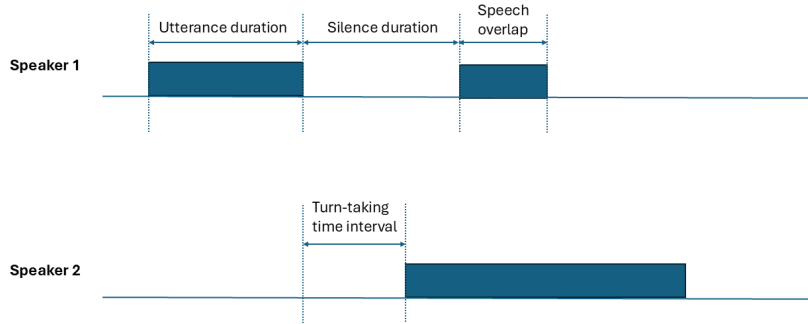


Figure 6: Schematic representation of a conversational timing measure between speaker 1 and speaker 2. Showing utterance duration, which is the total duration of a speaker's utterance, including any overlap with the other speaker. The speech overlap occurs when speaker 1 begins speaking before speaker 2 is finished. The turn-taking time interval is measured between the end of speaker 1's utterance and the start of the utterance of speaker 2. Silence duration refers to the period during which no one is speaking.

Gaze Measures

Gaze refers to the direction in which a person is looking. Several variables were captured from the gaze and timeline log to evaluate the visual attention of the participants during the conversation. The moments when eye contact was made were indicated when participants looked at the robot. This was determined by whether the yaw and pitch angles of the participants' head orientation met the eye contact thresholds of the experimental gaze detection model. The duration of eye contact was determined by the total amount of time participants maintained eye contact with the robot. Furthermore, the human-directed gaze showed the moments when participants directed their gaze towards the other participant instead of towards the robot. This was determined based on whether the gaze and pitch fall within the threshold for human-directed gaze. The participants' yaw and pitch angle of their head orientation was recorded to understand what happens with the gaze behavior of the participants during the multi-party interaction, which made it possible to analyse how gaze coordination relates to the timing of turn-taking. Figure 7 shows one scenario where the participants look at the robot, and when they avoid eye contact, it visualizes moments when participants look at the robot and how long they make eye contact. In cases where gaze is not directed at the robot, but the silence duration is long enough, the system labels the behavior as turn-yielding towards the other participant.

These measurements should give insights into how actively the robot participates during the conversations with two participants and how it can manage its turn-taking behavior.

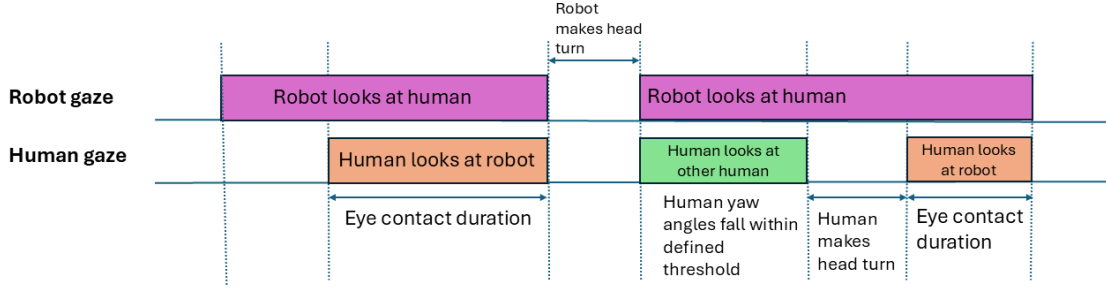


Figure 7: Schematic illustration of the gaze measure moments. It illustrates moments when the robot has eye contact with a human, and it shows when the human looks at the other human (defined by the thresholds on yaw angles).

Self-Reported Subjective Measures

The self-reported subjective experiences are collected through a questionnaire, which was developed based on validated questionnaires in the domain of human-robot interaction research. It had to capture the subjective experiences of the participants. The questionnaire consisted of a combination of the RoSAS questionnaire, developed by Carpinella et al. (2017), and the SASSI, which is developed by Hone and Graham (2000). These items captured experiences like perceived likability, social presence, naturalness of the interaction, and overall comfort. The complete questionnaire can be found in Appendix D.

Pre-Questionnaire

Before the participants started with the experiment, they had to complete a pre-questionnaire. This was a short questionnaire that collected demographic information, including gender and age. In addition to these basic demographics, a question was included to assess whether they had prior experience with a robot. This is done because it is shown that people who had interactions with robots before can influence how they experience the robot in the study (Rosén et al., 2024). Another question in the pre-questionnaire asked whether they had prior experience with speech interfaces, such as Siri and Google Assistant. Previous experiences with speech interfaces might influence the interaction with a speaking robot as well. Finally, the torso-to-eye length was collected in centimeters to adjust the robot’s gaze upward or downward to compensate for the different lengths of the participants. This could make the eye contact detection more accurate because, without this adjustment, the participant’s face might fall outside the detection range, leading to unreliable face detection.

RoSAS questionnaire

The participants had to complete a questionnaire after each of the two tasks. The first part of this questionnaire consisted of the Robotics Social Attributes Scale (RoSAS). This questionnaire focused on the social attributes of the robot. The questionnaire consisted of three dimensions: warmth, competence, and discomfort. A strength of the RoSAS was that it was validated and developed across robots in general, instead of only to a specific robot, image, or interaction scenario (Carpinella et al., 2017). The three dimensions were measured using descriptive adjectives. Each dimension captured a different aspect of the participants’ perception of the robot. The warmth dimension mainly focused on how friendly and approachable the robot was. This was relevant for conversational interactions where gaze and turn-taking might influence the perceived social connection. The competence dimension primarily concerned the robot’s capabilities and intelligence. The RoSAS part of the questionnaire also measured the discomfort dimension, which reflects how participants perceived the robot as awkward or uncomfortable. The dimensions had to be rated

on a 5-point Likert scale from 1 (not at all) to 5 (very much), indicating how well each adjective described the robot.

Adaptations to the RoSAS questionnaire

The RoSAS was a validated instrument to capture the social perceptions of robots. However, an adaptation was made to better align the questionnaire with the study. The dimension of "competence" was almost the same as the "perceived intelligence" in the Godspeed questionnaire from Bartneck et al. (2009). This measured if the robot responded normally and if it was capable of turn-taking. The remaining dimensions, warmth and discomfort, were useful as they captured complementary aspects of social perception. Gaze behavior and timings of turn-taking might influence how comfortable participants felt during the interaction (Parreira et al., 2022), and it was expected to influence the social connection because the warmth dimension included trust-related attributes, which were influenced by gaze (Saunderson & Nejat, 2019). All the items from these dimensions were used in the original form, except for the item "feeling". This item referred to the extent to which the robot was capable of experiencing emotions. The item 'feeling' was reformulated to 'seems to have feelings' to improve clarity and ensure consistent interpretation by participants. All the items were randomized to prevent order effects.

SASSI

The final component of the full questionnaire was the Subjective Assessment of Speech System Interfaces (SASSI). This assessment measured the user's subjective experience of speech interfaces. During the experiment, participants interacted with a speech system (Misty II), and SASSI was used to assess their subjective experience of this interaction, including aspects such as naturalness, usability, and system responsiveness, making it relevant for the study. SASSI consisted of six factors that had reasonable reliability. The factors were evaluated on a seven-point interval scale range (strongly disagree - strongly agree). The six factors were described as follows:

- **System Response Accuracy:** measured if the system response was accurate and did what the user expected. It was measured with nine items and was used to measure the gaze and speech behavior of the system, as the understanding of the conversation was done by the operator and not by the system.
- **Likeability:** consisted of nine items that were used to evaluate the user's opinion of the system.
- **Cognitive Demand:** reflected how demanding the participants found the interaction with the system. This was measured with five items.
- **Annoyance:** evaluated with five items what the negative experiences with the system.
- **Habitability:** used four items to describe how well users understood how to interact with the system and what the system was doing at any given moment.
- **Speed:** had only two items to describe the system's response time.

Adaptations to SASSI

To better fit this study, some adaptations were made to the SASSI. The dimensions of habitability and speed both had lower reliability scores compared to the other dimensions. The reliability score for the dimension habitability was 0.75, and the speed dimension had the lowest reliability of 0.69 (Hone & Graham, 2000). The low reliability of the factors speed and habitability could be improved by using more items to measure them. Spiegels (2015) used SASSI in his study about turn-taking and extended the factors with items from REVU-NL questionnaire (Report on the Enjoyment, Value, and Usability of Natural Language interaction) that was used in the report of Dzikovska et al. (2011). Spiegels (2015) added three questions from the REVU-NL questionnaire

to extend the SASSI, which should improve the reliability of this dimension. According to Knehta et al. (2019), it was preferred to use at least three items for each category to improve the reliability, which was why Spiegels (2015) added one item to habitability and two items to the factor speed. The item that was added and reworded to the habitability factor was: "I always knew what to say or do at each point to the system". The following two items were added to the factor speed: "The system took too long to respond" and "The system responded quickly". This resulted in four items for the factor of speed and five items for the factor of habitability, thereby improving the reliability of the factors by increasing the measured items. The full SASSI consisted of 37 items. All the items were randomized to prevent order effects.

Post-questionnaire

Participants were asked to complete the RoSAS and SASSI questionnaires after they finished each interaction with Misty II. The questionnaires were given to the participants immediately after the interaction to capture their impressions and reduce the influence of external factors and memory decay. In total, participants completed the questionnaires twice due to the within-subject design of the timing for gaze aversion behavior of the robot. Making it possible to compare within the same participant how they experienced the robot across different conditions. The complete questionnaire consisted of 57 items. The first question asked what task they performed, and the second question captured their participant ID. The remaining 55 items measured their experience during the interaction with the robot, based on the RoSAS and SASSI.

Procedure

The participants waited outside the lab before being picked up and escorted inside. They could store their belongings in the entrance room, and after that, they entered the experiment room. This room included a table with Misty II placed on top and two directional microphones. Before the experiment started, the participants had to fill in the pre-questionnaire on their own. They could ask any question regarding this questionnaire to the experimenter. At the start of this pre-questionnaire, the participant had to read and sign the consent form before they could continue. If they had any questions, they could ask them directly to the experimenter. The consent form can be found in Appendix E. The participants could measure their torso-to-eye length on the spot with a measuring tape if they did not know their length. They also received their own participant ID, which was needed for every questionnaire. When the participants were done with the pre-questionnaire, they were instructed to take a seat at the table with the robot. After participants were seated in their designated places, they were given time to get to know one another to create a more comfortable atmosphere for the experiment.

The experimenter will give them a detailed instruction about the task and instruct the participants to give their opinions and discuss the options on the asked questions. It was crucial to let the participants understand that they had to include the robot in the conversation and that they should not only talk with each other. Furthermore, during the discussion, participants were required to reach a mutual agreement on an answer before going to the next question. Participants were also informed that they had to complete two tasks, and that after finishing the first task, "The Holiday Planner," they would proceed to a second task. To familiarize participants with the first experiment, they were provided with example topics, such as holiday countries or activities. This was done in the setup shown in Figure 8.

To ensure the experiment was conducted under optimal conditions, participants were instructed to imagine going on a hypothetical holiday together and to plan it as if it were a real trip. Furthermore, the participants were instructed to remain seated in the same position and to limit their movements to head movements only, because this would optimize the eye contact detection. They were asked to lay their arms on the table to prevent large body movements. However,

small body movements were allowed, like head orientation. To have better conversations during the experiment, the participants were told to avoid short answers like yes or no and instead use short sentences. Further instructions specified that participants should work together to figure out what the robot said when they did not understand what was being told, because the robot could only repeat the question. The robot could only provide basic background information, so the participants were instructed not to ask complex and difficult questions to the robot. Also, they were asked not to touch the microphones in front of them, as these microphones will be calibrated from that position, and touching them could make loud noises that the robot could accidentally pick up as speech.

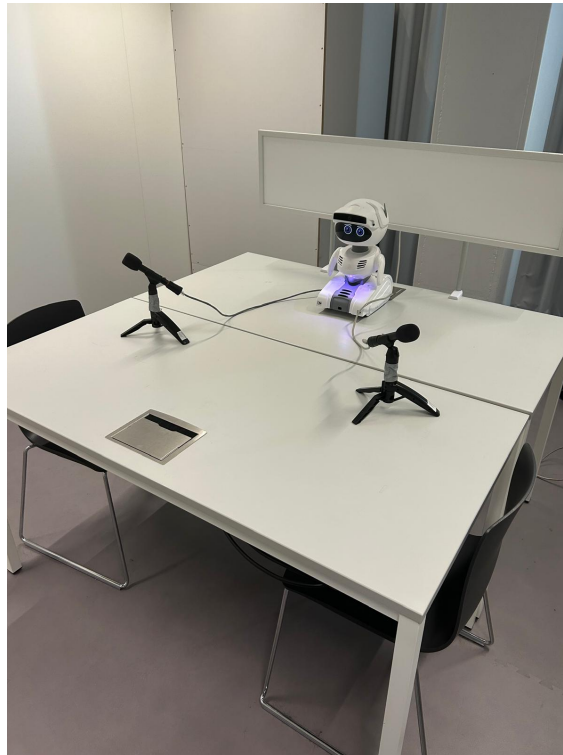


Figure 8: Setup in the experiment room

When everything was clear for the participants, the experimenter calibrated the microphones for both participants. This was done to minimize cross-detection. When everything was set up, the experimenter left the experiment room to enter the control room. From this room, the experimenter could watch the experiment room with the participants through several cameras. When the experimenter was ready, the program for the robot’s behavior was started. To maintain the flow of the conversation, the experimenter monitored the interaction and performed actions based on the current state of the participants and the discussion topics. This was done because the robot was not able to understand speech, so the experimenter had to make use of the Wizard of Oz technique. The experimenter gave the robot input through a keyboard. The experimenter monitored the conversation and based the keyboard input on the responses of the participants. The experimenter could select appropriate answer options that matched the mutual agreement of the participants, it could instruct the robot to repeat the question, and trigger predefined utterances to maintain the flow of the discussion. Furthermore, the experimenter could advance the interaction to the next question when the participants had already reached an agreement, preventing the robot from asking unnecessary follow-up questions regarding opinions or alternative answer options.

After the participants reached their final agreement and the robot registered their decision, the robot provided a summary of the conversation. The experimenter went back to the experiment room after the robot ended the interaction. Here, the experimenter instructed the participants to complete the questionnaire and told the participants that they could ask questions if they had any questions regarding the questionnaire. If they finished the questionnaire, the experimenter explained to them the second task they had to complete. Now the participants were informed that they would hypothetically live together, and therefore they had to design their dream house. First, the participants got some example topics, like the furniture they would want to have or the color of the house from the outside. The participants got the same instructions to prevent the use of large body movements, and they had to discuss and give their opinions on the different topics to come to a mutual agreement. If everything was clear, the experimenter calibrated the microphones and left the experiment room. From the control room, the experimenter started the second task and used again a Wizard of Oz technique to maintain the conversation between the robot and the participants. After the task was completed, the experimenter went back into the experiment room and guided the participants to the computer to fill in the last questionnaire, but now they had to complete this questionnaire based on their experiences from the second task. They could ask questions to the experimenter if they did not understand something.

The participants got debriefed about the purpose of the study when all the questionnaires were filled in. They got the time to ask additional questions about the study that they would like to know. Once everything was completed, they were thanked for their participation, received their compensation, and were escorted out.

Data Analysis

The analysis of the data is done in Stata 19. The descriptive statistics for the participants and behavioral measures were calculated. For the analysis of the data, mixed-effect models were used because the study had a mixed design with repeated observations within participant duos and repeated questionnaire measurements. Prior to the analysis, the data were cleaned by removing missing values, tracking errors, and extremes.

The model assumptions of normality of residuals, homoscedasticity, and normality of the random effects were evaluated through visual inspection of histograms, Q-Q plots, residual-versus-fitted plots, and the Shapiro-Wilk test. For the mixed-effects logistic regression, sufficient observations within outcome categories and model fit were evaluated. If assumptions were met, the linear mixed-effects models or the mixed-effect logistic regression were conducted, depending on the type of outcome (continuous or binary). Gaze detection model, robot gaze behavior, and their interaction, were used as fixed effects, while participant pair or participant ID was included as a random intercept when appropriate to account for repeated measures. If the assumptions were violated, appropriate transformations were applied to improve normality. If the assumptions remained violated, a non-parametric test was used. The Wilcoxon signed-rank tests were used for within-subject comparisons, and Mann-Whitney U tests were used for between-subject comparisons. Finally, for the questionnaire analyses, some items had to be reversed. Furthermore, Bonferroni corrections were applied separately to the RoSAS and SASSI dimensions to account for multiple testing.

Results

Demographics participants

A total of 46 participants took part in the experiment. During the experiment, some technical problems and unforeseen circumstances occurred, which led to the exclusion of six participants from the data. The participants had to come in pairs to the lab, where they got randomly put in one of the conditions. Table 3 shows the collected demographic characteristics of the participants. The participants had an average age of 24.35 years ($SD = 3.34$), with ages ranging from 20 to 35 years. Moreover, the torso-to-eye length was collected from the participants to adjust the pitch of the robot head to the participant’s height. However, the analysis showed several incorrect values for this data. That is why 8 values were excluded from the data before further analysis. These torso-to-eye lengths ranged between 67 and 86 centimeters. There was a nearly equal distribution of male and female participants across the experiments. Furthermore, the participants indicated if they had previous experience with robots, and of the 40 participants, 28 participants (70%) had already seen or played with a robot. Most participants (17.50%) have experience with a robotic vacuum cleaner, and 15% of the participants had an interaction with service robots. A bit more participants had experience with a speech interface (72.50%). From the speech interface, Siri (25%) and Google Home (22.50%) were the most used speech interfaces by the participants. In total, 35 participants (87.50%) have experience with either a speech interface or a robot.

Table 3: Participant demographics

Variable	Value
Sample size	$N = 40$
Age	$M = 24.35$, $SD = 3.34$
Torso-to-eye height (cm)	$N = 32$, $M = 77.70$, $SD = 5.31$
Gender	
Female	$n = 18$
Male	$n = 22$

Preliminary data analysis

The actual measurements were lower compared to the predicted measurements, with 894 turn-take measures recorded across all sessions. Even though the conversations were longer than previously thought ($M = 18.35$ minutes, $SD = 3.16$ minutes, $MIN = 13.11$ minutes, $MAX = 26.21$ minutes). The verdict command used by the Wizard helped to keep the interaction going by preventing the robot from asking redundant questions or asking for opinions that the participants had already given. This maintained the flow of the interaction, but resulted in variations in the number of robot turns across the interactions. This caused the lower amount of turn-taking measures than expected. The average turn takes, the duration of the conversations, the robot turn takes, and the robot turn takes per minute are shown in Table 4 together with the standard deviation. What stands out from Table 3 is that the number of turns per minute is almost the same for the condition with an extended gaze detection compared to the direct gaze detection model. Furthermore, the total turns are more in the direct gaze, which may have happened when people couldn’t get the robot’s attention due to the direct gaze model, and they tried to speak more to get its attention by calling it out. To keep the analysis simple, the robot’s gaze behaviors are labeled as static gaze aversion and timed gaze aversion. When the robot only gazed away and kept its head orientation towards the middle when it spoke was labeled as static gaze aversion. If the robot’s behavior used timings for gazing away and looking back and forth between the participants when it spoke, it was

RESULTS

Table 4: Descriptive statistics (means and standard deviations) for the robot turn takes and the duration of the interaction across the different robot behavior and gaze models.

Group	N	Turn takes	Dialogue duration (s)	Robot turn take	Robot turn take / 1 min
Robot Gaze Behavior					
Static Gaze Aversion	20	79.40 (35.30)	1074.79 (162.49)	21.15 (6.33)	1.20 (.35)
Timed Gaze Aversion	20	79.35 (37.77)	1127.22 (214.32)	19.40 (8.27)	1.03 (.37)
Gaze Detection Model					
Extended Gaze	20	75.45 (33.33)	1127.22 (187.30)	20.85 (7.71)	1.12 (.40)
Direct Gaze	20	83.30 (39.11)	1074.79 (193.01)	19.70 (7.06)	1.11 (.33)
Total	40	79.38 (36.08)	1101.01 (189.59)	20.28 (7.32)	1.11 (.36)

Note. Turn takes include both the participant turn takes and the robot turn takes.

labeled as timed gaze aversion. The total turns for the timed gaze aversion condition are similar to the robot’s static gaze aversion condition. The robot turns per minute differ between the gaze detection models, but it also differs between the robot’s gaze behavior. Showing that the robot got an average of one turn each minute in all experimental groups.

Pre-processing

Several steps are applied for cleaning the data before statistical analyses. This is done to improve the reliability of the turn-take measurements and reduce the noise in the data that comes from unintended events. For the analysis of the silence durations, speaking duration, and turn-taking timing, the interruptions and overlaps are excluded from the data. These are moments where participants spoke at the same time or where the human interrupted the robot, or the other way around. To keep in mind that when someone wants to address a person or robot in the conversation, they often gaze at them. However, very short gaze durations might be just a coincidence when someone quickly looks at another place. That is why short gaze events are filtered out. Gaze events shorter than 500 ms are excluded, as they would only be seen as a turn-take cue when it was longer than 500 ms. All speech durations lower than 500 ms are excluded because in the study of Spiegels (2015), a threshold of 250 ms was used, which was based on the minimum pronunciation time of short utterances such as “yes” and “no”. This present study made the threshold a bit stricter to reduce the inclusion of backchannels and brief acknowledgments that are unlikely to represent meaningful turns. The system also logged robot-to-robot transitions, which are excluded from the data for turn-taking analysis because they do not represent a transfer of the conversational floor between speakers. To evaluate the gaze models and yaw angles of the participants, the distributions were inspected. The observations followed the pattern as expected, with left-side participants exhibiting positive yaw angles and right-side participants exhibiting negative yaw angles. However, observations with the opposite yaw direction were also present. A lot of these observations were caused by the robot shifting its attention to the next participants, but in that motion still tracking the other participants’ yaw angle. To reduce the influence of these tracking errors, observations with yaw angles beyond the eye contact threshold and opposite to the expected participant position were excluded from further analyses. Observations were removed only when the yaw angle exceeded $\pm 25^\circ$ in the opposite direction, rather than at 0° , because small deviations around 0° may result from natural head movements while still falling within the eye contact region. Furthermore, the data of the participants were also excluded if eye contact was detected and their pitch angle was $< -20^\circ$, or $> +20^\circ$.

A detailed visualization of the trigger moment for the robot to take the turn is shown in Figure 9. This short timeline of a specific experiment session of two participants shows the speaking durations of the participants and the robot, but it also shows the moments of turn-takes. This timeline is created so that it becomes clear how the audio data in RMS values of two participants during an experiment session, together with the gaze model, trigger the robot to speak. The audio levels of the left participant (participant 37) are shown in blue, and the

audio levels for the participant sitting right from the robot (participant 38) are shown in red. The moments that the robot spoke are the long silences, as the microphones would not pick up sound when the robot produced an utterance. What can be seen is that when eye contact was detected together with a brief moment of silence, a longer moment of low audio levels occurred, indicating that the robot took the turn. It is also clear that when one person spoke, the microphone of the other participant did not detect that speech with high RMS values, so it minimized the moments of cross-detection. The moment the RMS values increased again shows a turn-yield from the robot to the participants.

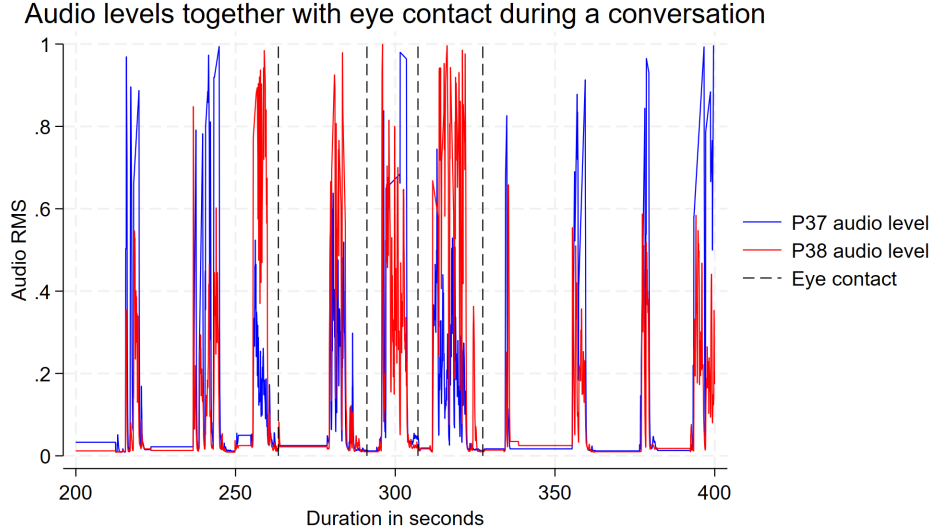


Figure 9: Audio levels (RMS) over time with the moments of eye contact detection for two participants that had the direct gaze model and the condition with the robot’s static gaze aversion behavior.

Successful turn takes by the robot

Robotic turn-taking was not always successful. The script and behavioral state machine are both built in a way to create as many eye contact moments and speaking moments for both participants as possible. This was successful as both participants had an equal amount of speaking moments because participants that were left of the robot had 47.07% (401/852) of the total speaking moments and participants on the right of the robot had 52.93% (451/852) of the speaking moments.

The real durations of silence before the robot took the turn during the conversations are shown in Figure 10. This is the duration between the end of the participant’s speech and the turn-taking of the robot. What is noticeable is that there is no silence of 1 second before the turn of the robot. Originally, the robot had to take a turn around 1 second when it detected both silence and eye contact. However, due to a programming error, it was delayed by 0.5 seconds. This resulted in the robot taking a turn around 1.5 seconds when both cues were detected. That is why a successful turn-take was determined when the silence duration was between 0.9 and 1.75 seconds, and a gaze of at least 500 ms was detected. Measurements outside this interval were classified as unsuccessful triggers. The vertical line at 4 seconds has many more detection points of silence than at 1.5 seconds. Indicating that the speaking turns of the robot often happened when the participants did not speak for four seconds. Furthermore, both the extended gaze models have high peaks at 1.5 seconds compared to the direct gaze models. The direct gaze models have higher peaks at four seconds, which can indicate that the extended gaze models triggered the robot earlier to take

the turn than the direct gaze model. There are some silence durations above four seconds, which could have occurred because the automatic turn-taking mechanism was interrupted by the Wizard and therefore in another state, or it could be delayed by the eye contact detection because in some cases, eye contact was still being detected by the system, preventing the robot from taking the turn until the no eye contact condition was satisfied. The total successful turns taken by the robot using both the silence and gaze cues for each experimental group can be seen in Table 5.

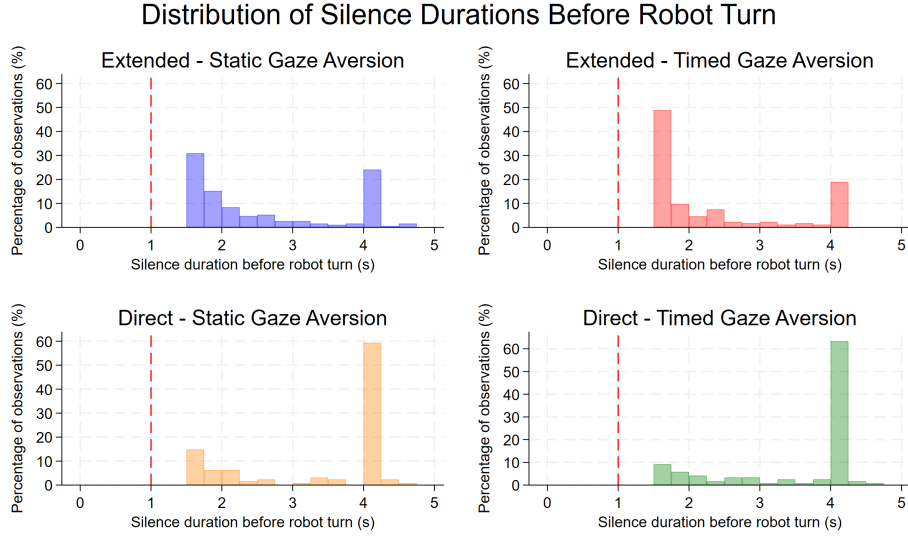


Figure 10: The silence duration in seconds before the robot’s turn-take for each of the four experimental models, and the vertical dashed line is the original planned silence before a robot turn.

What is interesting from Table 5 is that the extended gaze model with timed gaze aversion has the highest successful triggers of all experimental groups (49.42%), and that the direct gaze model with timed gaze aversion has the lowest amount of successful triggers (9.24%). In total, 28.45% were successful triggers (173/435). This indicates that the cues did not work as well as expected for the robot to take the turn.

Table 5: Successful robot turn-taking triggers by gaze model and robot’s gaze behavior.

Gaze Model	Robot Behavior	Unsuccessful	Successful	Total	Success Rate (%)
Extended	Static Gaze Aversion	132	58	190	30.53
Extended	Timed Gaze Aversion	87	85	172	49.42
Direct	Static Gaze Aversion	108	19	127	14.96
Direct	Timed Gaze Aversion	108	11	119	9.24
Total	–	435	173	608	28.45

To test whether the models have a significant effect on the number of successful triggers, a mixed-effects logistic regression model is used. For this model, the assumptions were evaluated and concluded that they were sufficiently met. The analyses showed that there is a significant effect of the gaze model on the amount of successful triggers, $B = -1.20$ log-odds, $SE = .60$, $z = -2.01$, $p = .045$. Meaning that the extended gaze model resulted in more successful triggers throughout the conversation. Also, a significant effect is found for the robot’s gaze behavior, indicating that timed gaze aversion got more successful triggers, $B = .97$ log-odds, $SE = .25$, $z = 3.92$, $p < .001$. Finally, a significant effect of the interaction between the gaze model and the robot’s gaze behavior was found, $B = -1.32$ log-odds, $SE = .49$, $z = -2.68$, $p = .007$, indicating

that the effect of the robot’s gaze behavior on trigger success differed between the extended and direct gaze models.

Conversation analysis

The interactions are analyzed at the moments that the robot took the turn, the participants spoke, and the moments of eye contact. Figure 11 shows how the behavioral state diagram operated in an experiment session where the participants were placed in the direct gaze detection and static gaze behavior design. The figure shows both the speaking turns of the robot and the participants seated left or right of the robot. Furthermore, it shows the moments when eye contact was made that triggered the robot. The participants alternated multiple speech turns between them and the robot, with little overlapping. It also shows that there were several moments when the robot took the turn, during which both the silence duration and eye contact were detected. Illustrating the role of these cues in the implemented turn-taking model. There are also moments when the robot took the turn without eye contact. This happened when the system did not detect speech for at least four seconds. This triggered the fallback state, in which the robot automatically took the turn to keep the conversation going.

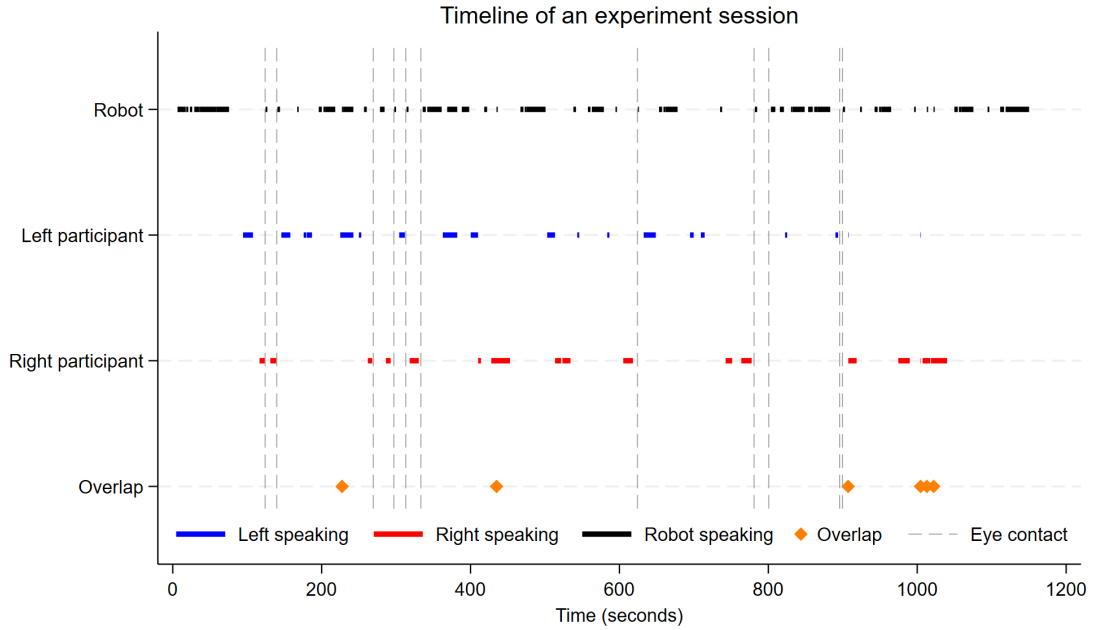


Figure 11: Timeline of an experiment session where two participants interacted with the robot using direct gaze detection for static robot gaze aversion. The moments when the robot, left participant, and right participant spoke are visualized. Also, the overlap moments and eye contact moments are shown.

A total of 3.354 turns have happened after excluding turn-take durations that were longer than 30 seconds. This resulted in 26.65% human-to-robot turns. Most of the participants switched between each other, accounting for 46.48% of all turns. The turn yield from the robot to the participants was 26.86% of the total. The final turns were between the robot itself. Human-robot and robot-human transitions together accounted for 53.52% of all turn transitions, indicating that the robot was involved in approximately half of the observed turn-taking events. All the amounts of turn types per experimental group can be found in Table 6.

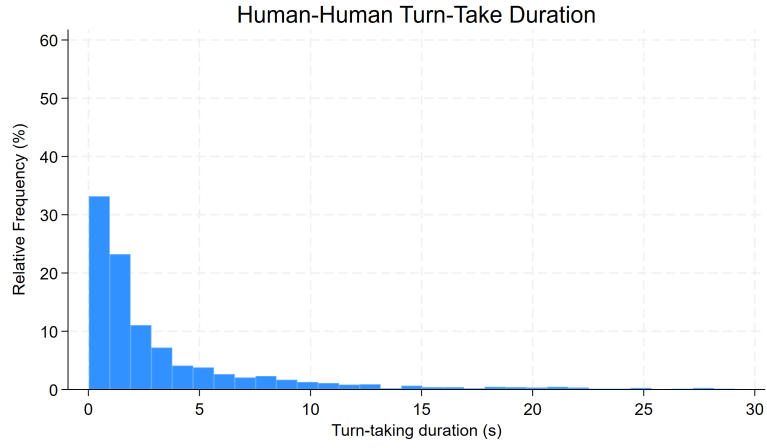
RESULTS

Table 6: Distribution of turn types across the four experimental groups with the percentages.

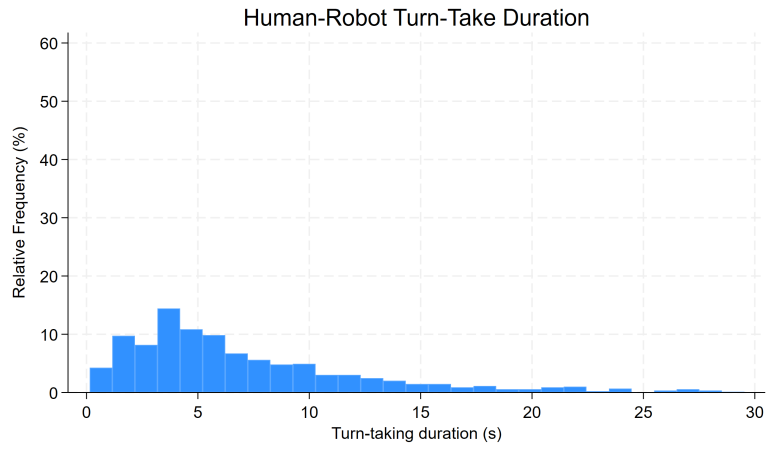
Experimental Group	Human-Human	Human-Robot	Robot-Human	Total
Extended, Static Gaze Aversion	314 (39.50%)	239 (30.06%)	242 (30.44%)	795
Extended, Timed Gaze Aversion	360 (45.06%)	218 (27.28%)	221 (27.66%)	799
Direct, Static Gaze Aversion	428 (49.03%)	222 (25.43%)	223 (25.54%)	873
Direct, Timed Gaze Aversion	457 (51.52%)	215 (24.24%)	215 (24.24%)	887
Total	1559 (46.48%)	894 (26.65%)	901 (26.86%)	3354

The turn-taking duration between humans or between the robot and humans is interesting, as these tell how quickly the robot responds to the humans compared to the timing between humans. The turn yield from human to robot has some outliers that can be dropped from the data. Looking at Figure 12, all turn-takes that are longer than 30 seconds will be considered outliers and are excluded from the data. Figure 12 shows three histograms with the frequency of every turn-taking duration across all conversations. Figure 12a and Figure 12c both show a peak at the first bin. The most turn-taking duration occurred in the 0–1 second interval for human-to-human turns. Approximately 19.1% of all human-human turn transitions occurred within 0.5 seconds. For the robot-to-human turns, this was in the 1–2 second interval. Comparing this to Figure 12b shows that the most turns for human-to-robot happen at the 3–4 second interval. Overall, Figure 12 indicates that the turns are earlier when humans take the turn compared to when the robot takes the turn. The median values support the same interpretation. The median duration for human-to-human turn-takes is 1.53 seconds, which is lower than that of both the median duration for robot-to-human transitions, 2.52 seconds, and the median duration for the turn from human-to-robot, which is 5.47 seconds. These results indicate that humans generally took turns more quickly than the robot.

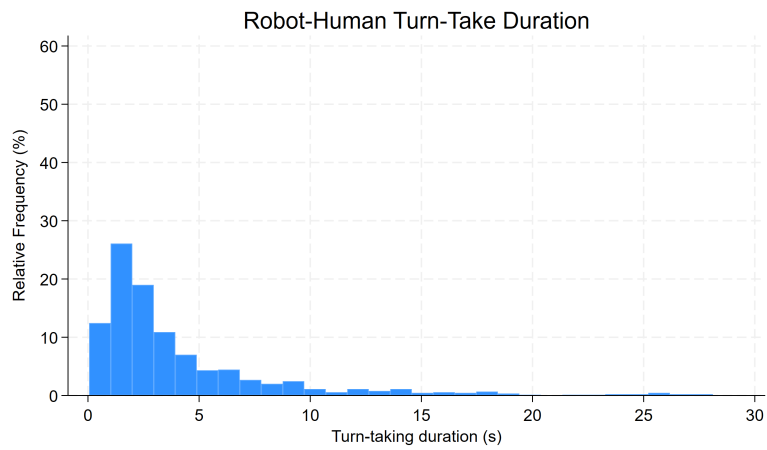
To test whether the gaze models, conditions, and their interaction influenced turn-taking duration, a rank-based non-parametric test was conducted. The assumption of normality could not be met with this distribution of turn-take durations. That is why a non-parametric test is used. The within-subject conditions: static gaze aversion behavior, and timed gaze aversion behavior are tested with a Wilcoxon signed-rank test. It showed no significant differences between the turn-take duration of both the robot’s gaze behavior, $z = .00$, $p = 1.000$. The between-subject effect is tested with the Mann-Whitney U test, which revealed no significant difference between the extended and direct gaze detection models, $z = .49$, $p = .626$. Finally, the interaction effect was examined by comparing the difference scores between the robot’s gaze behavior across the different gaze detection models, which was not significant, $z = -0.38$, $p = .706$. This suggests that the effect of the robot’s gaze behavior on turn-taking duration did not depend on the gaze model.



(a) Human-Human



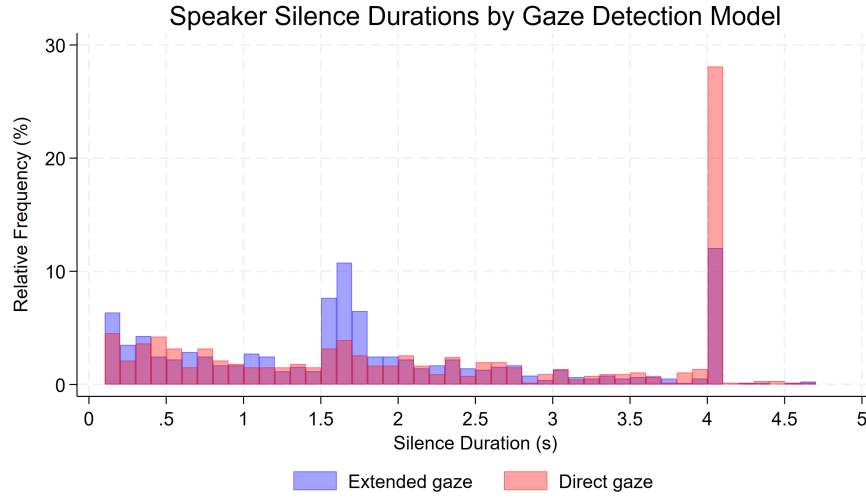
(b) Human-Robot



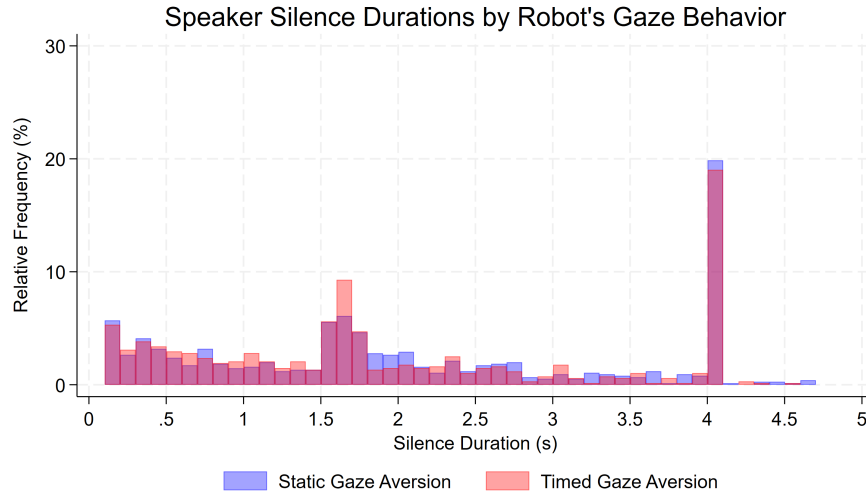
(c) Robot-Human

Figure 12: Distribution of turn-taking durations for the three turn-transition types: human-human (a), human-robot (b), and robot-human (c). The frequency (%) is displayed on the y-axis and the turn-taking duration (s) on the x-axis.

Speaker silences show how long someone was quiet before a new utterance began. Figure 13 shows the distribution of all the silence durations across both gaze models, including speaker hesitations, turn-taking moments, and speech planning pauses. It includes all silence, so silences for hesitations and silences between the onset and offset of speaker utterances. Most silences are quite short, with a lot beginning less than two seconds. There are two peaks in Figure 13a, one at 1.5 seconds and one at 4 seconds, which shows that the gaze model that included an extended view had more silence durations around 1.5 seconds, and the peak at 4 seconds was mostly by participants who had to face the robot directly. Participants had to wait quite long before a new utterance began, as in the direct gaze model. Figure 13b shows a similar distribution for the silence duration for the robot's gaze aversion behaviors. Also, this shows two peaks around 1.5 and 4 seconds, although the timed gaze aversion behavior showed a somewhat larger peak around 1.5 seconds, whereas around 4 seconds, both behaviors resulted in the same frequency of silence durations.



(a) Silence duration for extended versus direct gaze model



(b) Silence duration for conditions with robot gaze behavior (static vs. timed)

Figure 13: Distribution of all recorded silence durations. The figures include turn-taking gaps, hesitations, and speech-planning pauses occurring during the interaction.

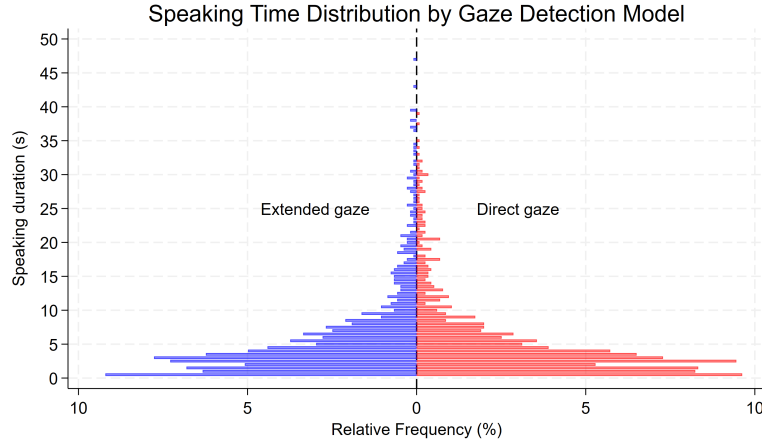
The descriptive statistics for the silence durations are shown in Table 7. It shows the difference between the two gaze detection models and the two conditions of the robot’s gaze behavior that are tested within. A total of 1433 silences are detected between .102 and 4.7 seconds with a mean of 2.04 seconds ($SD = 1.33$, $Mdn = 1.73$ s). From Table 7, it becomes clear that the direct gaze detection model ($M = 2.30$ s, $SD = 1.43$, $Mdn = 2.12$ s) had longer silences compared to the extended gaze detection model ($M = 1.82$ s, $SD = 1.20$, $Mdn = 1.66$ s). Looking at the within conditions of the robot’s gaze behavior, the difference between the silence durations was a bit shorter. For the behavior where the robot looked in the middle when speaking, the average silence was 2.09 s ($SD = 1.34$, $Mdn = 1.82$ s), which was similar to the average of the behavior where the robot switched between the participants when speaking ($M = 1.98$ s, $SD = 1.32$, $Mdn = 1.68$ s). Silence durations that are four seconds or longer occurred 194 times in the direct gaze detection model and 98 times in the extended gaze detection model. That means that the fallback silence of four seconds accounted for 29.31% of all silences for the direct gaze detection model, compared to 12.71% in the extended gaze detection model. For the robot’s gaze behavior condition, this was 159 (21.06%) for the static gaze aversion and 133 (19.62%) for the timed gaze aversion. To test for differences in the distribution between the four experimental groups, non-parametric tests were conducted. A comparison of the between-subject gaze models was done with a Mann-Whitney U test. This test showed a significant difference in the gaze detection model on the silence duration, $z = -2.42$, $p = .015$. The within-subject conditions, static gaze aversion and timed gaze aversion, were tested with a Wilcoxon signed-rank test, which showed no significant difference in median silence duration, $z = 0.43$, $p = .667$. Lastly, the interaction effect was tested by comparing difference scores between conditions, static gaze aversion, and timed gaze aversion across gaze detection models using a Mann-Whitney U test. The test revealed no significant interaction, $z = -0.30$, $p = .762$. Indicating that while silence durations differed between the extended and direct gaze models, the effect of the robot’s gaze behavior on silence duration did not depend on the gaze model.

Table 7: Descriptive statistics of silence durations (in seconds) by gaze detection model and robot’s gaze behavior.

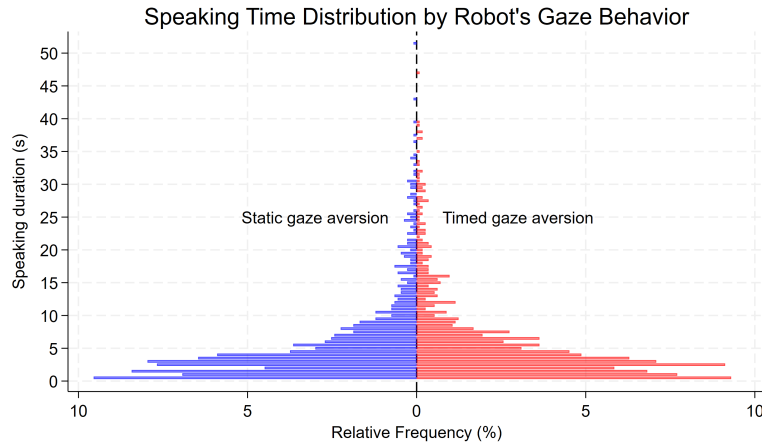
Group	N	Mean	SD	Median	Min	Max
<i>Gaze model</i>						
Extended	771	1.82	1.20	1.66	0.10	4.70
Direct	662	2.30	1.43	2.12	0.10	4.68
<i>Robot’s Gaze Behavior</i>						
Static Gaze Aversion	755	2.09	1.34	1.82	0.10	4.70
Timed Gaze Aversion	678	1.98	1.32	1.68	0.10	4.56
Total	1433	2.04	1.33	1.73	0.10	4.70

The utterance duration of the participants was examined for differences between the models. A linear mixed-effects model was used after the logarithmic transformation of utterance duration. The inspection of the residual and Q-Q plots showed minor deviations from normality. However, the Shapiro-Wilk test indicated a significant deviation from normality of the residuals ($W = .998$, $p = .018$). With the large sample size and no major deviations in the plots, the assumptions for the linear-mixed effects model with logarithmic transformation are considered sufficiently met. This analysis has the coefficient B, which represents the expected difference in the log-transformed speaking duration between groups. Where the negative values indicate shorter utterances and positive values indicate longer utterances. The analysis did not reveal a significant effect of the gaze model, $B = -.16$ log-seconds, $SE = .17$, $z = -.93$, $p = .350$, indicating that speaking durations did not differ significantly between the extended and direct gaze models. Furthermore, there was no significant difference between the conditions, $B = -.09$ log-seconds, $SE = .06$, $z = -1.65$, $p = .099$, suggesting that speaking durations were almost the same between the

robot's gaze behaviors. All utterance durations can be seen in Figure 14. The visualization also shows that both the gaze detection models and the robot's gaze aversions are comparable and that there are no big differences between the experimental conditions. The interaction between gaze model and condition was also not significant, $B = .11$ log-seconds, $SE = .08$, $z = 1.48$, $p = .140$. This indicates that the effect of the gaze model on speaking duration did not differ significantly across the two robot gaze aversions. Figure 14 shows that most utterances are 5 seconds or shorter, indicating that the moments of speech did not take very long.



(a) Speaker time in each gaze detection model



(b) Speaking time in each robot's gaze behavior.

Figure 14: Distributions of speaking durations across the four different conditions for all the participants together.

To gain insight into the opportunities to speak during a conversation for the participants. The utterance per minute of the participants was examined for differences between the models. A linear mixed-effects model was used on the square-root transformed utterances-per-minute variable. This transformation was done to meet the normality assumption, as that was the only one that was not met before the transformation. The Shapiro–Wilk test revealed that the residuals did not significantly deviate from normality, $W = .95$, $p = .075$. The model was used on the square-root transformation of the utterances-per-minute, so the reported regression coefficients (B) represent differences in square-root transformed utterances per minute. The analysis did not reveal a significant effect of the gaze detection model, $B = .16$, $SE = .16$, $z = .95$, $p = .340$, indic-

ating that the utterance per minute did not differ significantly between the extended and direct gaze models. Furthermore, there was no significant difference between the robot’s gaze behavior, $B = .03$, $SE = .10$, $z = .31$, $p = .757$, suggesting that the amount of speech was almost the same between static and timed gaze aversion. The utterances per minute for each experimental group can be seen in Figure 15. The visualization also shows that all conditions are comparable and that there are no big differences between the experimental groups. The interaction between gaze model and condition was also not significant, $B = -.07$, $SE = .14$, $z = -.47$, $p = .638$. This indicates that the effect of the gaze model on the number of utterances per minute did not differ significantly across the two task conditions. Figure 15 shows that participants produced approximately 3 utterances per minute on average across all experimental conditions, and that the amount of speaking moments per minute was similar for each experimental condition.

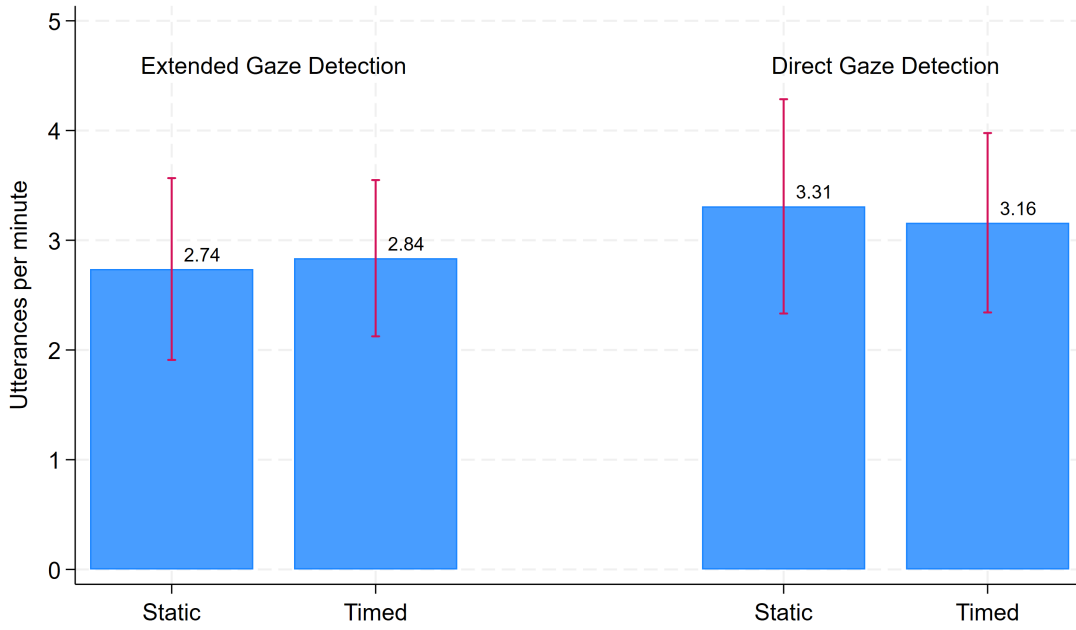


Figure 15: The average utterances per minute with the standard error for each of the experimental groups.

During the conversations, overlap in speech occurred. It could be when the human interrupted the other human, if the robot interrupted the participants, or if the human interrupted the robot. Cut-ins are moments when someone wants to speak, but before the other speaker gives their turn. Before the analysis could happen, some data was excluded. Some points were labeled as cut-ins even when the speaker was labeled the same as the speaker who gave the turn, and the turn from that speaker is the same (like the robot speaks, but got the turn from the robot). A mixed-effects logistic regression was conducted to examine whether the gaze model (extended vs. direct), condition (static gaze aversion vs. timed gaze aversion), and their interaction influenced the likelihood of cut-ins from the robot. First, the assumptions were tested and showed that they were not violated. The coefficient B represents the expected change in the log-odds of a robot interruption occurring. The test showed no significant effect of gaze model, $B = .06$, $SE = .48$, $z = .13$, $p = .895$. Furthermore, the effect of the robot’s gaze behavior on the robot cut-ins was not significant, $B = -.35$, $SE = .26$, $z = -1.38$, $p = .167$. There was a significant interaction between the gaze detection model and the robot’s gaze behavior condition, $B = 1.04$, $SE = .35$, $z = 3.00$, $p = .003$. The coefficient $B = 1.04$ corresponds to an odds ratio of $OR = 2.82$. Meaning that the robot interruptions were 2.82 times greater in the direct gaze model under timed gaze aversion than in the reference condition (extended gaze model under static gaze aversion). This indicates that

the effect of robot’s gaze behavior on the probability that an interruption was initiated by the robot rather than occurring as a human-human overlap differed between the extended and direct gaze models. The direct gaze model in the timed gaze aversion condition was associated with a higher proportion of robot-human interruptions (16.67%) compared to the other experimental conditions (11.27%-11.56%). Furthermore, it was tested if the total number of interruptions differed between the experimental groups. A square-root transformation was used to meet the assumptions for the linear mixed effects model. The Shapiro–Wilk test revealed that the transformation led to a normal distribution, $W = .96$, $p = .122$. The coefficient B for this model represents the estimated difference in the square-root transformed number of interruptions between groups, with positive values indicating more interruptions and negative values indicating fewer interruptions. There was no significant main effect of the gaze detection model, $B = 1.11$, $SE = 1.20$, $z = 0.92$, $p = .357$. The analysis showed no significant effect was found for the robot’s gaze behavior conditions, $B = 0.10$, $SE = 0.92$, $z = 0.11$, $p = .912$, and lastly, no significant interaction between gaze detection model and robot’s gaze behavior was found, $B = -0.82$, $SE = 1.30$, $z = -0.63$, $p = .528$. Comparing the total number of interruptions with the interruption-type analysis, it indicates that the manipulations did not influence how many interruptions happened, but rather the type of interruptions.

Gaze analysis

During the experiments, gaze logs were collected continuously. From the dataset, the following data is excluded: certain duos, extreme pitch angles that exceeded $\pm 20^\circ$, and eye contact durations longer than 20 seconds. The total gaze dataset after cleaning, contains 189,516 observations over the 20 participant duos. The gaze data is large because multiple observations per participant duo were collected throughout the interactions. They are aggregated to the session level before statistical analyses to account for the repeated measures.

Detecting eye contact was crucial for the robot to take the turn. The extended gaze model allowed the robot to detect eye contact within a range from -25° to $+25^\circ$. All the yaw angles from the experiment with participant 33 and participant 34 are visualized in Figure 16. This experiment was conducted when the robot was looking to the left and right when it was speaking (timed gaze aversion). What can be seen is that there are multiple moments of eye contact before the robot looks at the other participant. This happened as the eye contact had to be maintained for a certain period of time before the robot took the turn. An example is at 800 seconds in Figure 16, where it is visible that the left participant had eye contact for a couple of seconds, and then the robot looked at the right participant, detecting eye contact with that person. There are also some yaw angles between $30^\circ \leq |yaw| \leq 60^\circ$, which indicates the region that the participant was not looking at the robot, but at the other participant. According to Figure 16, this happened often during the experiment. The gap between $25^\circ \leq |yaw| \leq 30^\circ$ represents a neutral zone to reduce misclassifications near the boundary between the eye contact and human-directed region.

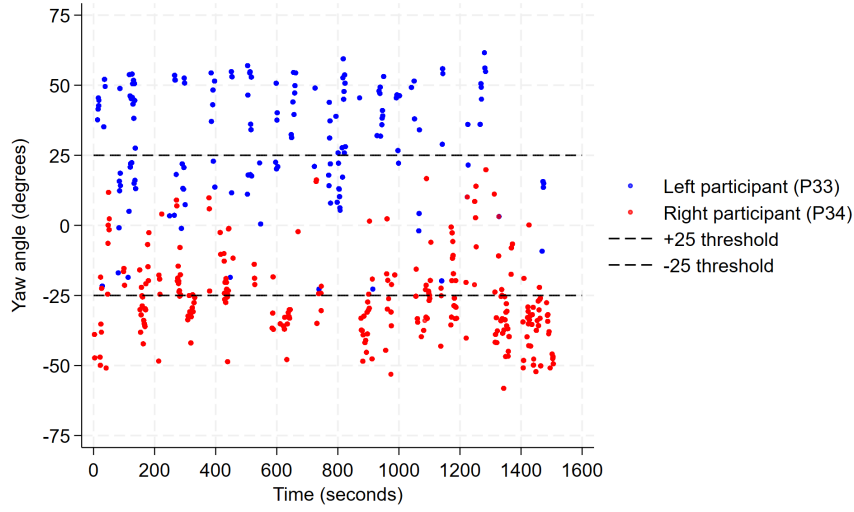


Figure 16: The yaw angles for the extended gaze detection model for participants 33 and 34. Eye contact could be detected between -25 and $+25$ degrees, and this was from the experiment with timed gaze aversion.

For the direct gaze model and static gaze aversion, where the robot kept looking in the middle when it was speaking, is visualized in Figure 17. It becomes clear that there are a lot less moments where the robot detected eye contact, making it harder for the participants to yield the turn to the robot. The yaw angles are often between $30^\circ \leq |yaw| \leq 60^\circ$, so the participants often looked at each other instead of looking at the robot.

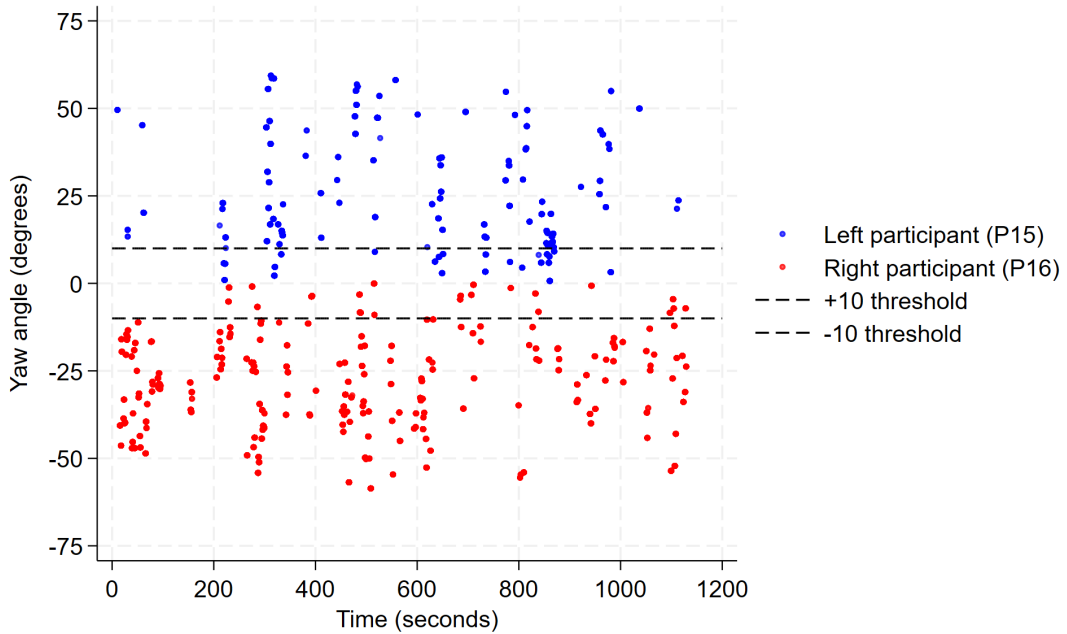
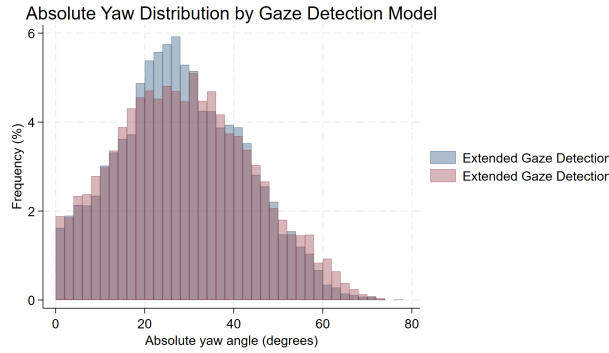


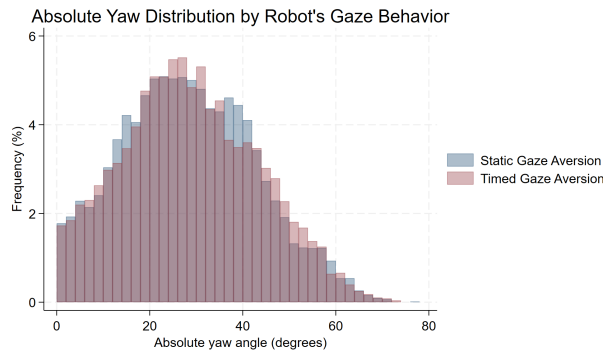
Figure 17: The yaw angles for the direct gaze model for participants 15 and 16. Eye contact could be detected between -10 and $+10$ degrees, and this was from the experiment with static gaze aversion.

Figure 18a shows clearly the distribution of the yaw angles between the two gaze detection models. It has a peak at around 20° . This indicates that in both conditions, the participants had the same gaze behavior. However, it can be seen that in the direct gaze detection condition, the bars are a bit higher at the lower yaw angles, and the extended gaze model has more yaw angles at 20° or higher. Figure 18b shows the yaw angle distribution between the static and timed gaze behavior of the robot. Both follow the same path with high peaks between $20^\circ \leq |yaw| \leq 30^\circ$. It indicates that participants looked a lot at the robot with a yaw angle of 20° in all conditions. The yaw angle distributions of the models are a bit wide towards the higher yaw angles, which can indicate that the participants looked a lot at each other during the experiment.

Looking at the individual gaze measurements, the extended gaze detection model had more eye contact detections with $M = .40$ ($SD = .49$) compared to the direct gaze detection model that had $M = .10$ ($SD = .30$). During the experiments, it was also logged when someone looked at the other participant. This happened more in the extended gaze model ($M = .11$, $SD = .31$) compared to the direct gaze model with $M = .06$ ($SD = .25$). There are only small differences between the conditions, as the moments of eye contact with the robot were quite similar ($M = .26$) and the eye contacts with the other participants are also quite the same, with the condition static gaze aversion $M = .10$ ($SD = .30$) and for the timed gaze aversion $M = .08$ ($SD = .27$). In total, 1837 eye contact moments were detected with an average duration of $M = 3.92$ ($SD = 3.77$). The gaze measurements are continuous and not independent, which is why the variables are aggregated to the session level for further analysis.



(a) Yaw angle distribution of the extended and direct gaze models. Negative angle values were mirrored to positive values, causing left and right gaze directions to overlap



(b) Yaw angle distribution of the two conditions with static and timed gaze aversion of the robot. Negative angle values were mirrored to positive values, causing left and right gaze directions to overlap

Figure 18: Yaw angle distributions between the four conditions.

Table 8 shows that eye contact in the extended gaze model condition was 42% of the observations in that group, which is the highest from all experimental conditions. Also, the mean absolute yaw is almost equal for all groups, indicating no difference in head turn behavior. Furthermore, timed gaze aversion has the highest amount of gaze frames, which was expected as the robot turned its head more often towards the participants, creating more opportunities to detect the faces.

Table 8: Session-level descriptive statistics for gaze measures by gaze model and condition with $n = 20$ per session and mean (standard deviation) for each column. The rates are defined as the proportion of gaze observations.

Variable	Extended Gaze	Direct Gaze	Static Gaze Aversion	Timed Gaze Aversion
Eye contact rate	.42 (.09)	.11 (.07)	.27 (.17)	.26 (.18)
Human-directed gaze rate	.09 (.09)	.06 (.05)	.09 (.10)	.07 (.07)
Mean absolute yaw ($^{\circ}$)	28.03 (3.88)	27.85 (6.51)	27.51 (5.92)	28.36 (4.70)
Mean pitch ($^{\circ}$)	-2.29 (-2.29)	-1.73 (3.76)	-2.11 (4.01)	-1.91 (3.90)
Number of gaze frames	5053 (2195)	4423 (1936)	3524 (1316)	5952 (1985)

The linear mixed-effects model is used to test the difference between the experimental groups. The assumptions for the linear mixed-effects model are visually checked on the histogram and Q=Q plot, and it can be concluded that they were met. For this analysis, the coefficient B represents the expected difference in eye-contact rate, where a value of 0.01 corresponds to a 1 percentage-point difference in eye contact. The analyses showed that there is a significant effect of gaze detection model on the amount of eye contact, $B = -.29$, $SE = .04$, $z = -8.05$, $p < .001$. The participants in the direct gaze detection group showed significantly fewer moments of eye contact. Furthermore, no significant effect was found within the conditions, $B \leq .01$, $SE = .03$, $z = -16$, $p = .875$, nor for the interaction between the gaze model and the conditions, $B = -.022$, $SE = .04$, $z = -.58$, $p = .560$.

The effects of the gaze detection model and the robot's gaze aversion behavior on the number of participants looking at the other person are also tested with a linear mixed-effects model, which met all the assumptions except the normality for the random effect. Some minor deviations from normality were observed, but they were not considered severe enough to affect the analysis. In this model, the coefficient B represents the expected difference in human-directed gaze rate between groups. However, no significant effect was found of the gaze detection model on the number of moments participants looked at the other person, $B = -.05$, $SE = .03$, $z = -1.35$, $p = .178$. Furthermore, no significant effect of the robot's gaze aversion was observed, $B = -.02$, $SE = .02$, $z = -1.11$, $p = .267$. The interaction between the gaze detection model and the robot's gaze aversion behavior was also not significant, $B \leq .01$, $SE = .02$, $z = .09$, $p = .925$. Finally, to check whether the gaze manipulation of the robot influences the gaze behavior of the participants, an analysis was conducted on the mean absolute yaw angle. For this analysis, a linear mixed-effects model was used. First, the assumptions are tested through visual inspections of the histogram and Q-Q plot. This inspection concluded that the assumptions are sufficiently met. The coefficient B is expressed in proportion units of human-directed gaze rate. This revealed no significant difference between the gaze detection models, $B = -.53$, $SE = 2.33$, $z = -.23$, $p = .819$, and no significant difference was found for the robot's gaze behavior, $B = .50$, $SE = 1.24$, $z = .40$, $p = .688$, or for their interaction, $B = .71$, $SE = 1.76$, $z = .40$, $p = .687$.

Subjective factors

There are self-reported subjective measurements collected during the experiment, next to the behavioral objective measurements. These measurements should give a better insight into how the participants felt during the conversation with the robot. The Cronbach's alpha reliability scores

are calculated for the RoSAS and SASSI questions, which should show whether the items of the questionnaires measure the same construct. Reliability scores $> .70$ are seen as acceptable. The RoSAS discomfort dimension showed the lowest reliability score ($\alpha = .5836$), which suggests low internal consistency among the items. It is checked if it is only one item from the dimension that causes the dimension to be low in reliability (aggressive with a low item-rest correlation of .118). However, removing the item that has a low internal consistency does not increase the dimension's reliability much. The RoSAS discomfort dimension is not excluded from analysis because this dimension is validated in the RoSAS questionnaire (Carpinella et al., 2017), and it can still give relevant information about the aspect of the perception of the robot from the participants. However, the results should be interpreted with caution. All the SASSI dimensions had reliability scores $> .70$. Table 9 has an overview of all the reliability scores of all the factors that are measured.

Table 9: Cronbach's alpha reliability scores for the RoSAS and SASSI subscales.

Questionnaire	Dimension	Cronbach's α
RoSAS	Warmth	.77
RoSAS	Competence	.87
RoSAS	Discomfort	.58
SASSI	Accuracy	.76
SASSI	Likeability	.83
SASSI	Cognitive Demand	.73
SASSI	Annoyance	.84
SASSI	Habitability	.84
SASSI	Speed	.87

The assumptions of the linear mixed-effects models were evaluated through visual inspection of histograms, Q-Q plots, and residual-versus-fitted plots. These inspections indicated that the assumptions were met for all dimensions. However, for the RoSAS discomfort and the SASSI habitability dimension, the Shapiro-Wilk tests for the RoSAS dimension resulted in $W = .95$, $p = < .01$, and for the SASSI habitability dimension, it resulted in $W = .94$, $p = < .01$. This indicates that both dimension significant deviate from normality. However, the histogram of the residuals, the Q-Q plots, and the residual-versus-fitted plot showed only small deviations from normality, and the sample size was large for a linear mixed-effects model to be robust against moderate violations of the normality assumption. That is why the assumptions for the linear-mixed effect model were considered sufficiently met.

To get more insights into the differences between the models, a linear mixed-effects model was conducted on the dimensions. Table 10 shows all the coefficients, standard errors, and p-values for all the dimensions. No significant effect is found for the self-reported subjective measurements, except for the SASSI likeability. The linear mixed-effects model has a coefficient B that represents the expected difference in score points between groups. Participants reported higher likeability scores in the condition with the timed gaze aversion, where the robot looked to the left and the right when speaking, compared to the condition with the static gaze aversion, $B = .30$, $SE = .11$, $z = 2.64$, $p = .008$. This means that participants rated the robot, on average, .30 points higher on the SASSI likeability scale when it used timed gaze aversion than when it had static gaze aversion while speaking. However, no significant differences are found on the effect of the gaze detection models, $B = -.08$, $SE = .24$, $z = -.35$, $p = .726$, indicating that participants did not perceive the robot as more or less likable in the direct gaze detection compared to the extended gaze detection. Furthermore, the participants did not perceive the robot as different with the interaction between the gaze detection model and the robot's gaze behavior, $B = -.04$, $SE = .16$, $z = -.26$, $p = .795$. Post-hoc Bonferroni-corrected pairwise comparisons showed that participants rated the robot on average, 0.30 as significantly more likeable in the extended gaze detection model with timed gaze aversion than in the extended gaze detection model with static gaze aversion. No other pairwise

RESULTS

differences reached statistical significance ($p > .05$). Although not statistically significant, a positive trend was observed for the SASSI habitability, suggesting that participants rated the robot somewhat more predictable and easier to interact with in the timed gaze aversion condition than in the static gaze aversion condition, $B = .24$, $SE = .13$, $z = 1.9$, $p = .061$. Three RoSAS and six SASSI subscales were analyzed. To correct for multiple tests, a Bonferroni correction was applied to each questionnaire separately. The RoSAS got an adjusted significance threshold of $\alpha = .0167$, and the SASSI questionnaire got an adjusted significance threshold of $\alpha = .0083$. None of the RoSAS items was significant after the correction. Only the SASSI likeability factor was significant after the Bonferroni correction ($p = .008$).

Table 10: Mixed-effects model results for the RoSAS and SASSI subscales for the gaze detection model, robot's gaze behavior, and the interaction. Showing the coefficient (estimated effect), standard error, test-statistic, and p -value.

Outcome	Gaze Detection Model				Robot's Gaze Behavior				Interaction			
	B	SE	z	p	B	SE	z	p	B	SE	z	p
RoSAS Warmth	.02	.21	.08	.94	.08	.09	.83	.409	-.10	.13	-.78	.436
RoSAS Competence	.08	.25	.30	.767	-.04	.15	-.28	.780	-.23	.21	-1.07	.287
RoSAS Discomfort	.03	.17	.20	.843	-.03	.11	-.30	.761	.21	.16	1.34	.179
SASSI Accuracy	-.09	.18	-.52	.602	.15	.13	1.18	.237	-.06	.18	-.31	.757
SASSI Likeability	-.08	.24	-.35	.726	.30	.11	2.64	.008	-.04	.16	-.26	.795
SASSI Cognitive Demand	.06	.24	.25	.806	-.09	.16	.73	.583	.17	.23	.73	.464
SASSI Annoyance	.27	.28	.97	.334	-.10	.15	-.67	.504	-.13	.21	-.61	.539
SASSI Habitability	-.163	.209	-.78	.438	.244	.130	1.87	.061	-.238	.184	-1.29	.197
SASSI Speed	.19	.25	.74	.460	.08	.14	.55	.585	-.13	.20	-.64	.520

The patterns of the higher likeability scores are consistent with the results of the linear mixed-effects model, which can be seen in Figure 19. The mean score increased from 2.91 in the condition with static gaze aversion to 3.21 in the condition with timed gaze aversion, and was similar in the direct gaze detection model, where the mean increased from 2.83 in static gaze aversion to a mean of 3.08 in timed gaze aversion. The means are similar between the gaze models, which is supported by the non-significant effect from the analyses.

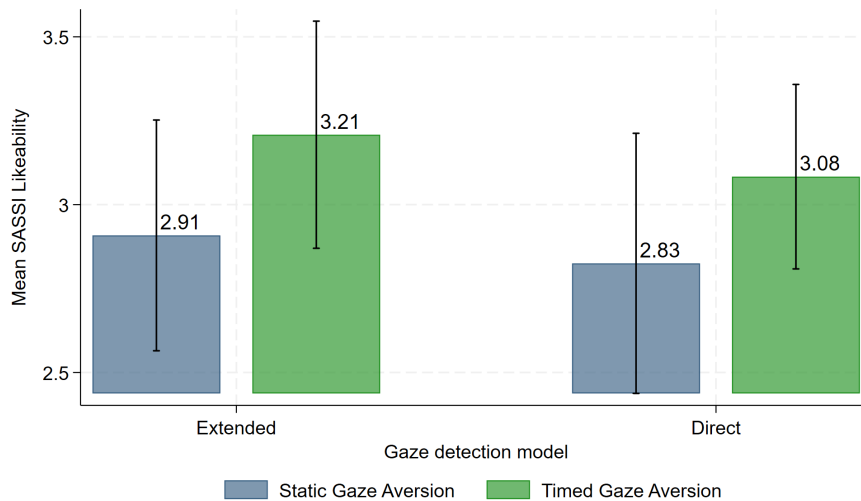


Figure 19: The mean SASSI Likeability factor scores and the 95% confidence intervals for each condition and gaze model.

The perception of the robot during the interaction can be influenced by other factors. The age, gender, previous interaction with robots, and previous interaction with speech interfaces can have an influence on the participant's judgment. A moderator was added to the linear mixed-effects model to analyse if these factors have an effect on the SASSI likeability factor, as this was the only factor to be significant. Age, previous robot and speech interface experience did not have a significant effect on the likeability scores. The analysis revealed only one significant main effect after the Bonferroni correction, which was the interaction between gender and the robot's gaze behavior after the Bonferroni correction, $B = .40$, $SE = .15$, $z = 2.66$, $p = .008$, indicating that female participants reported approximately 0.40 points higher on the likeability scale compared to male participants. Moreover, further analysis of the role of gender on the participants' perception of the robot is out of scope for this study.

At the end of the questionnaire, the participants could give additional comments on the experiment. Some participants mentioned that the robot was socially engaging, but others mentioned that the robot was too limited in its speech capabilities. It felt that the robot did not accurately understand them. Furthermore, the short acknowledgments were sometimes disruptive and interrupted the conversation. One participant said: "It felt like Misty is not really able to take relative answers or a combination or middle way between answers. It had to be either A or B, but could not really feel like a combination of things.". This shows that participants are often irritated by the dialog quality, which could have influenced their perception of the robot.

Discussion

This study aimed to get better insights into the role of eye contact in a multi-party human-robot interaction. A model was developed that enabled the robot to improve the conversational turn-taking with the use of eye contact and silence cues to enhance the conversational fluency and the perceived interaction quality. Next to this, the robot used different gaze behaviors during its own speech to influence the user's perceptions. To answer these questions, an eye contact detection model from Johnson and Cuijpers (2013) was implemented into a turn-taking state machine. Furthermore, two different gaze behaviors for the robot are developed to investigate the differences in the perceived quality of the interaction. The results will be discussed, the study's implications explored, and recommendations for future research provided.

Effects of Gaze on Turn-Taking

It was hypothesized that the extended eye contact region would better reflect the natural human gaze behavior compared to a strict gaze region. Analyzing the data showed that this hypothesis was supported; it also showed that participants almost did not look directly at the robot, but mostly between the robot and the other participant. The extended gaze detection model resulted in more successful triggers for the robot to take the turn when comparing it to the direct gaze detection model. This indicates that the robot detected more moments to take the conversational floor. The extended gaze detection model might have increased the false positive eye contact detection; it also increased the amount of successfully detected turn-taking opportunities, which allowed the robot to respond more frequently to conversational cues. However, this increase in successful triggers was not reflected in participants' subjective evaluations, as no significant effects of the gaze detection model were found on the RoSAS or SASSI dimensions. The eye contact rate for the extended gaze detection model was 44%, showing that from all the eye contact moments during the experiments, almost half of it was detected by the extended gaze detection model, and only 12% by the direct gaze detection model.

This finding is consistent with the earlier research from Johnson and Cuijpers (2013) that showed that eye contact can not solely be determined by head orientation. Humans in a multi-party setting have to divide their attention among several people. This was also seen in this study, that participants shifted their gaze between the robot and the other participant often. A possible explanation for the lower eye contact detection is that the direct gaze model is that the participants often maintained head orientations that helped them to simultaneously monitor both interaction partners. The extended boundaries of the extended gaze model appear to better accommodate this natural gaze behavior. That is why a direct gaze threshold might be too restrictive, especially in a setting with multiple participants, and a more flexible interpretation of eye contact should be used by robots. This means that the extended region of eye contact better fits the human gaze behavior. There was also a significant interaction effect between the gaze model and the robot's gaze behavior when taking the turn. This suggests that the effectiveness of the gaze detection model also depended on whether the robot would look away when speaking or when it looked at the participants when speaking.

Even though the extended gaze model had a significant effect, the overall rate of successful triggers for the robot to take the turn was still low. Only 28.45% of the robot's turn-takes were opportunities that were classified as successful. Meaning that the combination of silence and eye contact is still insufficient to predict when the robot should take the turn. It was shown that the human-human turn transitions happen fast, with around 19% of the human-to-human turn takes happening within 500 ms with a median of 1.53 seconds, which aligns with the research of Skantze (2021) that showed that humans predict turn endings before they occur. The distribution turn-take duration showed that human-robot turn-takes were slower by approximately 4 seconds based on the

medians (5.47 s vs. 1.53 s). The robot has to do additional processing to detect the cues, interpret the state, and process its speech. However, no significant effect was found of the gaze detection model on the turn-take durations or the amount of turn distributions. This may reflect general differences between human-human and human-robot interaction rather than the manipulation of this study. Detecting eye contact is one part of the conversational coordination, but the robot should also respond at the right time and in the right way that fits the conversation. Therefore, improving detection of the cues alone does not automatically lead to smoother turn-taking. This finding supports previous research suggesting that turn-taking relies on multiple simultaneous cues rather than a single signal Bohus and Horvitz (2011). What is also interesting is that almost half of the turn yields happened between the participants, and fewer turn transitions included the robot. Approximately 47% of turn transitions occurred between the two participants, whereas only 26.65% involved transitions from humans to the robot and 26.86% involved transitions from the robot to humans. This shows that the participants primarily interacted with each other and excluded the robot often. This reduced the influence of the robot's behavior on the interaction. According to Mutlu et al. (2009), gaze cues play an important role in establishing and maintaining participant roles during multi-party conversations. In the present study, the robot's gaze was used as a cue for turn-taking and for indicating attention towards participants. The robot worked as a moderator, which could have made the participants feel like collaborating directly with each other. This might have resulted in participants perceiving each other as the primary addressees, leading to a high proportion of human-human turn transitions. This is supported by the turn-taking duration results, which showed that the robot generally entered the conversation later than humans.

Effects of Gaze Detection Thresholds

It was expected that the direct gaze model would reduce the inappropriate turn-taking moments, as the robot would only respond when the participants clearly looked at it. However, this claim was not clearly supported, and it lowered the proportion of successful turn-taking opportunities. The success rate ranged from only 9.24% to 14.96% for the direct gaze detection model, compared to 30.53% to 49.42% for the extended gaze detection model. This suggests that many conversational cues were missed when eye contact was defined too narrowly.

The robot missed moments when the participants tried to include the robot. As was visualized in Figure 17, it became clear that yaw angles were close to the strict threshold, but just outside the threshold. These are not detected by the strict gaze detection model, but would have been detected as eye contact when the extended gaze detection model was used. Given that the distribution of yaw angles did not differ between the extended and direct gaze detection model, it shows that participants did not really adapt their looking behavior, which means that the robot most likely missed potential turn-yields. This is also supported by the silence durations per group. The silence duration was longer when people faced the robot with the strict gaze detection model. The reason for this is that the strict gaze model did not detect the turn yield from the participant to the robot, which caused the robot to do nothing for four seconds before it automatically took the turn. It suggests that the stricter gaze boundaries frequently prevented the robot from detecting earlier turn-yield cues. As a result, the robot often relied on the fallback silence threshold, 194 times (29.31%) of all silences in the direct gaze detection model, rather than responding immediately to participants' gaze behavior. The extended gaze model had less amounts of silence durations that were four seconds or longer silence 98 times (12.71%), compared to the direct gaze model. The extended gaze detection model resulted in significantly shorter silence durations compared to the direct gaze detection model, improving a behavioral indicator of conversational fluency.

From the distribution of the yaw angles, it became clear that the participants did not appear to change their gaze behavior when they noticed the robot would not take the turn after the participants were quiet. The distribution shows a low of yaw angles around 20°, and the scatter plot

shows a consistent gaze behavior from the participants that did not change to make more direct eye contact. This suggests that participants maintained their natural gaze patterns regardless of how the robot detected or produced gaze cues. Furthermore, no significant effect of the gaze detection model on the behavior of the participants was found during the interaction. The distribution of speech duration showed that most utterances were 5 seconds or shorter, which may indicate that the participants did not actively change their behavior in response to the turn-taking decisions of the robot. They would continue the conversation normally, or they would be quiet till the moment the robot would take the floor automatically. In a multi-party setting, the participants may rely more on the behavior of the other humans in the conversation to decide whether to continue speaking. This would lead to less influence of the robot's behavior on the conversational quality. The direct gaze detection model was developed to structure the interaction by limiting the robot's response to direct attention from the participants. However, this was not seen in their speaking behavior. The participants appeared to give their opinion in the same utterance duration across the different experimental groups. No significant difference was found for how long participants spoke between the different gaze detection models and the robot's gaze behavior. As earlier shown, the gaze detection models significantly change the duration of the silence and the amount of successful triggers, but it does not affect the speaking duration.

The amount of cut-ins (interruptions) was also analyzed to see whether it would differ between the experimental groups. No effect was found for gaze direction alone, so the gaze detection model did not significantly influence the turn interruptions. However, a significant interaction between the gaze detection model and the robot's gaze behavior was observed. The number of interruptions was affected by the type of gaze behavior from the robot, which depended on the gaze detection model. Specifically, the combination of the direct gaze detection model and timed gaze aversion was associated with a higher likelihood of cut-ins compared to the other experimental conditions. This could be a result of the fact that, in timed gaze aversion, the robot looked around when speaking, given the perception of turn availability. If the timed gaze behavior is in combination with the direct gaze detection model, it would not detect the turn transitions as easily as the other gaze detection model. Other explanations are also possible. The timed gaze aversion could be interpreted as a social signal rather than a turn-taking signal, which could have led to different responses from the participants. However, it could also be a result of delayed cue detections and inaccuracies in timings, rather than gaze behavior alone.

Overall, the results show that robots should take into account eye movements and not solely base the eye contact on head orientation alone. Therefore, the findings suggest that eye contact detection in conversational robots should account for natural variation in the gaze behavior of humans. The results show that the extended gaze detection is necessary for fluent conversation, as it results in more successful turn-take triggers and reduces the silence durations during the conversation.

Effect of robot gaze behavior

The third hypothesis was that the timed gaze behaviors of the robot during its own speech would improve the quality of the conversation. From the results, it became clear that timed gaze aversion had more successful triggers compared to static gaze aversion. This shows that the robots' own gaze behavior affects how participants produce detectable turn cues. The subjective experiences show almost no effect for the RoSAS and SASSI dimensions. However, a significant effect was found for the SASSI likeability factor. The participants rated the robot more likeable in the timed gaze aversion condition, indicating that when the robot changes its gaze when it is speaking makes it more socially engaging. It could be explained by the fact that participants noticed the head movements of the robot and interpreted them as more interactive. Participants might base their judgments on the robot's speech and responsiveness rather than on gaze behavior alone, as it was not strong enough to change their overall perception of the interaction.

The combination of the extended gaze model together with timed gaze aversion showed the highest successful trigger rate. This suggests that using a flexible eye-contact detection model that includes eye movements with a timed gaze behavior from the robot itself creates the most effective turn-taking model in terms of trigger success. This indicates that gaze detection and gaze production should not be seen as separate events. It not only needs to detect the gaze from the participants, but it should also produce an accurate gaze for itself to structure the interaction. In the timed gaze aversion, the robot's gaze shifts may have created more opportunities for participants to respond to the robot or yield the turn back to it. However, no significant difference in the number of utterances per minute was detected, meaning that during the conversation, the number of utterances was the same for the timed gaze aversion condition as in the static gaze aversion condition. The combination of the extended gaze model together with timed gaze aversion improves the cue detection, but it does not necessarily enhance the full experience during the conversation.

There can be several explanations for the non-significant results of the subjective experience. It was clear that the participants got irritated that the robot had limited speech capabilities, even when the instructions were clear that the robot was limited and could only provide basic information and repeat the questions. A reason could be that the speech quality and inaccurate understanding of the conversation had a greater influence on the impressions of the participants than the subtle changes in the gaze behavior of the robot. Also, it could be that participants prioritized the task completion over the evaluation. When generating ideas and discussing, the gaze behavior of the robot could have been less noticeable. This interpretation is consistent with the finding that participants primarily interacted with each other and that the gaze behavior manipulation was less clear. It also showed that it did not significantly affect the perceived interaction quality and the conversational fluency. Suggesting that the timed gaze behavior of the robot does not change the subjective smoothness experience of the conversation, and it does not affect the overall interaction experience. An explanation can be that the participants did not consciously notice the differences between the gaze-aversion behaviors or did not consider them important when evaluating the interaction. However, the use of timed gaze behavior did significantly improve the likeability of the robot. Participants rated the robot as more likeable when the timed gaze aversion behavior was used.

Most of the findings are in contrast with the work by Andrist et al. (2014), who suggested that the gaze behavior of the robot would affect the interaction quality. It could be that the timings from that study will only work correctly in a human-robot interaction, and that a multi-party setting needs different timings for the gaze behavior of the robot when it takes the turn. In the multi-party setting, gaze helps to regulate turn-taking and indicate who is addressed. As a result, gaze timings that are effective in a one-on-one interaction may not have the same effect in a group setting. Participants may therefore have relied less on the robot's gaze cues when evaluating the interaction. However, this may be less likely than the influence of the robot's speech limitations. As several participants commented on this, whereas no participants mentioned the gaze aversion behavior. Overall, timed gaze aversions in a robot should be implemented to improve the technical effectiveness of turn-taking and increase the robot's perceived likeability.

Methodological Considerations

During the experiment, several issues happened that might have influenced the study's outcomes. First, a programming issue affected the turn-taking duration by the robot. Originally, it was designed to detect both the gaze and silence cues and take the turn after 1 second. However, an error in the program caused this turn-take behavior to be delayed by 0.5 seconds. This resulted in the robot's turn-taking duration of 1.5 seconds. Affecting the responsiveness of the robot to the participants. The successful triggers worked how they should have worked; the 0.5-second delay

caused it to change the intended behavior of the robot. Another issue that occurred during the experiment was that sometimes the face detection model had difficulty detecting the faces. This resulted in fewer detection points and increased the missed potential eye contact moments. This negatively affects the interaction as the turn yields from participants to the robot are missed, making the robot less responsive. The last issue is the limited speech capabilities of the robot. The participants became irritated when the robot could not answer their questions or when the robot asked for one answer instead of a combination. This made them feel that the robot did not actually understand them and influenced the naturalness of the interaction.

Recommendations

This study focused on a turn-take model that implemented a face-detection model and two gaze behaviors for the robot. This is a base for future work, which could be improved to make the conversations in a multi-party setting with a robot.

Firstly, future studies can investigate turn-taking models that not only consider gaze and silence cues, but also take into account contextual factors and group composition. The robot could benefit from adaptive systems that adjust the detection of eye contact based on the interaction flow and characteristics of the participants. This would improve the robot's ability to detect appropriate moments to take the floor in different social settings. Information about the last speaker, the duration of the utterance, and the state of the conversation is useful for robots to create better turn-taking moments. This would make the robot more robust against variations and reduce the incorrect turn-takes.

Secondly, an important section that should be focused on in future studies is to improve the responsiveness of the robot turn-taking system. Although eye contact and silence cues could be detected, delays between cue detection and the robot's response may have reduced the effectiveness of the implemented turn-taking mechanism. Removing this delay and accurately programming the robot to take the turn as soon as both one second of silence and eye contact have been detected. This would allow for more accurate evaluation of the contribution of gaze and silence cues to turn-taking performance.

Furthermore, future studies should implement a large language model instead of using pre-defined scripts like the holiday planner and building a dream house. If a large language model is used, the robot would better understand the conversation that is held by the participants and could give answers to all their questions. Given the impression that the robot accurately understands what they are talking about. This would also lower the influence of the limited ability in speech on the participants' perceptions.

Future studies can also look into the different timings of the gaze behavior for the robot itself when it has the turn. The results have shown that the likeability of the robot will increase when using timed gaze behavior, but it would also be good to understand which timing is the most appropriate. Getting a better understanding if different timings would have different effects on the conversation quality will be a useful insight into conversational robots.

Finally, future studies could implement a better and more accurate face detection model. The face detection model in this study missed important eye contact moments, which could have affected the responsiveness of the robot. Implementing a detection model that continuously detects the face and combining it with the linear relationship of head orientation and gaze from Johnson and Cuijpers (2013) would improve the detection moments of eye contact. This would help the robot to reduce the missed moments of eye contact and improve its own behavior and turn-taking.

Implications

The findings of this study have implications for both the design of conversational robots and the broader understanding of human-robot interaction. The technical perspective shows that detecting eye contact should not be based on only direct head orientation. Conversational robots should have an extended eye contact region that accounts for the relationship between gaze direction and head orientation when detecting turn-taking opportunities in a multi-party setting. The reason for this is that humans do not perfectly align their gaze with the robot's views during conversations, so the robot has to account for this. The societal relevance of this is that robots are used more and more in collaborative workspaces, education, and other sectors that have multiple participants who work together, where effective turn-taking is crucial for natural and efficient communication between humans and robots.

Secondly, the study highlights the importance of robot gaze behavior during its own speech. The results indicate that timed gaze behavior that resembles human gaze aversion patterns can improve the likeability of the robot. However, the overall interaction quality did not increase solely by gaze behavior, indicating that gaze alone is insufficient to create natural conversations. Future systems should use timed gaze behavior for the robot that is combined with better conversational understanding and speech processing.

The study also contributes to the scientific perspective. It expands the literature by showing that the findings from a single human-robot interaction do not immediately generalize to a multi-party setting. Gaze functions as a cue for turn-taking and as a social signal that influences the perception of the robot. Moreover, the findings show that the ability to detect cues better does not immediately improve the perceived interaction quality. It highlights that technical and psychological perspectives in human-robot interaction should be combined and that they together contribute to the development of conversational robots that are not only effective, but also socially acceptable in a multi-party setting.

Conclusion

This study examined the role of eye contact on turn-taking in a multi-party conversation consisting of a robot and two humans. More specifically, investigating the effect of implementing a gaze detection framework that accounts for uncertainty in eye movements. In addition, it examines whether a robot that uses timed head orientation towards the participants while speaking is perceived more positively compared to using static gaze aversion. These manipulations are tested for their effects on interaction quality and fluency and participants' perceptions of the robot's likeability, comfort, accuracy, speed, annoyance, cognitive demand, warmth, competence, habitability, and speed.

To address the research questions, an extended and direct gaze detection model was created. In addition, the robot maintained a static head orientation towards the middle while speaking, or it would look back and forth between the participants (timed gaze aversion). The results showed that the extended gaze detection model detected significantly more eye contact compared to the direct gaze detection model. This indicates that the detection of eye contact in a multi-party setting may require an extended interpretation of gaze behavior, including the uncertainty for eye movements, compared to solely relying on direct alignment with the robot. Furthermore, the results suggest that the timing of the robot's gaze influences how humans perceive it. Even though no significant effects were found on conversational measures, such as speech duration and the duration needed for turn-takes, participants evaluated the robot as more likeable when it resembled more natural human behavior. It highlights how important it is to implement appropriate nonverbal behavior in robots. The combination of the extended gaze model with the timed gaze behavior of the robot resulted in the highest rate of successful turn-taking triggers, suggesting that future research should further investigate how this interaction works best in a conversational robot.

The study contributes to the field of human-robot interaction by providing insights into the role of gaze in a multi-party human-robot conversation. The results showed that extended gaze detection and the implementation of a timed gaze behavior into the robot increased the likeability of the robot and improved indicators that are associated with the fluency of the conversation, such as successful turn-taking triggers and reduced silence durations. However, these improvements did not affect the overall interaction quality. This highlights that implementing human-like gaze in conversational robots is still difficult and emphasizes the need to integrate both the detection of human gaze and the production of robot gaze behavior more accurately in a multi-party interaction.

References

- Admoni, H., & Scassellati, B. (2017). Social eye gaze in human-robot interaction: A review. *Journal of Human-Robot Interaction*, 6(1), 25–63.
- Andrist, S., Mutlu, B., & Gleicher, M. (2013). Conversational gaze aversion for virtual agents. *International workshop on intelligent virtual agents*, 249–262.
- Andrist, S., Tan, X. Z., Gleicher, M., & Mutlu, B. (2014). Conversational gaze aversion for human-like robots. *Proceedings of the 2014 ACM/IEEE international conference on Human-robot interaction*, 25–32.
- Awang, S., Hassan, W., Abdullah, N., & Zawawi, M. Z. M. (2022). Use of fillers as a communication strategy in oral interactions. *International Journal of Academic Research in Business and Social Sciences*, 12(7), 1024–1037.
- Barthel, M., Meyer, A. S., & Levinson, S. C. (2017). Next speakers plan their turn early and speak after turn-final “go-signals”. *Frontiers in psychology*, 8, 393.
- Bartneck, C., Kulić, D., Croft, E., & Zoghbi, S. (2009). Measurement instruments for the anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety of robots. *International journal of social robotics*, 1(1), 71–81.
- Blik, A. (2020). *Backchanneling in human-robot interaction* [Doctoral dissertation].
- Bohus, D., & Horvitz, E. (2010). Facilitating multiparty dialog with gaze, gesture, and speech. *International Conference on Multimodal Interfaces and the Workshop on Machine Learning for Multimodal Interaction*, 1–8.
- Bohus, D., & Horvitz, E. (2011). Decisions about turns in multiparty conversation: From perception to action. *Proceedings of the 13th international conference on multimodal interfaces*, 153–160.
- Butter, J. (2025, June). The effect of timed gaze shifts on engagement and turn-taking in multiparty human-robot interaction [Bachelor End Project, Eindhoven University of Technology, Department of Industrial Engineering & Innovation Sciences].
- Carpinella, C. M., Perez, M. A., Wyman, A. B., & Stroessner, S. J. (2017). The robotic social attributes scale (rosas): Development and validation. *Proceedings of the 2017 ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, 254–262. <https://doi.org/10.1145/2909824.3020208>
- Cuijpers, R. H., & Van Der Pol, D. (2013). Region of eye contact of humanoid nao robot is similar to that of a human. *International Conference on Social Robotics*, 280–289.
- Dzikovska, M. O., Moore, J. D., Steinhauser, N., & Campbell, G. (2011). Exploring user satisfaction in a tutorial dialogue system. *Proceedings of the SIGDIAL 2011 Conference*, 162–172.
- Faul, F., Erdfelder, E., Lang, A.-G., & Buchner, A. (2007). G* power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior research methods*, 39(2), 175–191.
- Hara, K., Inoue, K., Takanashi, K., & Kawahara, T. (2019). Turn-taking prediction based on detection of transition relevance place. *Interspeech*, 4170–4174.
- Heinz, B. (2003). Backchannel responses as strategic responses in bilingual speakers’ conversations. *Journal of pragmatics*, 35(7), 1113–1142.
- Hessels, R. S. (2020). How does gaze to faces support face-to-face interaction? a review and perspective. *Psychonomic bulletin & review*, 27(5), 856–881.
- Hietanen, J. K. (1999). Does your gaze direction and head orientation shift my visual attention? *Neuroreport*, 10(16), 3443–3447.
- Hone, K. S., & Graham, R. (2000). Towards a tool for the subjective assessment of speech system interfaces (sassi). *Natural Language Engineering*, 6(3-4), 287–303.
- Hoque, M. M., Onuki, T., Kobayashi, Y., & Kuno, Y. (2011). Controlling human attention through robot’s gaze behaviors. *2011 4th International Conference on Human System Interactions, HSI 2011*, 195–202.

- Johnson, D. O., & Cuijpers, R. H. (2013). Predicting gaze direction from head pose yaw and pitch. *Human-Technology Interaction, Eindhoven University of Technology*.
- Jokinen, K. (2009). Gaze and gesture activity in communication. *International Conference on Universal Access in Human-Computer Interaction*, 537–546.
- Jokinen, K. (2010). Non-verbal signals for turn-taking and feedback. *LREC*.
- Jokinen, K., Furukawa, H., Nishida, M., & Yamamoto, S. (2013). Gaze and turn-taking behavior in casual conversational interactions. *ACM Transactions on Interactive Intelligent Systems (TiiS)*, 3(2), 1–30.
- Knekta, E., Runyon, C., & Eddy, S. (2019). One size doesn't fit all: Using factor analysis to gather validity evidence when using surveys in your research. *CBE—Life Sciences Education*, 18(1), rm1.
- Köhler, C. (2025). *Evaluating the timing of eye-contact and gestures in turn-taking behavior for a human-robot interaction* [Department of Innovation Sciences].
- Koller, M., Weiss, A., Hirschmanner, M., & Vincze, M. (2023). Robotic gaze and human views: A systematic exploration of robotic gaze aversion and its effects on human behaviors and attitudes. *Frontiers in Robotics and AI*, 10, 1062714.
- Misty Robotics. (n.d.). Misty ii — misty robotics. <https://www.mistyrobotics.com/misty-ii>
- Muckler, F. A., & Seven, S. A. (1992). Selecting performance measures: "objective" versus "subjective" measurement. *Human factors*, 34(4), 441–455.
- Mutlu, B., Kanda, T., Forlizzi, J., Hodgins, J., & Ishiguro, H. (2012). Conversational gaze mechanisms for humanlike robots. *ACM Transactions on Interactive Intelligent Systems (TiiS)*, 1(2), 1–33.
- Mutlu, B., Shiwa, T., Kanda, T., Ishiguro, H., & Hagita, N. (2009). Footing in human-robot conversations: How robots might shape participant roles using gaze cues. *Proceedings of the 4th ACM/IEEE international conference on Human robot interaction*, 61–68.
- Nordmark, L. (2025). Uncovering gaze aversion: How cognitive load and arousal influence visual disengagement.
- Palinko, O., Rea, F., Sandini, G., & Sciutti, A. (2016). Robot reading human gaze: Why eye tracking is better than head tracking for human-robot collaboration. *2016 IEEE/RSJ international conference on intelligent robots and systems (IROS)*, 5048–5054.
- Parreira, M. T., Gillet, S., Vázquez, M., & Leite, I. (2022). Design implications for effective robot gaze behaviors in multiparty interactions. *2022 17th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, 976–980.
- Patamia, R. A., Dinh, H. P. T., Liu, M., & Cosgun, A. (2025). Turn-taking modelling in conversational systems: A review of recent advances.
- Patel, S. D., & Rollins, P. R. (2025). Leveraging mutual gaze to facilitate social attention in autistic children. *Journal of Autism and Developmental Disorders*, 1–12.
- Pinto, M. J., & Belpaeme, T. (2024). Predictive turn-taking: Leveraging language models to anticipate turn transitions in human-robot dialogue. *2024 33rd IEEE International Conference on Robot and Human Interactive Communication (ROMAN)*, 1733–1738.
- Rosén, J., Lindblom, J., Lamb, M., & Billing, E. (2024). Previous experience matters: An in-person investigation of expectations in human-robot interaction. *International Journal of Social Robotics*, 16(3), 447–460.
- Sacks, H., Schegloff, E. A., & Jefferson, G. (1974). A simplest systematics for the organization of turn-taking for conversation. *language*, 50(4), 696–735.
- Saunderson, S., & Nejat, G. (2019). How robots influence humans: A survey of nonverbal communication in social human-robot interaction. *International Journal of Social Robotics*, 11(4), 575–608.
- Shokirjonovna, T. F., & Muxtarovna, S. N. (2025). Hesitation and turn-taking in spoken communication. *YANGI O 'ZBEKISTON, YANGI TADQIQOTLAR JURNALI*, 3(7), 392–394.
- Skantze, G. (2021). Turn-taking in conversational systems and human-robot interaction: A review. *Computer Speech & Language*, 67, 101178.

- Skantze, G., & Irfan, B. (2025). Applying general turn-taking models to conversational human-robot interaction. *2025 20th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, 859–868.
- Skantze, G., Johansson, M., & Beskow, J. (2015). Exploring turn-taking cues in multi-party human-robot discussions about objects. *Proceedings of the 2015 ACM on international conference on multimodal interaction*, 67–74.
- Spiegels, M. G. H. H. (2015). *Evaluating multiparty gaze-based turn-taking behavior for a robot in situated dialog* [Doctoral dissertation, University of Technology Eindhoven].
- Stanton, C. J., & Stevens, C. J. (2017). Don't stare at me: The impact of a humanoid robot's gaze upon trust during a cooperative human-robot visual task. *International Journal of Social Robotics*, 9(5), 745–753.
- Van Rossum, A. (2025). *The impact of backchannel type and timing on perceived likability in human-robot interaction* [Bachelor thesis]. Eindhoven University of Technology [Department of Innovation Sciences].
- Weldon, C. F., Gillet, S., Cumbal, R., & Leite, I. (2021). Exploring non-verbal gaze behavior in groups mediated by an adaptive robot. *Companion of the 2021 ACM/IEEE International Conference on Human-Robot Interaction*, 357–361.
- Xu, T., Zhang, H., & Yu, C. (2016). See you see me: The role of eye contact in multimodal human-robot interaction. *ACM Transactions on Interactive Intelligent Systems (TiiS)*, 6(1), 1–22.

APPENDIX A: Task Dialog Scripts

D.1 Holiday Planner Script

```
1 import random
2 import msvcrt
3 CONTINENT = [ ' ' ]
4 CITY = [ ' ' ]
5 PERIOD = [ ' ' ]
6 DURATION = [ ' ' ]
7 HOLIDAYTYPE = [ ' ' ]
8 THINGSTODO = [ ' ' ]
9 MOBILITY = [ ' ' ]
10 BUDGET = [ ' ' ]
11 ACCOMODATION = [ ' ' ]
12 OPT = [ ' ' ]
13 CURRENTNAME = [0]
14 OTHERNAME = [0]
15 RIGHTWRONG = [0]
16 max_options = 9
17
18 introduction = ["For this conversation the main goal is to figure out what a
19     holiday should be like if you "+
20     "have to travel and spend the entire vacation together. "+
21     "I am going to ask you some questions about what your ideal holiday
22     is. "+
23     "Since the two of you are going on a hypothetical holiday together,
24     "+
25     "ask for each others opinion. "+
26     "Are you ready to begin?"]
27
28 #1 CONTINENT
29 #2 CITY
30 #3 PERIOD
31 #4 DURATION
32 #5 HOLIDAYTYPE
33 #6 THINGSTODO
34 #7 ACCOMODATION
35 #8 MOBILITY
36 #9 BUDGET
37
38 ending = [[
39     "Well thats about it. With all the information combined, you have arranged
40     yourselves a holiday for "
41 ] + DURATION + [
42     " that will bring you to the continent of "
43 ] + CONTINENT + [
44     " in the city "
45 ] + CITY + [
46     " , during the "
47 ] + PERIOD + [
48     " . Once arrived you will have a typical "
49 ] + HOLIDAYTYPE + [
50     " vacation primarily going to "
51 ] + THINGSTODO + [
52     " . You will stay in a comfortable "
53 ] + ACCOMODATION + [
54     " . To explore your surroundings you will mainly go by "
55 ] + MOBILITY + [
56     " , which means the estimated budget will be "
57 ] + BUDGET + [
58     " . Thanks for having participated in our dialogue , the experiment will now
59     continue to the next phase."
60 ]]
```

```

57 holidaytype = [
58     "Active",
59     "Relaxing",
60     "Partying"
61 ]
62
63 thingstodo = {
64     holidaytype[0]: [
65         "visiting famous architectural buildings",
66         "seeking adventure and entertainment"
67     ],
68     holidaytype[1]: [
69         "exploring green areas of local nature",
70         "simply enjoying good sunny weather"
71     ],
72     holidaytype[2]: [
73         "simply enjoying good sunny weather",
74         "visiting popular clubs and enjoying nightlife"
75     ]
76 }
77
78 mobility = [
79     "Public transfer",
80     "Hitchhiking",
81     "Car",
82     "Foot",
83     "Bike"
84 ]
85
86 budget = [
87     "less than 1000 euros",
88     "up to 1500 euros",
89     "up to 2000 euros",
90     "more than 2000 euros"
91 ]
92
93 accomodation = [
94     "Hostel",
95     "Camping",
96     "Bed and Breakfast",
97     "Hotel",
98     "Holiday resort",
99     "Local residence"
100 ]
101
102 continent = ["Asia",
103              "Africa",
104              "United States",
105              "Central South America",
106              "Europe",
107              "Australia"]
108
109 city = {continent[0]: ["Singapore", "Hong Kong", "Bali", "Tokyo", "Maldives"],
110         continent[1]: ["Cape Town", "Cairo", "Marrakech", "Tanzania", "Seychelles"],
111         continent[2]: ["San Francisco", "New York", "Las Vegas", "New Orleans", "Miami"],
112         continent[3]: ["Buenos Aires", "Rio de Janeiro", "Argentina", "Santiago", "Costa Rica"],
113         continent[4]: ["Berlin", "Paris", "Barcelona", "Rome", "Copenhagen"],
114         continent[5]: ["Darwin", "Brisbane", "Sydney", "Melbourne", "Perth"]}
115
116 travelperiod = ["Summer",
117                 "Autumn",
118                 "Winter",
119                 "Spring"]
120

```

```

121
122 tripduration = ["a few days",
123                 "a few weeks",
124                 "a few months"]
125
126 def build_holiday_question(topicID=0, alt=0):
127     global option
128
129     if alt == 0:
130         if topicID == 1:
131             option = [None] + continent
132         elif topicID == 2:
133             chosen_continent = CONTINENT[0] if CONTINENT[0] else continent[0]
134             option = [None] + city[chosen_continent]
135         elif topicID == 3:
136             option = [None] + travelperiod
137         elif topicID == 4:
138             option = [None] + tripduration
139         elif topicID == 5:
140             option = [None] + holidaytype
141         elif topicID == 6:
142             chosen_type = HOLIDAYTYPE[0] if HOLIDAYTYPE[0] else holidaytype[0]
143             option = [None] + thingstodo[chosen_type]
144         elif topicID == 7:
145             option = [None] + accomodation
146         elif topicID == 8:
147             option = [None] + mobility
148         elif topicID == 9:
149             option = [None] + budget
150         else:
151             raise ValueError(f"Invalid topicID: {topicID}")
152
153         option += [''] * (max_options - len(option) + 1)
154
155     def format_options(opts):
156         opts = [str(x) for x in opts if x]
157         if len(opts) == 0:
158             return ""
159         elif len(opts) == 1:
160             return opts[0]
161         elif len(opts) == 2:
162             return f"{opts[0]} or {opts[1]}"
163         else:
164             return ", ".join(opts[:-1]) + f" or {opts[-1]}"
165
166     current_continent = CONTINENT[0] if CONTINENT[0] else ""
167     current_city = CITY[0] if CITY[0] else ""
168     current_period = PERIOD[0] if PERIOD[0] else ""
169     current_type = HOLIDAYTYPE[0] if HOLIDAYTYPE[0] else ""
170
171     topicoptions = {
172         1: [
173             "What continent or worldpart would you prefer to visit, and what are
174             your reasons for choosing it? Please discuss your ideas together
175             and try to reach an agreement on whether it should be "
176             ] + [format_options(option[1:len(continent)+1])] + ["."],
177         2: [
178             "What city in "
179             ] + [current_continent] + [
180             " would you like to see? There are several options that are available,
181             and each city has it own culture and experiences. The options are "
182             ] + [format_options([x for x in option[1:] if x])] + ["."],
183         3: [
184             "During which time period of the year would you think is best to be in
185             "

```

```

184         ] + [current_city] + [
185             "? Considering factors like the weather and the atmosphere. As these
                factors can have a positive or negative impact on your holiday.
                Would you prefer "
186         ] + [format_options([x for x in option[1:] if x])] + ["?"],
187
188         4: [
189             "This is going great, only a few more steps are needed to complete your
                holiday definition. "
190             "Also important to determine is the total duration of your holiday stay
                . Staying longer gives you opportunities to see more of the city,
                but shorter holidays have their own advantages, such as being more
                flexible and less demanding. "
191             "Would you prefer to go for "
192         ] + [format_options([x for x in option[1:] if x])] + ["?"],
193
194         5: [
195             "Now that you have decided how long you will be staying on holiday,
                which was",
196         ] + [current_period] + [
197             ", what type of vacation would you prefer to have? Different types of
                vacations can give you different experiences of the city. Would you
                like an ",
198         ] + [format_options([x for x in option[1:] if x])] + [
199             " vacation?"],
200
201         6: [
202             "What would you mainly like to do during this ",
203         ] + [current_type] + [
204             " holiday? Would you prefer ",
205         ] + [format_options([x for x in option[1:] if x])] + [
206             ", and what kind of activities or experiences are you most interested
                in while you are there?"
207         ],
208
209         7: [
210             "What type of accommodation would fit this holiday best? Would you
                prefer ",
211         ] + [format_options([x for x in option[1:] if x])] + [
212             ". Will you go for more comfort, or would you rather keep the costs as
                low as possible and give up some comfort?"
213         ],
214
215         8: [
216             "How would you mainly like to travel around at the destination? You can
                choose from the following options, each offering a different way
                to explore the area: ",
217         ] + [format_options([x for x in option[1:] if x])],
218
219         9: [
220             "And finally, what budget would suit this holiday best? Would you
                prefer ",
221         ] + [format_options([x for x in option[1:] if x])] + [
222             ", and how does that fit with what you would like to do and experience
                during your trip?"
223         ],
224
225         10: ending
226     }
227
228     try:
229         output_line = topicoptions[topicID]
230     except:
231         output_line = "???"
232
233     return output_line
234

```

```

235 def set_derived_holiday_details():
236
237     chosen_type = HOLIDAYTYPE[0]
238
239     if not THINGSTODO[0]:
240         if chosen_type in thingstodo:
241             THINGSTODO[0] = random.choice(thingstodo[chosen_type])
242         else:
243             THINGSTODO[0] = "enjoying the destination"
244
245     if chosen_type == "Active":
246         if not MOBILITY[0]:
247             MOBILITY[0] = random.choice(["Foot", "Bike", "Car"])
248         if not BUDGET[0]:
249             BUDGET[0] = random.choice(["up to 1500 euros", "up to 2000 euros"])
250         if not ACCOMODATION[0]:
251             ACCOMODATION[0] = random.choice(["Hostel", "Camping", "Local residence"
252 ])
253
254     elif chosen_type == "Relaxing":
255         if not MOBILITY[0]:
256             MOBILITY[0] = random.choice(["Public transfer", "Foot", "Car"])
257         if not BUDGET[0]:
258             BUDGET[0] = random.choice(["less than 1000 euros", "up to 1500 euros"])
259         if not ACCOMODATION[0]:
260             ACCOMODATION[0] = random.choice(["Hotel", "Holiday resort", "Bed and
261 Breakfast"])
262
263     elif chosen_type == "Partying":
264         if not MOBILITY[0]:
265             MOBILITY[0] = random.choice(["Public transfer", "Car"])
266         if not BUDGET[0]:
267             BUDGET[0] = random.choice(["up to 1500 euros", "more than 2000 euros"])
268         if not ACCOMODATION[0]:
269             ACCOMODATION[0] = random.choice(["Hotel", "Local residence"])
270
271     if not CONTINENT[0]:
272         CONTINENT[0] = random.choice(continent)
273
274     if not CITY[0]:
275         chosen_continent = CONTINENT[0] if CONTINENT[0] else continent[0]
276         CITY[0] = random.choice(city[chosen_continent])
277
278     if not PERIOD[0]:
279         PERIOD[0] = random.choice(travelperiod)
280
281     if not DURATION[0]:
282         DURATION[0] = random.choice(tripduration)
283
284     if not HOLIDAYTYPE[0]:
285         HOLIDAYTYPE[0] = random.choice(holidaytype)
286
287     if not MOBILITY[0]:
288         MOBILITY[0] = random.choice(mobility)
289
290     if not BUDGET[0]:
291         BUDGET[0] = random.choice(budget)
292
293     if not ACCOMODATION[0]:
294         ACCOMODATION[0] = random.choice(accomodation)
295
296     breaksilence = ["So who has any ideas?",
297                     "So what do you both think?",
298                     "Who of you can say something about it?",
299                     "Let us try to share some ideas.",
300                     "There must be some opinion about it."]

```

```

300 | direct_turntake = ["Why do you think that?",
301 |                    "Why do you see it that way?",
302 |                    "Please elaborate a bit more.",
303 |                    "Any more comments?",
304 |                    "What would be another reason?",
305 |                    "And what other reason would there be?",
306 |                    "What could be a counterargument?",
307 |                    "What would be a statement against?",
308 |                    "What would your conversation partner most likely think?",
309 |                    ["What would "]+[OTHERNAME]+["'s opinion be?"],
310 |                    ["What do you think will be the opinion of "]+[OTHERNAME]+["?"],
311 |                    ["Why could "]+[OTHERNAME]+[" think you are "]+[RIGHTWRONG]+["?"
312 |                    ],
313 |                    ["Why could your partner think you are "]+[RIGHTWRONG]+["?" ]]
314 | switch_turntake = ["To what degree would this also be your opinion?"],
315 |                    ["To what extent do you agree with "]+[CURRENTNAME]+["?" ],
316 |                    ["That is interesting, what about you?"],
317 |                    ["Ok fair enough, how about you?" ],
318 |                    ["And what is your view?" ],
319 |                    ["How do you feel about that?" ],
320 |                    ["Anything to add on "]+[CURRENTNAME]+["'s opinion?" ],
321 |                    ["Given what "]+[CURRENTNAME]+[" said, how would you comment on
322 |                    that?" ],
323 |                    ["Given what "]+[CURRENTNAME]+[" said, what would be your
324 |                    opinion?" ]]
325 |
326 | OpenPrefix = ["I can tell you that ",
327 |               "Interesting to know is that ",
328 |               "Did you know that ",
329 |               "According to my information, ",
330 |               "It is often said that ",
331 |               "I like to share with you that "]
332 |
333 | TT_NoneFirstOPT = ["How about "]+[OPT]+["?" ],
334 |                    ["What about "]+[OPT]+["?" ],
335 |                    ["What do you think of "]+[OPT]+["?" ],
336 |                    ["What is your opinion about "]+[OPT]+["?" ],
337 |                    ["What can you say about "]+[OPT]+["?" ],
338 |                    ["Any other judgement about "]+[OPT]+["?" ],
339 |                    ["What is your mind on "]+[OPT]+["?" ]]
340 |
341 | SpeedUP = [
342 |     "You both agree to the same choice?",
343 |     "So we have a mutual agreement?",
344 |     "I guess the answer is clear then?",
345 |     "Is it clear then what option it should be?",
346 |     "It seems the answer is obvious then?",
347 |     "No need to discuss more options, I guess?"
348 | ]
349 |
350 | info = {
351 |     continent[0]: "Asias culture is rich in every sense, from heritage, architecture
352 |     , and way of life , to the intense spirituality of the people. This means a
353 |     trip there can feel both culturally immersive and very different from what
354 |     you may be used to.",
355 |     continent[1]: "Travelling to Africa offers many vacation options. From a boat
356 |     tour and safari in the jungle, to a relaxing spa and viewing the wildlife.
357 |     It can therefore be a destination that combines adventure, nature, and
358 |     relaxation in a unique way.",
359 |     continent[2]: "The USA is a versatile land, which is not surprising when nearly
360 |     4800 kilometres separate people on the west coast from those on the east
361 |     coast. Because of this, travelling there can offer very different holiday
362 |     experiences depending on where you choose to go.",
363 |     continent[3]: "America is a super colourful continent, meaning that the people,
364 |     their clothes, the music, and just life itself there is very diverse. That

```

diversity can make the holiday feel lively, surprising, and full of different cultural influences.”,

354 continent[4]:”With 27 countries located within the European Union alone, Europe offers a big cultural variety of travel experiences. This allows you to choose between atmospheres and traditions that lies relatively closeby.”,

355 continent[5]:”With its vast and varied landscapes, unique wildlife, and white sand beaches, Australia is one of the most interesting continents around. It provides a lot of different cultures and cities to visit, where you can combine nature and city life.”,

356

357 city[continent[0]][0]:”You can enjoy both urban and natural attractions in the mega metropolis Singapore. This makes it a destination where modern city life and calm green spaces can both be part of the same holiday experience.”,

358 city[continent[0]][1]:”It is said that Hong Kong will no doubt surprise you, and that there is an inspiring view of the Symphony of the Stars lightshow from the promenade. Hong Kong has an amazing skyline and gives you an modern and luxury experience.”,

359 city[continent[0]][2]:”No matter which resort in Bali you would choose, it will most likely boast a beautiful beach, an exotic spa, and an array full of dining options. This makes Bali especially suitable for travellers who enjoy comfort, relaxation, and a tropical atmosphere.”,

360 city[continent[0]][3]:”No trip to Tokyo would be complete without visiting some of the Buddhist and Shinto temples and shrines. At the same time, the city is very modern and known for its anime, which is a style of animation.”,

361 city[continent[0]][4]:”Despite the numerous options for things to do in the Maldives, most visitors simply lounge on the palatial resort island of their choice. This would be very relaxing, laying on the beach and escaping the daily stress.”,

362

363 city[continent[1]][0]:”Your could start a day in Cape Town with a morning trip up the Table Mountain from where you will be able to enjoy spectacular views of the city. It is a place where you can also book day trips to see wild life.”,

364 city[continent[1]][1]:”Many visitors of Cairo go for a tour to the Pyramids of Giza, and see more of its ancient Egyptian ruins. This makes the city especially attractive for people who are interested in history and see one of the seven wonders of the world.”,

365 city[continent[1]][2]:”If you like history you can spend most of your time in or around the Medina, Marrakechs fortified old city. The city is often appreciated for its atmosphere, markets, and strong sense of local culture.”,

366 city[continent[1]][3]:”Tanzania is mainly known for Serengeti National Park, which houses a huge population of wildlife large mammals. The people are very kind and the local food is something you have to try.”,

367 city[continent[1]][4]:”Famous for its white idyllic beaches, even the most popular stretches of sand in Seychelles are never crowded. Making it a calm and peaceful destination”,

368

369 city[continent[2]][0]:”The Golden Gate Bridge is a must see in San Francisco, just like a visit to Alcatraz Island to tour the infamous federal prison. The city provides several experiences that has to be visited once in your life.”,

370 city[continent[2]][1]:”You will be surprised by New Yorks flourishing art, night life scenes, and the many huge skyscrapers and monuments. It is a really big city with a lot of people, and you will see extremely much yellow cabs driving around.”,

371 city[continent[2]][2]:”A visit to Las Vegas will most likely revolve around the Strip, this is the place where you will find all the iconic neon lights and famous sights. The city is especially suitable for travellers looking for entertainment, nightlife, but it is mostly known for gambling.”,

372 city[continent[2]][3]:”Night-life and rolling good times are the main attractions in New Orleans, plentiful live music clubs of nearly every style. This gives the city a lively and distinctive atmosphere that stands out from many other destinations.”,

373 city[continent[2]][4]:”Relaxing at the beach is truly the best free activity possible in Miami. At the same time, the city also offers nightlife, warm

weather, and a very recognisable holiday vibe.”,

city[continent[3]][0]:”Buenos Aires has much to offer like boutique—shopping, opera—watching, and tango—dancing. It is a place that loves football and a must do is visiting one of the matches.”,

city[continent[3]][1]:”If it is your first trip to Rio, you will want to savour a chilled coconut as you survey Copacabana beach. You can travel through one of the many favelas with a gids, seeing how the local live overthere.”,

city[continent[3]][2]:”Whale watching and horseback riding are for the adventurous traveller ways you can get acquainted with Argentine. It is therefore a destination that can appeal strongly to people who want outdoor activities and variety.”,

city[continent[3]][3]:”Impressive skyscrapers, colonial architecture and spectacular peaks all jockey for your attention in Santiago. Climbing the Cerro Santa Luc a has to be on your bucket list.”,

city[continent[3]][4]:”Costa Ricas strikingly diverse terrain of forests, wildlife reserves, and tropical beaches, offers something for every traveller. A place where backpacking is very common.”,

city[continent[4]][0]:”Berlins history of battling ideologies makes for some of the most fascinating sightseeing in Europe. Besides its history, the city is also known for its creative atmosphere and the main music is techno.”,

city[continent[4]][1]:”If it is your first time to Paris, you will probably want to spend some time at the Eiffel Tower. Paris is seen as a romantic city, taking a ferry over the seine will give you the perfect view over the city.”,

city[continent[4]][2]:”You do not want to miss out on seeing Gaudis La Sagrada Familia in Barcelona. The city is often appreciated for its mix of beach life and city life. Eating paella and drinking sangria gives you an impression of their local dishes.”,

city[continent[4]][3]:”A must—see in ancient Rome on many travellers agenda is the Trevi Fountain. Rome can offer a strong sense of history as the Colloseum is located there as well.”,

city[continent[4]][4]:”You should definitely visit the Tivoli gardens in Copenhagen located nearby the Central Train Station. The city is very expensive, but it gives a modern look.”,

city[continent[5]][0]:”Quite fascinating to see in Darwin are the big termite mounds in Litchfield natural park. This makes the area especially appealing for travellers who enjoy unusual natural sights and warm outdoor settings.”,

city[continent[5]][1]:”If you are not afraid to get wet feet, maybe rent a kayak to paddle across the twisty river of Brisbane. Giving adventures experiences when visiting this place.”,

city[continent[5]][2]:”In Sydney you should make time for the beach, Bondi and Coogee beach are favourites. Sydney also has the Opera House which has to be visited when you are there.”,

city[continent[5]][3]:”If you are a sports fan, visiting the Cricket Ground in Melbourne is essential. This is one of the main sports in Australia.”,

city[continent[5]][4]:”Rottnest Island in Perth is a protected Class A nature reserve, perhaps nice to enjoy a little nature. This will give you views of wild animals.”,

travelperiod[0]+continent[0]:”In the summer, Asia is for a large part pretty hot, muggy, and typhoon—prone. That means this period may be less comfortable for travellers who prefer mild weather and predictable conditions.”,

travelperiod[1]+continent[0]:”It is a good period to enjoy daytime temps of around thirty degrees with below average room rates in autumn. This combination can make autumn attractive for travellers who want pleasant weather and better value.”,

travelperiod[2]+continent[0]:”While cool temperatures during winter will discourage some travellers, maybe you will actually think it is ok. For some people this can be a welcome change from extreme heat and humidity.”,

travelperiod[3]+continent[0]:”If you wish to avoid both winters climate and summers humidity, spring is an exceptional time to visit. It is often seen as one of the more balanced periods of the year.”,

397
398 travelperiod[0]+continent[1]: "Spending summertime in a desert climate is not
 really advised for travellers. The heat can strongly affect the activities
 you can do.",

399 travelperiod[1]+continent[1]: "Late fall marks a sweet spot in the tourism
 calendar, the summer heat retreats and the crowds have yet to arrive. This
 can make travelling feel more relaxed and manageable.",

400 travelperiod[2]+continent[1]: "Winter is prime tourist season in Africa, with
 visitors hoping to pair sightseeing with pleasant weather. For many
 travellers this makes it one of the most convenient times to go.",

401 travelperiod[3]+continent[1]: "Springtime is a great time to visit Africa since
 the winter crowds are waning and the weather is gorgeous. This can create a
 nice balance between comfort and a less crowded experience.",

402
403 travelperiod[0]+continent[2]: "People from all over the country are drawn by the
 hope for nice weather and the promise of summertime activities in autumn.
 This can make it an appealing period, although busier in some destinations
 and more expensive.",

404 travelperiod[1]+continent[2]: "Fall marks a sweet spot for North Americas
 tourism. Believe it or not, the weather is often warmer now than it is in
 the summer. It can therefore be a very comfortable and attractive time to
 travel.",

405 travelperiod[2]+continent[2]: "If you do not mind the chilly winds, you will
 find that winter is a great time to spend in the United States. This period
 may especially suit travellers who enjoy seasonal atmosphere and lower
 crowds making it less expensive as well.",

406 travelperiod[3]+continent[2]: "You can beat the tourist rush by visiting the USA
 in the spring, when the weather is mild and hotel prices have yet to rise.
 This makes spring one of the best periods to go.",

407
408 travelperiod[0]+continent[3]: "South America winter season is great if you want
 to meet more locals that enjoy the moderate weather. This can make the
 destination feel lively without the discomfort of extreme heat.",

409 travelperiod[1]+continent[3]: "South America spring is an ideal time for seeking
 sun and adventure. It is often a period in which the weather supports both
 exploration and outdoor activities.",

410 travelperiod[2]+continent[3]: "Peak season is autumn in South America, hotel
 prices can be inflated during these months. This means it may be attractive
 , but possibly less ideal for travellers with a tighter budget.",

411 travelperiod[3]+continent[3]: "Crowds and hot summer weather dissipate in May,
 but still expect high humidity. Depending on your preferences, this can
 feel either manageable or slightly uncomfortable.",

412
413 travelperiod[0]+continent[4]: "Be aware that summer forms the tourist season
 with high temperatures, high humidity and high prices for everything.
 Summer provides a lot of sun and almost no rain, but it will be busier and
 more expensive.",

414 travelperiod[1]+continent[4]: "In autumn tourist season slows and hotel rates
 fall a little bit while still having comfortable temperatures. This period
 provides a good balance between comfort and costs.",

415 travelperiod[2]+continent[4]: "You will find some great deals if you travel
 during the winter season, but it will be a little chilly. This period is
 more suitable if budget matters.",

416 travelperiod[3]+continent[4]: "Spring season is possibly the ideal time to
 travel in Europe due to low prices and pleasant temperatures. It is often
 considered one of the most attractive periods if you want to have a holiday
 for sightseeing.",

417
418 travelperiod[0]+continent[5]: "Although wintertime in Australia, do not let that
 label fool you since the calendar is filled with mostly sunny days. This
 can make the season feel much milder and more pleasant than expected.",

419 travelperiod[1]+continent[5]: "While autumn season here, the springtime in
 Australia is marked by warm days and breezy nights with an occasional
 serious rainfall. Many travellers may find this a pleasant and lively time
 to visit.",

420 travelperiod[2]+continent[5]: "Australias wet, humid summer season comes with
 temperatures reaching up to thirty degrees. This may be ideal for some sun-

```

421         seeking travellers , but less comfortable for others." ,
travelperiod[3]+continent[5]: "There is no need to pack anything more than a
    light jacket if you visit Australia during autumn. This can make travelling
    feel easy and comfortable without extreme weather conditions." ,
422
423     tripduration[0]: "Sometimes shorter vacations make a more memorable experience.
    They can feel intense , efficient , and easier to fit into a busy schedule." ,
424     tripduration[1]: "Going for a few weeks will allow for more extensive
    sightseeing. It also gives you more flexibility to combine relaxation with
    multiple activities." ,
425     tripduration[2]: "Going away for a few months can really change your perspective
    on things. A longer stay often allows you to settle in more and experience
    the destination in a deeper way." ,
426
427     holidaytype[0]: "Being active will make sure you experience a lot on your
    vacation. It can be a good fit if you prefer movement, variety , and making
    the most of your time there." ,
428     holidaytype[1]: "Relaxing is a good way to clear your head from stress and your
    day-to-day life. It may suit you best if you want comfort , calmness , and
    time to unwind." ,
429     holidaytype[2]: "It will be an opportunity to meet a lot of new people and make
    friends. This type of holiday can be especially attractive if you enjoy
    energy , nightlife , and social experiences." ,
430 }
431
432 def get_verdict_utterance(chosen_option , verdict_key="1" , verdict_step=1):
433     info_text = info.get(chosen_option , "")
434     prefix = f"{random.choice(OpenPrefix)}{info_text} " if info_text else ""
435
436     # support further discussion
437     if verdict_step == 1:
438         if verdict_key == '2':
439             return (
440                 f"It sounds like multiple options could work. "
441                 f"What do you both think?"
442             )
443
444         elif verdict_key == '3':
445             return (
446                 f"It seems none of the options fully fit your preferences. "
447                 f"What would you suggest?"
448             )
449
450         elif verdict_key == '4':
451             return (
452                 f"I am not completely sure what your final preference is. "
453                 f"What do you both think about {chosen_option}?"
454             )
455
456         else:
457             return f"It sounds like you both agreed on {chosen_option}. Is that
                correct?"
458
459     # final decision
460     else:
461         if verdict_key == '2':
462             return (
463                 f"It sounds like multiple options could work. "
464                 f"{prefix}"
465                 f"For now, I will select {chosen_option}. Is that okay?"
466             )
467
468         elif verdict_key == '3':
469             return (
470                 f"It seems none of the options fully fit your preferences. "
471                 f"{prefix}"

```

```

472         f"I will choose {chosen_option} as the closest option. Is that okay
473         ?"
474     )
475     elif verdict_key == '4':
476         return (
477             f"I am not completely sure what your final preference is. "
478             f"{prefix}"
479             f"I think {chosen_option} might fit best. Is that correct?"
480         )
481
482     else:
483         return f"It sounds like you both agreed on {chosen_option}. Is that
484         correct?"
485
486 def listtostr(listobj):
487     if listobj != None:
488         string = str(listobj)
489         string = string.replace("'", ["'", ""])
490         string = string.replace("['", " ")
491         string = string.replace("']", " ")
492         string = string.replace(", '", " ")
493         string = string.replace(", '", " ")
494         string = string.replace("['", " ")
495         string = string.replace("']", " ")
496     else:
497         string = ''
498     return string

```

D.2 Dream House Script

```

1  import random
2  HOUSELOCATION = [ '']
3  HOUSETYPE = [ '']
4  HOUSESIZE = [ '']
5  OUTSIDE_SPACE = [ '']
6  HOUSESTYLE = [ '']
7  INTERIORSTYLE = [ '']
8  SUSTAINABILITY = [ '']
9  NEIGHBORHOOD = [ '']
10 HOUSEBUDGET = [ '']
11 OPT = [ '']
12 CURENINAME = [0]
13 OTHERNAME = [0]
14 RIGHTWRONG = [0]
15 max_options = 9
16
17 introduction = [
18     "For this conversation the main goal is to figure out what your ideal dream
19     house should be like if you "
20     "have to live there together. "
21     "I am going to ask you some questions about what your ideal dream house is. "
22     "Since the two of you are designing a hypothetical dream house together, "
23     "ask for each others opinion. "
24     "Are you ready to begin?"
25 ]
26
27 # 1 LOCATION
28 # 2 HOUSE TYPE
29 # 3 HOUSE SIZE
30 # 4 OUTSIDE SPACE
31 # 5 HOUSE STYLE
32 # 6 INTERIOR STYLE
33 # 7 SUSTAINABILITY

```

```

33 # 8 NEIGHBORHOOD
34 # 9 BUDGET
35
36 ending = [[
37     "Well thats about it. With all the information combined, you have arranged
        yourselves a dream house located in the "
38 ] + HOUSELOCATION + [
39     ", and it will be a "
40 ] + HOUSETYPE + [
41     " with "
42 ] + HOUSESIZE + [
43     ". The outdoor area will include "
44 ] + OUTSIDE.SPACE + [
45     ", while the overall architectural style will be "
46 ] + HOUSESTYLE + [
47     ". Inside, the house will have a "
48 ] + INTERIORSTYLE + [
49     " atmosphere. In terms of sustainability, it will feature "
50 ] + SUSTAINABILITY + [
51     ", and it will be situated in a "
52 ] + NEIGHBORHOOD + [
53     ". Finally, the house will fit within a "
54 ] + HOUSEBUDGET + [
55     ". Thanks for having participated in our dialogue, the experiment will now
        continue to the next phase."
56 ]]
57
58 houselocation = [
59     "a City",
60     "a Village",
61     "the Countryside",
62     "the Coast"
63 ]
64
65 housetype = [
66     "an Apartment",
67     "a Terrace house",
68     "a Semi-detached house",
69     "a Detached house"
70 ]
71 housesize = [
72     "a 2 bedrooms",
73     "a 3 bedrooms",
74     "a 4 bedrooms",
75     "more than 4 bedrooms"
76 ]
77
78 outside_space = [
79     "no backyard",
80     "a balcony",
81     "a backyard with terrace",
82     "a backyard with swimming pool"
83 ]
84
85 housestyle = [
86     "Minimalistic",
87     "Traditional",
88     "Industrial",
89     "Farmhouse"
90 ]
91
92 interiorstyle = [
93     "cozy and warm",
94     "modern and sleek",
95     "luxurious and elegant",
96     "creative and colorful"
97 ]

```

```

98
99 sustainability = [
100     "solar panels",
101     "a heat pump",
102     "good insulation",
103     "smart energy management"
104 ]
105
106 neighborhood = [
107     "a friendly neighborhood",
108     "an urban neighborhood",
109     "a green environment",
110     "a neighborhood close to work and facilities"
111 ]
112
113 housebudget = [
114     "a low budget",
115     "a medium budget",
116     "a high budget",
117     "a luxury budget"
118 ]
119
120 def build_dream_house_question(topicID=0, alt=0):
121     global option
122
123     if alt == 0:
124         if topicID == 1:
125             option = [None] + houselocation
126         elif topicID == 2:
127             option = [None] + housetype
128         elif topicID == 3:
129             option = [None] + housesize
130         elif topicID == 4:
131             option = [None] + outside_space
132         elif topicID == 5:
133             option = [None] + housestyle
134         elif topicID == 6:
135             option = [None] + interiorstyle
136         elif topicID == 7:
137             option = [None] + sustainability
138         elif topicID == 8:
139             option = [None] + neighborhood
140         elif topicID == 9:
141             option = [None] + housebudget
142         else:
143             raise ValueError(f"Invalid topicID: {topicID}")
144
145         option += [ ' ' ] * (max_options - len(option) + 1)
146
147     def format_options(opts):
148         opts = [str(x) for x in opts if x]
149         if len(opts) == 0:
150             return ""
151         elif len(opts) == 1:
152             return opts[0]
153         elif len(opts) == 2:
154             return f"{opts[0]} or {opts[1]}"
155         else:
156             return ", ".join(opts[:-1]) + f" or {opts[-1]}"
157
158     topicoptions = {
159         1: [
160             "First, it is important to know at what location your dream house would  

             be in? It should fit your lifestyle and routine. Please discuss  

             whether it should be in "
161             ] + [format_options(option[1:len(houselocation)+1])] + [". "],
162

```

```

163     2: [
164         "Now it is time to decide what type of house would your dream house be?
165         There are several options, so would you like to live together in "
166     ] + [format_options(option[1:len(housetype)+1])] + ["?"],
167
168     3: [
169         "What size would you like your dream house to be, looking at the number
170         of bedrooms? How much space do you think you would need for
171         comfort and guests? Would you prefer "
172     ] + [format_options(option[1:len(housesize)+1])] + ["?"],
173
174     4: [
175         "This is going great, only a few more steps are needed to complete your
176         dream house. "
177         "Also important to determine is how the outside area of your dream
178         house should look. "
179         "Would you like "
180     ] + [format_options(option[1:len(outside_space)+1])] + ["?"],
181
182     5: [
183         "Now what kind of appearance would make your house feel alive? Decide
184         what style your dream house has to be? Would you like your house to
185         be "
186     ] + [format_options(option[1:len(housestyle)+1])] + ["?"],
187
188     6: [
189         "What kind of interior atmosphere, think of mood or feelings, would fit
190         your dream house best? Would you prefer something "
191     ] + [format_options(option[1:len(interiorstyle)+1])] + ["?"],
192
193     7: [
194         "Should your house be environmental friendly or future-proof? What
195         sustainability feature would you most like to include? Would you
196         prefer "
197     ] + [format_options(option[1:len(sustainability)+1])] + ["?"],
198
199     8: [
200         "What kind of neighborhood would you like to live in? What will make
201         your everyday life more pleasant and enjoyable? Would you like it
202         to be in "
203     ] + [format_options(option[1:len(neighborhood)+1])] + ["?"],
204
205     9: [
206         "And finally, what budget would fit your decisions of comfort and
207         features for this dream house? Would you prefer "
208     ] + [format_options(option[1:len(housebudget)+1])] + ["?"],
209
210     10: ending,
211 }
212
213 try:
214     output_line = topicoptions[topicID]
215 except Exception:
216     output_line = "???"
217
218 return output_line
219
220 breaksilence = ["So who has any ideas?",
221                 "So what do you both think?",
222                 "Who of you can say something about it?",
223                 "Let us try to share some ideas.",
224                 "There must be some opinion about it."]
225
226 direct_turntake = ["Why do you think that?",
227                    "Why do you see it that way?",
228                    "Please elaborate a bit more.",
229                    "Any more comments?",

```

```

217         "What would be another reason?",
218         "And what other reason would there be?",
219         "What could be a counterargument?",
220         "What would be a statement against?",
221         "What would your conversation partner most likely think?",
222         ["What would "]+[OTHERNAME]+["s opinion be?"]],
223         ["What do you think will be the opinion of "]+[OTHERNAME]+["?"]],
224         ["Why could "]+[OTHERNAME]+[" think you are "]+[RIGHTWRONG]+["?"]
225         ],
226         ["Why could your partner think you are "]+[RIGHTWRONG]+["?"] ]
227
228 switch_turntake = [ ["To what degree would this also be your opinion?"],
229                     ["To what extent do you agree with "]+[CURRENTNAME]+["?"]],
230                     ["That is interesting, what about you?"],
231                     ["Ok fair enough, how about you?"],
232                     ["And what is your view?"],
233                     ["How do you feel about that?"],
234                     ["Anything to add on "]+[CURRENTNAME]+["s opinion?"],
235                     ["Given what "]+[CURRENTNAME]+[" said, how would you comment on
236                     that?"],
237                     ["Given what "]+[CURRENTNAME]+[" said, what would be your
238                     opinion?" ]]
239
240 OpenPrefix = ["I can tell you that ",
241              "Interesting to know is that ",
242              "Did you know that ",
243              "According to my information, ",
244              "It is often said that ",
245              "I like to share with you that "]
246
247 TT_NoneFirstOPT = [ ["How about "]+[OPT]+["?"]],
248                     ["What about "]+[OPT]+["?"]],
249                     ["What do you think of "]+[OPT]+["?"]],
250                     ["What is your opinion about "]+[OPT]+["?"]],
251                     ["What can you say about "]+[OPT]+["?"]],
252                     ["Any other judgement about "]+[OPT]+["?"]],
253                     ["What is your mind on "]+[OPT]+["?"] ]
254
255 SpeedUP = [
256     "You both agree to the same choice?",
257     "So we have a mutual agreement?",
258     "I guess the answer is clear then?",
259     "Is it clear then what option it should be?",
260     "It seems the answer is obvious then?",
261     "No need to discuss more options, I guess?"
262 ]
263
264 info = {
265     houselocation[0]: "Living in the city provides easy access to public
266     transportation, shopping centers, and cultural attractions, making it a
267     convenient choice for those who enjoy a vibrant lifestyle.",
268     houselocation[1]: "A village setting offers a quieter lifestyle with a close-
269     knit community feel, often surrounded by nature and less crowded than the
270     city. It is a great option if you value peace and familiarity.",
271     houselocation[2]: "The countryside provides a peaceful and spacious environment
272     , ideal for those who enjoy nature, farming, or simply a more relaxed pace
273     of life. It provides more space and privacy.",
274     houselocation[3]: "Living near the coast means being close to the beach,
275     offering a scenic environment and potential for water activities, as well
276     as fresh seafood. The views can give a holiday vibe in your everyday life."
277     ,
278     housetype[0]: "An apartment is a compact living space in a multi-unit building,
279     usually offering amenities such as security, gyms, and sometimes a shared
280     garden. This can be a practical choice if you do not want high maintenance.
281     "

```

housetype[1]: "A terrace house is a row of identical houses sharing side walls, popular in urban areas, offering a balance between apartment living and standalone houses. Mostly close to the city and cheaper than a standalone house.",

housetype[2]: "A semi-detached house is connected to another house on one side but offers more space and privacy compared to apartments or terrace houses. Often affordable and less trouble with the neighbors.",

housetype[3]: "A detached house is a standalone building that provides the most privacy and space, ideal for families who prefer a larger living area and possibly a garden. It will be less dependent from others and gives you enough space to life.",

housesize[0]: "A 2-bedroom house provides a bit more space, which can be used for guests, a home office, or a small family. Mostly for starters, but also useful for people that do not want to have a big house.",

housesize[1]: "A 3-bedroom house is a common choice for families, offering enough space for children or additional rooms for hobbies or work. Provides more space and it is a balance between manageable and having enough room.",

housesize[2]: "With 4 bedrooms, there is ample space for a larger family, guests, or even creating specialized rooms like a gym or studio. It is suitable for extra flexibility and gives room for hobby's.",

housesize[3]: "More than 4 bedrooms provide significant living space, ideal for large families, multi-generational households, or those who need extra rooms for various purposes. Fits people who dream of having a vary spacious home with enough room for everything.",

outside_space[0]: "Having no backyard may be typical for city apartments, but it often means less maintenance and more time to enjoy other activities, but it removes your option to sit outside in your own space when it is sunny.",

outside_space[1]: "A balcony provides a small outdoor space to enjoy fresh air, grow some plants, or have a cup of coffee in the morning. It is a good option when you want to live close to the city.",

outside_space[2]: "A backyard with a terrace gives space for outdoor activities like gardening, dining, or playing with pets or children. This is more affordable and gives extra flexibility.",

outside_space[3]: "A backyard with a swimming pool offers luxury and relaxation, ideal for hot summers and entertaining guests. It makes the house feel more exclusive and create possibilities for social activities.",

housestyle[0]: "A minimalistic style emphasizes simplicity, with clean lines and minimal decorations, creating a calm and clutter-free living space. Mostly appreciated by people who value calmness and order.",

housestyle[1]: "Traditional style houses are known for their classic architecture and cozy interiors, often featuring wood and other natural materials. It makes the house feel watm and timeless, which is attractive to many people.",

housestyle[2]: "An industrial style is characterized by raw, unfinished materials like exposed brick and metal, giving a modern, edgy look. People choose for this when they have bold design choices. ",

housestyle[3]: "A farmhouse style combines rustic charm with modern comforts, often featuring wooden beams, spacious kitchens, and comfortable living areas. Creating a welcoming atmosphere with a lot of character.",

interiorstyle[0]: "A cozy and warm interior style has warm colors and soft materials making it a comfortable place. It would be a good choice if you want your home to feel inviting.",

interiorstyle[1]: "A modern and sleek interior is a style that uses open spaces and simple features. A clean and elegant interior style makes the house feel organized and spacious.",

interiorstyle[2]: "A luxurious and elegant interior uses good quality materials and uses aesthetic features. Making it stylish and focuses more on comfort and appearance.",

interiorstyle[3]: "A creative and colorful interior has artistic expression, and playful design choices. This interior style makes the house feel more lively and personal.",


```

292 sustainability[0]: "Solar panels lowers the energy bills , making the house more
293     environmentally friendly. It helps to save energy costs, while also
294     reducing their impact on the environment.",
295 sustainability[1]: "A heat pump is great to produce heat and cool down a home
296     without using gas. This makes it a modern and efficient option for people
297     who want a more sustainable home system.",
298 sustainability[2]: "Excellent insulation improves the climate inside the house
299     and lowers overall energy consumption. An example is that it can make the
300     house feel warmer in the winter and cooler in the summer.",
301 sustainability[3]: "Smart energy management helps monitor and optimize energy
302     use in the house. This can make daily living more efficient and give better
303     control over how the home performs.",
304
305 neighborhood[0]: "A friendly neighborhood provides a place with peace, safety
306     and a strong sense of community. People who value connection and a safe
307     environment often choose this option.",
308 neighborhood[1]: "An urban neighborhood makes it easy to go to shops,
309     restaurants, and social activity. It makes life more convenient, everything
310     is closeby and it can save you energy and time.",
311 neighborhood[2]: "A green environment has beautiful views and easy access to
312     outdoor activities. It creates a peaceful environment and a living
313     experience, making it easier to enjoy nature.",
314 neighborhood[3]: "Living close to work and facilities saves travel time and
315     make daily life more convenient. It might reduce your stress and makes it
316     easier to manage your day.",
317
318 housebudget[0]: "A modest budget asks to make practical choices and
319     prioritizing of features. Meaning you have to decide on what matters most
320     because you can not have everything. ",
321 housebudget[1]: "A medium budget gives a good balance between comfort, quality,
322     and affordability. It is the most realistic option for attractive choices
323     without becoming excessive.",
324 housebudget[2]: "A high budget allows for more customization, extra space, and
325     premium materials. This gives more freedom to shape the house according to
326     your personal wishes and preferences.",
327 housebudget[3]: "A luxury budget offers the freedom to do what you want, to let
328     you choose for extensive space, and exclusive features. There are no
329     limitations and makes it possible to create your ideal house."
330 }
331
332 def set_derived_dream_house_details():
333     if not HOUSELOCATION[0]:
334         HOUSELOCATION[0] = random.choice(houselocation)
335
336     if not HOUSETYPE[0]:
337         HOUSETYPE[0] = random.choice(housetype)
338
339     if not HOUSESIZE[0]:
340         HOUSESIZE[0] = random.choice(housesize)
341
342     if not OUTSIDE.SPACE[0]:
343         OUTSIDE.SPACE[0] = random.choice(outside_space)
344
345     if not HOUSESTYLE[0]:
346         HOUSESTYLE[0] = random.choice(housestyle)
347
348     if not INTERIORSTYLE[0]:
349         INTERIORSTYLE[0] = random.choice(interiorstyle)
350
351     if not SUSTAINABILITY[0]:
352         SUSTAINABILITY[0] = random.choice(sustainability)
353
354     if not NEIGHBORHOOD[0]:
355         NEIGHBORHOOD[0] = random.choice(neighborhood)
356
357     if not HOUSEBUDGET[0]:

```

```

335     HOUSEBUDGET[0] = random.choice(housebudget)
336
337 def get_verdict_utterance(chosen_option, verdict_key="1", verdict_step=1):
338     info_text = info.get(chosen_option, "")
339     prefix = f"{random.choice(OpenPrefix)}{info_text} " if info_text else ""
340     # support further discussion
341     if verdict_step == 1:
342         if verdict_key == '2':
343             return (
344                 f"It sounds like multiple options could work. Let me give you some
345                 useful information. "
346                 f"{prefix}"
347                 f"What do you both think?"
348             )
349         elif verdict_key == '3':
350             return (
351                 f"It seems none of the options fully fit your preferences. Let me
352                 give you some useful information. "
353                 f"{prefix}"
354                 f"What would you suggest?"
355             )
356         elif verdict_key == '4':
357             return (
358                 f"I am not completely sure what your final preference is. Let me
359                 give you some useful information. "
360                 f"{prefix}"
361                 f"What do you both think about {chosen_option}?"
362             )
363         else:
364             return f"It sounds like you both agreed on {chosen_option}. Is that
365                 correct?"
366     # final decision
367     else:
368         if verdict_key == '2':
369             return (
370                 f"It sounds like multiple options could work. "
371                 f"For now, I will select {chosen_option}. Is that okay?"
372             )
373         elif verdict_key == '3':
374             return (
375                 f"It seems none of the options fully fit your preferences. "
376                 f"I will choose {chosen_option} as the closest option. Is that okay
377                 ?"
378             )
379         elif verdict_key == '4':
380             return (
381                 f"I am not completely sure what your final preference is. "
382                 f"I think {chosen_option} might fit best. Is that correct?"
383             )
384         else:
385             return f"It sounds like you both agreed on {chosen_option}. Is that
386                 correct?"
387
388 def listtostr(listobj):
389     if listobj is not None:
390         string = str(listobj)
391         string = string.replace("]", ["'", "'"])
392         string = string.replace("['", "'")
393         string = string.replace("'", "'")
394         string = string.replace("'", "'")
395         string = string.replace("'", "'")

```

```
396         string = string.replace('"', '"', '"')
397         string = string.replace('"', '"', '"')
398         string = string.replace('"', '"', '"')
399         string = string.replace('"', '"', '"')
400     else:
401         string = ''
402     return string
```

APPENDIX B: RedNet X2P Audio Interface Setup

In the experiment, both participants had their own directional microphone. The audio from these microphones had to be transformed into a digital signal that could be used on the HP laptop. This setup needed to handle a stereo signal, which provided two separate audio channels. This would not be possible when it was recorded as one combined signal (mono). The RedNet X2P audio interface was used, which could do this. This audio interface could capture two input channels simultaneously and configure them individually. The audio interface was very suitable for the experimental setup that was used during the experiment because the audio interface supported stereo input. This meant that it could distinguish if the signals came from microphone one (left participant) or from microphone two (right participant). The RedNet X2P worked with Ethernet cables. Both the power supply, which was done through power over Ethernet (PoE), and the input of the microphones. The audio interface also had an audio output that could be connected to a headphone. The headphone was used to test whether the microphones worked and to test whether the audio signals had the right sound volume. Furthermore, the RedNet X2P had several buttons on the interface that could be used to control each input channel. Each microphone input had its own knob to regulate the sound level. The microphones only worked when the 48V power buttons on the interface were turned on. This was also separate for the inputs, making it possible to turn one microphone off and the other one on. The last button that was used was the high-pass filter. This filter already removed low-frequency sounds like table vibrations, footsteps, or ventilation systems, making the signal clean and more reliable.



Figure B.1: Schematic overview of the microphone setup and network connections between the experiment room, network infrastructure, and the RedNet X2P audio interface.

The microphones in the experiment room had to be connected to the RedNet X2P audio interface in the control room. All the connections that were made to set up the microphones are illustrated in Figure B.1. The lab had several wall-mounted boxes with Ethernet ports that connected the experiment room with the control room (green lines). The wall-mounted connection box was linked through other Ethernet cables to a network patch panel in the control room. More Ethernet cables connected the appropriate ports to the microphone inputs of the audio interface. The router, which is also in the network rack, connected the HP laptop with the audio interface through Ethernet (blue line). This connection transported the audio signals to the laptop. Power was provided through the Ethernet network switch (orange line), and the whole system received internet as it is connected to the building's network infrastructure (red line).

APPENDIX C: System Logging Overview

gaze_log.csv	
Variable name	Description
timestamp	Timestamp of the logged gaze sample.
direction	Gaze direction based on yaw angle: left, middle, or right.
yaw	Estimated horizontal head rotation angle in degrees.
pitch	Estimated vertical head rotation angle in degrees.
robot_eye_contact	Indicates whether eye contact with the robot was detected.
human_directed_gaze	Indicates whether the participant looked toward the other participant.
total_eye_contact_time	Total duration of detected robot eye contact during the session.
dialogstage	Current stage of the dialog sequence.
eye_contact_duration_individually	Duration of a single uninterrupted eye-contact episode.
log_facepose_all.txt	
Variable name	Description
timestamp	Timestamp of the logged face-pose frame.
pitch_deg	Estimated vertical head rotation angle in degrees.
yaw_deg	Estimated horizontal head rotation angle in degrees.
direction	Classified face/gaze direction based on yaw: left, middle, or right.
eye_contact	Indicates if head orientation was within the eye-contact threshold
operator_log.csv	
Variable name	Description
state	Finite-state-machine state at the moment of operator input.
key	Keyboard input pressed by the operator.
timestamp	Timestamp of the operator action.
head_position	Robot head orientation at the time of the operator action.
rms_log.csv	
Variable name	Description
timestamp	Timestamp of the audio sample.
left_rms	RMS value for the left participant microphone.
right_rms	RMS value for the right participant microphone.
left_threshold	Speech detection threshold for the left participant microphone.
right_threshold	Speech detection threshold for the right participant microphone.
speech_log.csv	
Variable name	Description
timestamp_start	Indicating the start of a speech or turn event.
timestamp_end	Indicating the end of a speech or turn event.
speaker	Speaker associated with the event.
event_type	Type of logged conversational event.
speaking_time	Duration of the speaking turn.
silence_before	Silence duration preceding the detected turn.
turn_from	Previous speaker.
turn_to	Next speaker.
offset_to_onset_interval	Time between the end of one turn and the start of the next turn.
is_turn_take	Indicates whether the event represents an actual turn transition.
gaze_before_silence	Indicates that the gaze cue occurred before the silence cue.
silence_before_gaze	Indicates that the silence cue occurred before the gaze cue.
required_threshold	Required silence threshold for the robot to initiate turn-taking.
actual_trigger_silence	Actual silence duration of robot turn-take.
dialogstage	Current dialog stage when the speech event was logged.
time_since_gaze_start	Time since the start of the gaze cue.
time_since_silence_start	Time since the start of the silence cue.
interruption_type	Type of detected interruption.
trigger_source	Source that initiated the robot action.
summary_log.csv	
Variable name	Description
metric	Name of the aggregated event.
value	Total value of the corresponding metric.
timeline_log.csv	
Variable name	Description
timestamp	Timestamp of the timeline sample.
speaker	Active speaker.
silence_duration	Detected silence duration in seconds.
yaw	Estimated horizontal head rotation angle in degrees.
pitch	Estimated vertical head rotation angle in degrees.
eye_contact	Indicates whether eye contact was made with the robot.
	Indicates whether the participant was looking toward the other participant.
human_directed_gaze	
head_position	Current robot head orientation.
state	Current finite-state-machine state.
dialogstage	Current stage of the dialog sequence.

APPENDIX D: Questionnaires

Pre-experiment Questionnaire

This pre-experiment questionnaire collects background information to support the study. Please answer the following questions as accurately as possible.

What is your participant ID?

What is your gender?

Male

Female

Prefer not to say

Other: _____

What is your age?

*

Do you have any prior experience with robots?

(e.g., humanoid robots, vacuum cleaners, or service robots)

*

Do you have any prior experience with speech interfaces?

(e.g., Siri or Alexa)

*

What is your torso-to-eye height (in centimeters, measured from the place you sit to your eyes)?

*

Post-experiment questionnaire

Thank you for taking part in this study.

This questionnaire asks about your experience interacting with Misty. Please respond as honestly as possible based on your own impressions.

1. Which task did you work on?

- Planning a holiday
- Designing a dream house

2. What is your participant ID?

Use the scale to reflect how strongly each characteristic applies to Misty, based on your interaction.

(1 = Not at all | 2 = Slightly | 3 = Moderately | 4 = Quite a bit | 5 = Very much)

- | | |
|----------------------------|-----------|
| 3. Competent | 1 2 3 4 5 |
| 4. Capable | 1 2 3 4 5 |
| 5. Strange | 1 2 3 4 5 |
| 6. Happy | 1 2 3 4 5 |
| 7. Aggressive | 1 2 3 4 5 |
| 8. Knowledgable | 1 2 3 4 5 |
| 9. Responsive | 1 2 3 4 5 |
| 10. Awkward | 1 2 3 4 5 |
| 11. Compassionate | 1 2 3 4 5 |
| 12. Dangerous | 1 2 3 4 5 |
| 13. Emotional | 1 2 3 4 5 |
| 14. Interactive | 1 2 3 4 5 |
| 15. Social | 1 2 3 4 5 |
| 16. Reliable | 1 2 3 4 5 |
| 17. Awful | 1 2 3 4 5 |
| 18. Organic | 1 2 3 4 5 |
| 19. Scary | 1 2 3 4 5 |
| 20. Seems to have feelings | 1 2 3 4 5 |
-

Please indicate to what extent you agree with the following statements about your interaction with Misty.

(1 = Strongly disagree | 2 = Disagree | 3 = Neutral | 4 = Agree | 5 = Strongly agree)

- | | |
|-----------------------|-----------|
| 21. Misty is pleasant | 1 2 3 4 5 |
|-----------------------|-----------|

22. The interaction with Misty is frustrating	1 2 3 4 5
23. I always knew what to say or do at each point when interacting with Misty	1 2 3 4 5
24. Misty is accurate	1 2 3 4 5
25. The interaction with Misty is repetitive	1 2 3 4 5
26. Misty responded quickly	1 2 3 4 5
27. I felt tense while interacting with Misty	1 2 3 4 5
28. The interaction with Misty is efficient	1 2 3 4 5
29. Misty is too inflexible	1 2 3 4 5
30. I was not always sure what Misty was doing	1 2 3 4 5
31. The interaction with Misty is boring	1 2 3 4 5
32. Misty makes few errors	1 2 3 4 5
33. I felt confident using Misty	1 2 3 4 5
34. The interaction with Misty is unpredictable	1 2 3 4 5
35. Misty took too long to respond	1 2 3 4 5
36. It is clear how to speak to Misty	1 2 3 4 5
37. Misty is unreliable	1 2 3 4 5
38. I enjoyed interacting with Misty	1 2 3 4 5
39. The interaction with Misty is consistent	1 2 3 4 5
40. Misty is useful	1 2 3 4 5
41. I felt calm while interacting with Misty	1 2 3 4 5
42. The interaction with Misty is irritating	1 2 3 4 5
43. Misty did not always do what I wanted	1 2 3 4 5
44. I was able to recover easily from errors when interacting with Misty	1 2 3 4 5
45. It is easy to lose track of where you are in the interaction with Misty	1 2 3 4 5
46. Misty is friendly	1 2 3 4 5
47. The interaction with Misty is fast	1 2 3 4 5
48. I sometimes wondered if I was using the right words when interacting with Misty	1 2 3 4 5
49. Misty is easy to use	1 2 3 4 5
50. Misty did not always do what I expected	1 2 3 4 5
51. A high level of concentration is required when interacting with Misty	1 2 3 4 5
52. I always knew what to say to Misty	1 2 3 4 5
53. I would use Misty again	1 2 3 4 5
54. Misty is dependable	1 2 3 4 5
55. Misty responds too slowly	1 2 3 4 5
56. I felt in control of the interaction with Misty	1 2 3 4 5
57. It is easy to learn how to use Misty	1 2 3 4 5

This was the end of the experiment. Thank you for participating. If you have any comments, please share them here.

APPENDIX E: Informed Consent Form



Information sheet for research project “Turn-Taking in a multi-party human-robot setting”

1. Introduction

Eindhoven University of Technology (TU/e) invites you to take part in this research project which aim is to investigate how eye contact can influence turn-taking in conversations between humans and a social robot. The data that will be collected through observations and questionnaires will be used for analysis in a master’s thesis, a presentation, and a potential scientific publication. You are receiving this invitation because your details are listed in the JSF database. For your participation you will receive a compensation of 12 euro’s when you are a student at the TU/e or Fontys or 15 euro’s if you do not study at the TU/e or Fontys as a sign of our appreciation.

Joining this research project is your choice. Your participation is completely voluntary and does not pose any physical, legal or economic risks. You are not obliged to answer questions you are uncomfortable with, and you can withdraw from the research at any time without explaining why. Declining or withdrawing will not have negative impact for you.

Throughout the research, you might be asked personal questions related to the delicate topic of turn-take behavior with the robot. During the study, audio and video recordings will be made of the interaction. Furthermore responses to the questionnaires will be collected and analyzed. Additionally, participants may be asked personal questions related to the research topic. Noteworthy, the recordings and responses will be used for the research purpose only. We ask these questions only for the benefit of the research project.

Before you decide, please read the following information to understand what the project is about, what we expect from you and how we handle your personal data. After reading, you can sign up by completing the attached form.

If you have questions, feel free to contact us (contact details below). You can also discuss this information with people you trust.

2. Who are we?

This research project is conducted by:

Technische Universiteit Eindhoven (TU/e) De Groene Loper 3 5612 AE Eindhoven	Data controller
--	-----------------

How to contact us?

Cas Crijns	c.e.j.r.crijns@student.tue.nl
First supervisor: Raymond Cuijpers	R.H.Cuijpers@tue.nl
Second supervisor: Chao Zhang	C.Zhang.5@tue.nl

3. What will taking part in the research project involve?

In the research project we will collect your personal data using the following methods:

- Participating in a laboratory experiment in which you engage in a conversation with a robot (Misty II) and another participant.
- Completing a pre-questionnaire containing questions about your demographic information and prior experience with robots.
- Completing a post-questionnaire about your experience and perception of the interaction with the robot.
- Taking part in a collaborative task where you have to plan a hypothetical holiday.

- Audio and video recordings will be made of the conversation to analyze the speech, turn-taking behavior and head movements.

4. What personal data from you do we gather and process?

We collect and process the following personal data which is necessary for the project purpose:

Category and type	Personal data	Purpose	Retention period
Contact data (regular personal data)	Name, e-mail, home address, city of residence	To communicate with you and invite you to events	6 months after data collection

Your data is retained only for the time period as specified in the table. Keeping your data for this period helps us to comply with scientific principles, such as producibility and verification.

After this period, your personal data will be deleted or anonymized to ensure it can no longer be linked to you. Unless you explicitly agree to the use of your identifiable information in publications (for example your quotes or your name).

5. Stopping your participation and your rights

If you end your participation in the research we will not use your data anymore from that moment on.

For questions, ending your participation or complaints, please contact the researcher via the contact details as provided in the table under section 2.

For concerns or questions about the handling of personal data, you can e-mail the data protection officer as indicated in the table under section 2. You can also file a complaint with the Dutch data protection authority: the Autoriteit Persoonsgegevens.

You have the right to request access, rectification, objection, erasure or adaptation of your data. Submit your request through Cas Crijns via the contact details as provided in the table under section 2.

6. Legal basis for processing your personal data

The TU/e processes your personal data to conduct scientific research, which is the university's public task as stated in article 1.3 of the Dutch Wet Hoger onderwijs en Wetenschappelijk onderzoek. The TU/e always follows the applicable codes of conduct for research integrity and the scientific standards when conducting research.

7. Who has access to your personal data?

Only authorized employees involved in the research have access to your personal data if this is necessary for their tasks. The authorized employees will keep your personal data confidential.

Other parties/processors that have access to the data are listed in the table below:

Party/processor	Why access?	Category	Processing within the European Economic Area?
SURF Research Drive / Microsoft (Netherlands)	Storage solution		Yes

TU/e has a suitable agreement with these parties to protect your personal data. We will not share your personal data with any other party, unless we are required to do so by law.

TU/e will process your personal data within the European Economic Area (EEA) by storing your data on a server inside the EEA. In addition, TU/e has implemented appropriate technical and organizational

measures to protect your personal data. These measures include using centrally managed and verified research- and storage tools.

8. Future research

During the research project, your personal data may also prove useful for new, socially important research. In such case, we would like to reuse your data for the new research. This will only happen if (1) your personal data is truly necessary, (2) the recognized ethical standards for scientific research are followed, and (3) the new research objectives align with the current research objectives. If your personal data is used in future research, we will take all reasonable steps to inform you about this. You can object to the use of your personal data for new research.

We might use anonymized data for new purposes such as research or education. We will ensure the data cannot be linked to you and we will not disclose anything that makes you identifiable.

This research has been assessed and approved by the ethical committee of Eindhoven University of Technology.

By signing this form, I confirm:

1. I have enough information about the research project from the separate information sheet. I have read it and I had the chance to ask questions, which have been answered to my satisfaction.
2. I take part in this research project voluntarily. There is no explicit or implicit pressure for me to take part in this research project and I understand I can stop my participation at any moment, without explaining why. I do not have to answer any question I do not want to answer.
3. I know my personal data will be collected and used for the research, as explained to me in the information sheet.

Furthermore, I consent to the following parts of the research project:

4. I consent to my personal data, such as footage taken of me, audio taken of me, answers or quotes I gave during the research, to be used by the researcher in publications and/or in presentations – without including my name.

YES ☐ NO ☐

Name of Participant:

Name of researcher: Cas Crijns

Signature:

Signature:

Date:

Date:

APPENDIX F: Dialog Manager Constructs

Construct name	Description
Dialogue Structure	
introduction	Opening utterance explaining the task and setting expectations
topicoptions	Utterances introducing each decision topic and options
topic sequence	Fixed sequence of decision topics guiding the interaction
ending	Final summary combining all selected options into one holiday
Dialog Stages Holiday vs Dream House	
0. INTRODUCTION	Gives a short explanation of the task
1. CONTINENT / LOCATION	Destination
2. CITY / HOUSE TYPE	Specific destination (depends on continent) / Specifies the structural type of the house
3. PERIOD / HOUSE SIZE	Time of travel / How much space inside
4. DURATION / OUTSIDE SPACE	Length of stay / How much space in the backyard
5. HOLIDAYTYPE / HOUSE STYLE	Overall style of the holiday / Overall style of the house
6. THINGSTODO / INTERIOR STYLE	Activities (dependent on holiday type) / Defines the look and feel of the interior
7. ACCOMMODATION / SUSTAINABILITY	Type of stay / Environmentally friendly features
8. MOBILITY / NEIGHBORHOOD	Transport / Describes the surrounding environment
9. BUDGET / BUDGET	Financial constraint
10. END	Final summary
Interaction Constructs	
breaksilence	Utterances used when participants remain silent for at least 4 seconds
direct_turntake	Prompts encouraging elaboration from current speaker
switch_turntake	Prompts shifting the turn to the other participant
TT_NoneFirstOPT	Prompts listener to speak up their opinion about the topic
SpeedUP	Lets the participant wrap up to speed up the conversation
longturnindicator	Acknowledgments during long speaking turns
Information & Framing Constructs	
info	Information about topics
OpenPrefix	Phrases introducing informational utterances
option formatting	Formatting options into readable phrases
Verdict Utterance	
Verdict - Single	A mutual agreement is made
Verdict - Multiple	Multiple options, one is selected by operator
Verdict - Unclear	No agreement, one is selected by operator
Verdict - None	All options rejected, one is selected by operator
Dependencies in Holiday Planner	
CITY	Depends on CONTINENT
THINGSTODO	Depends on HOLIDAYTYPE
Variables	
set_derived_holiday_details()	When skipping questions this function automatically fills in the dialogs, so the end summary does not have missing values
get_verdict_utterance()	Decides whether there is a mutual agreement otherwise it provides extra information, If there is not a mutual agreement the second time for the same dialog state the operator chooses the topic

APPENDIX G: Use of AI

For this Master's Thesis Project, generative AI, specifically OpenAI ChatGPT, is used. This OpenAI is used to improve grammar and spelling mistakes. Furthermore, it is used to help with programming problems in Python and Stata, and extending the scripts for the experiment.